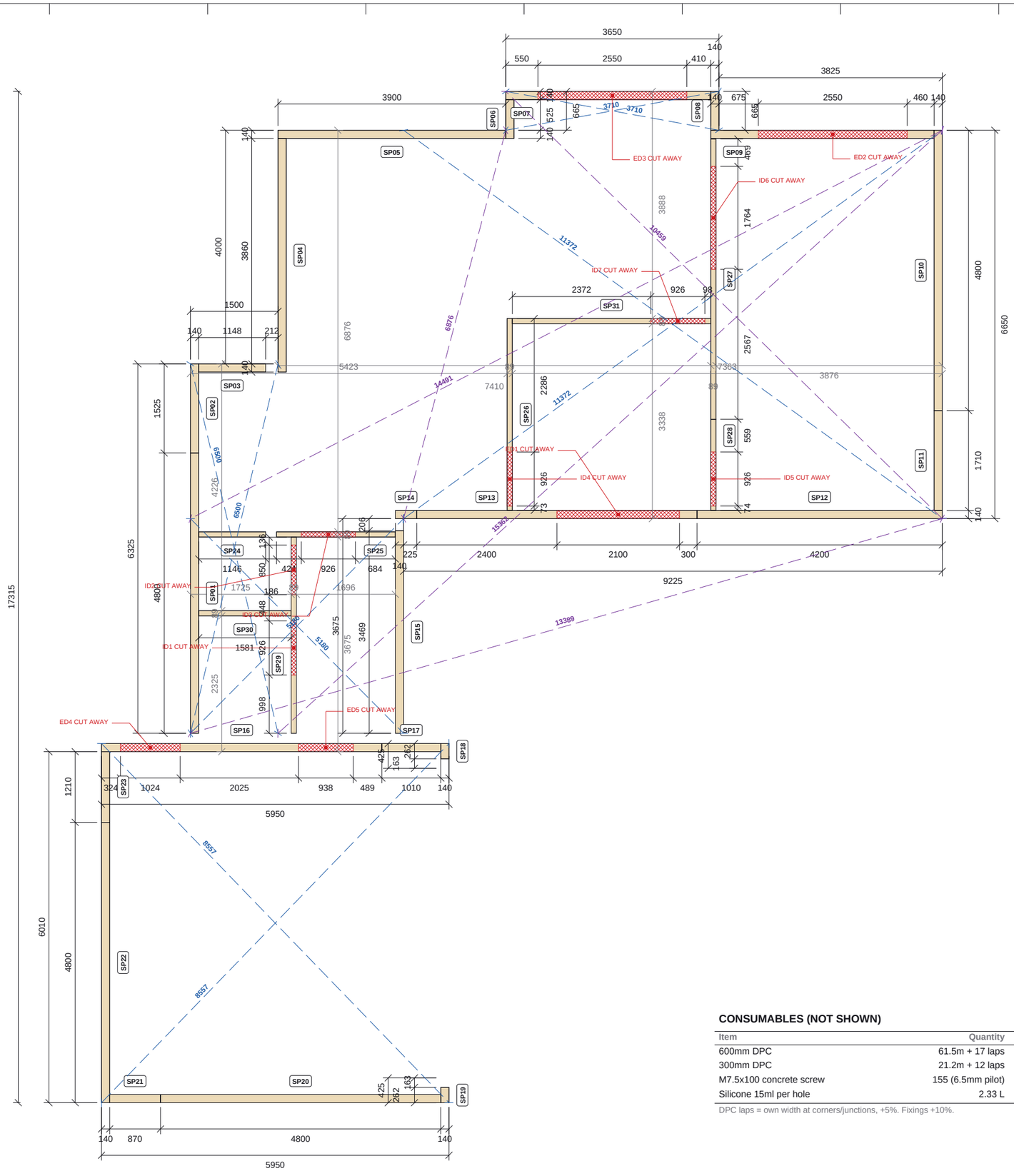


GROUND FLOOR SOLE PLATE SETTING OUT PLAN

Scale 1:75 @ A3 | Hatched red = sole plate cut away on site at door openings

Do not scale. All dimensions in mm. Check all dimensions on site.

SOLE PLATE SCHEDULE

SOLE PLATES

Section	Pieces	Run m	4.8m lengths
38x140 C16	23	61.4	14
38x89 C16	8	21.2	7

Cut-optimised 4.8m stock, 3mm kerf, +5%.

SOLE PLATE CUT LIST

Mark	Section	Length	Cut-away
SP01	38x140	4800	
SP10	38x140	4800	
SP13	38x140	4800	ED1
SP16	38x140	4800	ED4, ED5
SP20	38x140	4800	
SP22	38x140	4800	
SP12	38x140	4200	
SP04	38x140	4000	
SP05	38x140	3900	
SP09	38x140	3825	ED2
SP07	38x140	3510	ED3
SP15	38x140	3469	
SP11	38x140	1710	
SP02	38x140	1525	
SP23	38x140	1210	
SP03	38x140	1148	

Mark	Section	Length	Cut-away
SP17	38x140	1010	
SP21	38x140	870	
SP06	38x140	665	
SP08	38x140	665	
SP14	38x140	365	
SP18	38x140	262	
SP19	38x140	262	
SP27	38x89	4800	ID6
SP31	38x89	3396	ID7
SP29	38x89	3358	ID1, ID2
SP26	38x89	3285	ID4
SP25	38x89	2034	ID3
SP30	38x89	1581	
SP28	38x89	1559	ID5
SP24	38x89	1146	

Door cut-aways are within the piece: fix the plate, then cut the door section away. All C16.

CHECK DIAGONALS

Block	Size	Diag 1	Diag 2
Main building	9225 x 6650	11372	11372
Block 2	3650 x 3675	5180	5180
Block 3	1500 x 6325	6500	6500
Block 4	3650 x 665	3710	3710
Main to Block 2	corner-corner	14491	13389
Main to Block 3	corner-corner	15362	13389
Main to Block 4	corner-corner	10459	6876
Block 5	5950 x 6150	8557	8557

Outer plate corners. Block diagonals equal; purple = main body to pop-out, match figures.

NOTES

1. Sole plates: 38x140 C16 under external and internal load-bearing walls; 38x89 C16 under internal partitions. Outer face of sole plate = outer face of timber frame.
2. DPC under all sole plates: 600mm wide centred under external walls, 300mm wide under internal walls. Lap DPC by its own width at every corner and junction.
3. Fix sole plates with M7.5x100 concrete screws at max 600mm centres, one 50mm from each door opening (retained side) and one at each plate end at corners and junctions.
4. Pre-drill each hole with a 6.5mm bit and fill with 15ml silicone before fixing.
5. Sole plate across door openings (hatched red) is left in for erection and CUT AWAY ON SITE.
6. Check each block for square before fixing: its two diagonals (blue dashed, length figure on the line) must be equal; see the Check diagonals schedule.
7. Dimensions are to sole plate faces. Read with P4-GA-01 and engineer's drawing 26-431-S-0010.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

CONSUMABLES (NOT SHOWN)

Item	Quantity	Order
600mm DPC	61.5m + 17 laps	3 x 30m roll
300mm DPC	21.2m + 12 laps	1 x 30m roll
M7.5x100 concrete screw	155 (6.5mm pilot)	2 box of 100
Silicone 15ml per hole	2.33 L	10 x 280ml

DPC laps = own width at corners/junctions, +5%. Fixings +10%.

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

GROUND FLOOR SOLE PLATE SETTING OUT

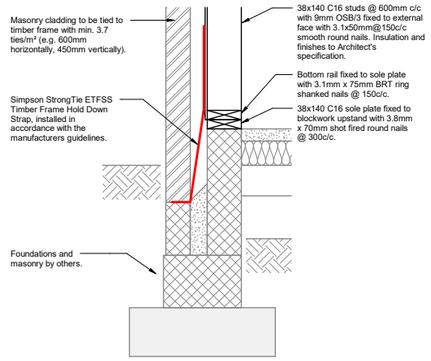
DPC, fixings and silicone are scheduled - not shown

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

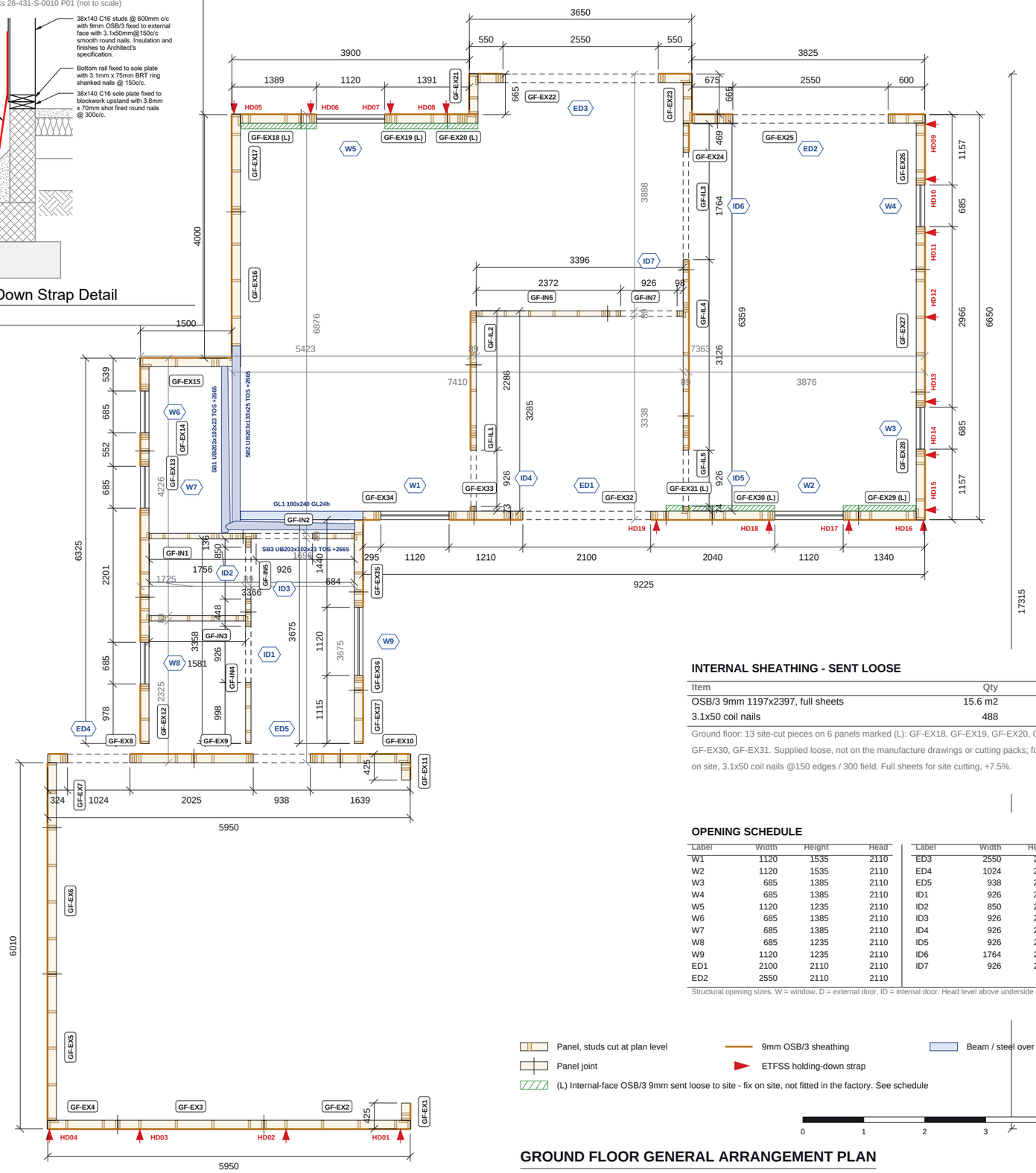
STATUS	DRAWING No.	REV
CONSTRUCTION	P4-SP-01	C03

DETAIL 1 - ETFSS HOLDING DOWN STRAP

Engineer's detail reproduced from Skyhooks 26-431-S-0010 P01 (not to scale)



1 Plot 4 - Holding Down Strap Detail
1 : 10



INTERNAL SHEATHING - SENT LOOSE

Item	Qty	Order
OSB/3 9mm 1197x2397, full sheets	15.6 m2	6 sheets
3.1x50 coil nails	488	1 x box 2,500

Ground floor: 13 site-cut pieces on 6 panels marked (L): GF-EX18, GF-EX19, GF-EX20, GF-EX29, GF-EX30, GF-EX31. Supplied loose, not on the manufacture drawings or cutting packs; fix to the panel studs on site, 3.1x50 coil nails @150 edges / 300 field. Full sheets for site cutting, +7.5%.

OPENING SCHEDULE

Label	Width	Height	Head
W1	1120	1535	2110
W2	1120	1535	2110
W3	685	1385	2110
W4	685	1385	2110
W5	1120	1235	2110
W6	685	1385	2110
W7	685	1385	2110
W8	685	1235	2110
W9	1120	1235	2110
ED1	2100	2110	2110
ED2	2550	2110	2110

Label	Width	Height	Head
ED3	2550	2110	2110
ED4	1024	2110	2110
ED5	938	2110	2110
ID1	926	2050	2050
ID2	850	2050	2050
ID3	926	2050	2050
ID4	926	2050	2050
ID5	926	2050	2050
ID6	1764	2050	2050
ID7	926	2050	2050

Structural opening sizes. W = window, D = external door, ID = internal door. Head level above underside of sole plate.

Panel, studs cut at plan level
Panel joint
(L) Internal-face OSB/3 9mm sent loose to site - fix on site, not fitted in the factory. See schedule
9mm OSB/3 sheathing
ETFSS holding-down strap
Beam / steel over

0 1 2 3 4 5m

GROUND FLOOR GENERAL ARRANGEMENT PLAN

Scale 1:75 @ A3 | Plan cut at 1200 above slab

GROUND FLOOR SCHEDULES

HEAD BINDER

Section	Pieces	Run m	4.8m lengths
38x89 C16	3	9.6	4
38x140 C16	25	61.3	14

External and load-bearing walls only. Cut-optimised 4.8m stock, 3mm kerf, +5%.

TEMPORARY BRACING (NOT SHOWN)

Item	No.	Order
Brace 4x1 treated rough-sawn, 2.7m	24	13 x 5.4m
Noggin 38x89 C16, 600	24	5 x 4.8m
M7.5x100 concrete screw (2 per noggin)	48	1 box of 100

One brace per panel joint on external / load-bearing walls (24 joints). 2 braces per 5.4m; +5%.

WINDOW BATTENS 38X43

Item	Qty	Order
Battens (head, sill, jambs)	51 / 72.5m	18 x 4.8m 38x43
Ripped from 38x89 C16 4.8m	2 per length	9 x 4.8m 38x89

External openings only. Head/sill = width + 76; jambs = height. Cut-optimised, 3mm kerf, +5%.

HOLDING-DOWN STRAPS (ETFSS)

Mark	Wall	At	On
HD01	W1/1	159	junction
HD02	W1/1	2038	stud
HD03	W1/1	4438	stud
HD04	W1/1	5923	end stud
HD05	W5/3	33	end stud
HD06	W5/3	1294	king W5
HD07	W5/3	2604	king W5
HD08	W5/3	3518	end stud
HD09	W5/8	159	end stud
HD10	W5/8	1062	king W4

Mark	Wall	At	On
HD11	W5/8	1937	king W4
HD12	W5/8	3326	stud
HD13	W5/8	4713	king W3
HD14	W5/8	5588	king W3
HD15	W5/8	6491	junction
HD16	W5/9	20	end stud
HD17	W5/9	1245	king W2
HD18	W5/9	2555	king W2
HD19	W5/9	4405	king ED1

19 No. Simpson Strong-Tie ETFSS, engineer 26-431-S-0010 Detail 1. 'At' = from the wall's start corner along the external frame face.

NOTES

1. Panels: 38x140 C16 studs at 600 centres, 9mm OSB/3 to the external face (external and load-bearing walls); 38x89 C16 partitions. Panel marks refer to the manufacture drawings.
2. Head binder 38x140 C16 over external and load-bearing walls only; joints staggered from panel joints.
3. Temporary bracing (not shown): one 2.7m 4x1 treated brace at every panel joint of external and load-bearing walls, through a 600 38x89 noggin fixed with 2 No. M7.5x100 concrete screws.
4. Window battens 38x43 to external openings: head and sill run over the jambs by 38; doors head and jambs.
5. ETFSS holding-down straps (red) on the stud scheduled, to engineer's Detail 1 and Simpson guidance.
6. Beams and steel over (dark blue) set out as dimensioned; TOS = top of steel above underside of sole plate. Dimensions to timber frame faces. Read with P4-SP-01, P4-FL-01 and engineer's drawing 26-431-S-0010.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

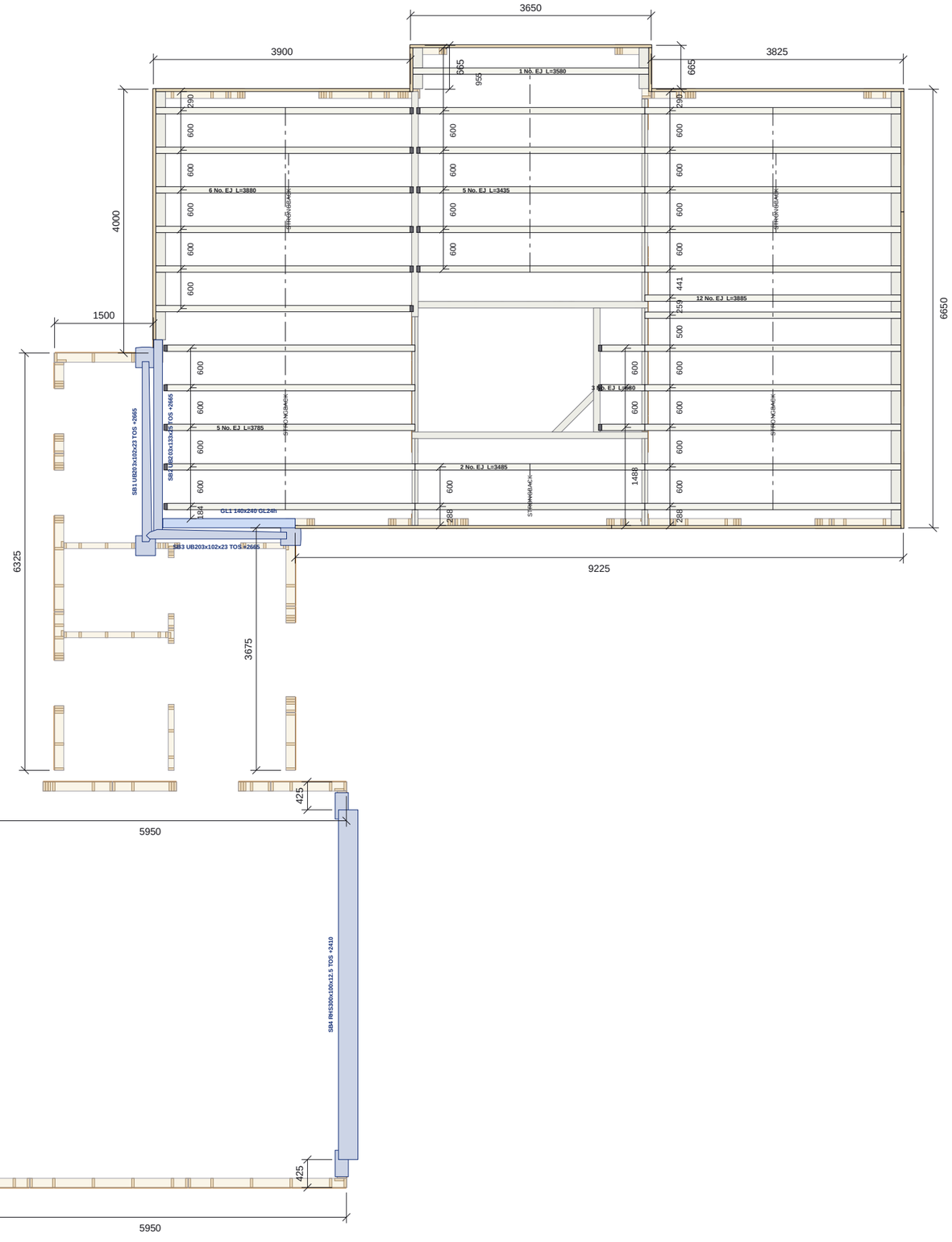
GROUND FLOOR GENERAL ARRANGEMENT

Panels, openings, holding-down straps, beams and steel over

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-GA-01	C03

Do not scale. All dimensions in mm. Check all dimensions on site.



FIRST FLOOR STRUCTURE PLAN - INDICATIVE

Scale 1:75 @ A3 | Ground floor walls below as on the GA (studs cut at 1200); joists by Acorn Timber Engineering

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

FIRST FLOOR STRUCTURE SCHEDULES

BEAMS

Mark	Section	Length	No.
GL1	140x240 GL24h	2008	1
SB1	UB203x102x23 S355JR (TLS)	2729	1
SB2	UB203x133x25 S355JR (TLS)	2955	1
SB3	UB203x102x23 S355JR (TLS)	2126	1
SB4	RHS300x100x12.5 S355J2H (TLS)	5300	1
PK	14mm marine ply packer, 100 wide	15.0m	6

PK: 14mm marine ply fixed to the top of each 240mm glulam to give the full 254mm floor depth (240 + 14 = 254, flush with the joists and edgebinder). Glulam and steel supplied by TLS; set out to faces as dimensioned; bearings to engineer. Steel TOS above underside of sole plate.

METAL-WEB JOISTS - BY ACORN

Type	Section	Length	No.
EJ	97x254 metal web	680	3
EJ	97x254 metal web	3435	5
EJ	97x254 metal web	3485	2
EJ	97x254 metal web	3580	1
EJ	97x254 metal web	3785	5
EJ	97x254 metal web	3880	6
EJ	97x254 metal web	3885	12

Acorn Timber Engineering supply, with 19 hangers, 5 strongback runs and noggins. Not in TLS procurement. Indicative; Acorn's layout supersedes.

EDGEBINDER, WAIST BAND, BREATHER

Item	Quantity	Order
Edgebinder 254 (45x195 + 45x59 rip)	32.3m	9 x 4.8m 45x195
45x59 C24 rips, 3 per 45x195	32.3m	3 x 4.8m 45x195
45x195 C24 total		12 x 4.8m
OSB/3 waist band 290x9	14 strips	4 x 1197x2397
Breather 450 x 2.7m strips, 150 lap	13 strips	14 (1 roll)

Run = modelled edgebinders. Timber cut-optimised +5%; OSB +7.5%; membrane laps +5%.

FLOOR DECKING

Item	Area	Order
22mm Egger P5 600x2400	73.6 m2	55 sheets
D4 adhesive 1L (per 6.5 sheets)		9 x 1L

Laid area from the model (27 full sheets + 49 cut pieces) = 51.1 sheets, +7.5%.

NOTES

1. INDICATIVE ONLY. Metal-web joists, hangers, strongbacks and noggins are designed and supplied by Acorn Timber Engineering; this drawing is superseded by Acorn's engineered floor layout on receipt.
2. Glulam beams 140x240 GL24h (dark blue) set out to the faces dimensioned, bearing on multi-ply studs to engineer's drawing 26-431-S-0010. 14mm marine ply fixed to the top of each 240mm glulam to give the full 254mm floor depth. Steel RHS300x100x12.5 / UB203x102x23 / UB203x133x25 beams and SHS100x100x6.3 / SHS120x120x10 posts supplied by TLS.
3. Edgebinder 254mm = 45x195 C24 with a 45x59 C24 rip fixed on top (rips cut 3 per 45x195, 3mm kerf).
4. OSB/3 waist band 290x9 to the outer face of the edgebinder, strips ripped 4 per 1197x2397 sheet.
5. Breather membrane over the waist band in 450 deep x 2.7m strips, each lapped 150 over the next.
6. 22mm Egger P5 flooring glued with D4 adhesive, fixed to Acorn's specification. Dimensions to frame and beam faces.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

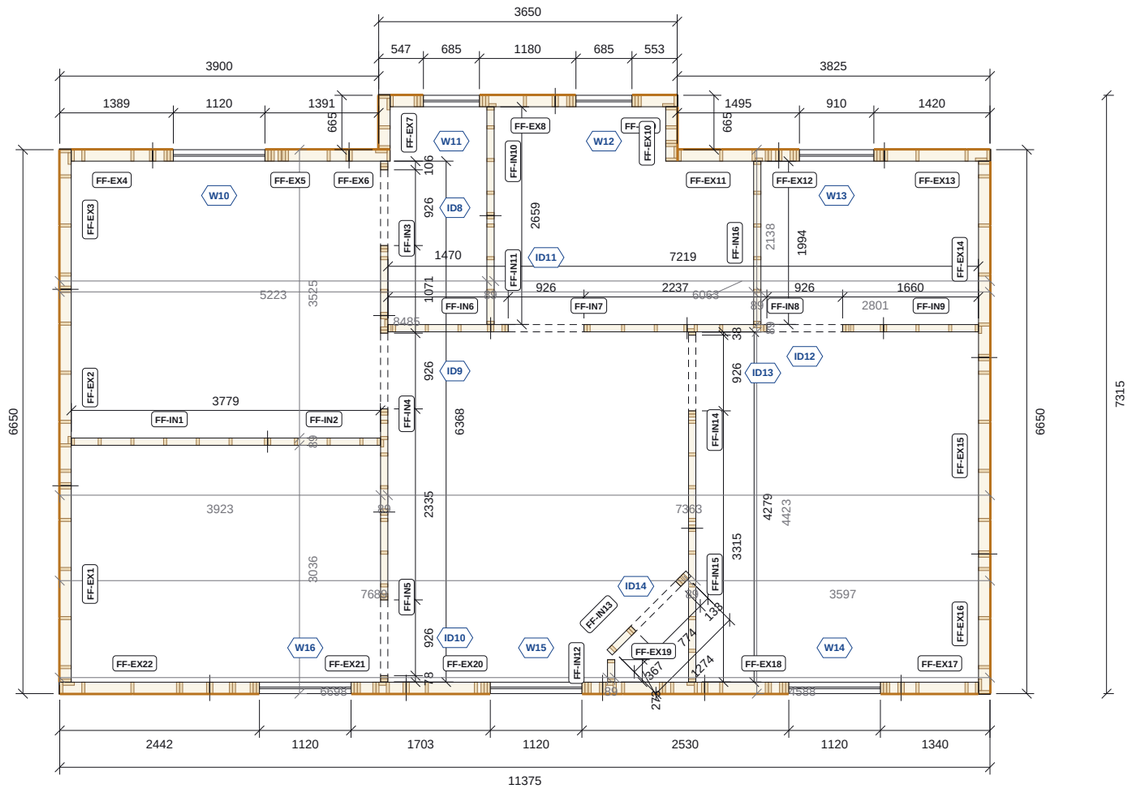
CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE
FIRST FLOOR STRUCTURE - INDICATIVE

Beams, set-out and metal-web joist positions; Acorn layout supersedes

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-FL-01	C03



FIRST FLOOR GENERAL ARRANGEMENT PLAN

Scale 1:75 @ A3 | Plan cut at 1200 above floor deck

Panel, studs cut at plan level

Panel joint

9mm OSB/3 sheathing

Beam / steel over

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

FIRST FLOOR SCHEDULES

HEAD BINDER

Section	Pieces	Run m	4.8m lengths
38x140 C16	12	36.8	10

External and load-bearing walls only. Cut-optimised 4.8m stock, 3mm kerf, +5%.

TEMPORARY BRACING (NOT SHOWN)

Item	No.	Order
Brace 4x1 treated rough-sawn, 2.7m	14	8 x 5.4m
Noggin 38x89 C16, 600	14	3 x 4.8m

One brace per panel joint on external / load-bearing walls (14 joints). 2 braces per 5.4m; +5%.

WINDOW BATTENS 38X43

Item	Qty	Order
Battens (head, sill, jambs)	28 / 32.8m	9 x 4.8m 38x43
Ripped from 38x89 C16 4.8m	2 per length	5 x 4.8m 38x89

External openings only. Head/sill = width + 76; jambs = height. Cut-optimised, 3mm kerf, +5%.

OPENING SCHEDULE

Label	Width	Height	Head
W10	1120	1235	2073
W11	685	1235	2073
W12	685	1235	2073
W13	910	1235	2073
W14	1120	1385	2073
W15	1120	1385	2073
W16	1120	1385	2073

Label	Width	Height	Head
ID8	926	2050	2050
ID9	926	2050	2050
ID10	926	2050	2050
ID11	926	2050	2050
ID12	926	2050	2050
ID13	926	2050	2050
ID14	774	2050	2050

Structural opening sizes. Head level above underside of panel bottom rail.

NOTES

1. Panels: 38x140 C16 studs at 600 centres, 9mm OSB/3 to the external face (external and load-bearing walls); 38x89 C16 partitions. Panel marks refer to the manufacture drawings.
2. Head binder 38x140 C16 over external and load-bearing walls only; joints staggered from panel joints.
3. Temporary bracing (not shown): one 2.7m 4x1 treated brace at every panel joint of external and load-bearing walls, through a 600 38x89 noggin nailed to the deck.
4. Window battens 38x43 to external openings: head and sill run over the jambs by 38; doors head and jambs.
5. Steel over (dark blue, TLS supply) set out as dimensioned; TOS = top of steel above top of 22mm deck. Attic trusses and cut roof members over by Acorn Timber Engineering, see P4-RF-02. Dimensions to timber frame faces. Read with P4-FL-01 and engineer's drawing 26-431-S-0010.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

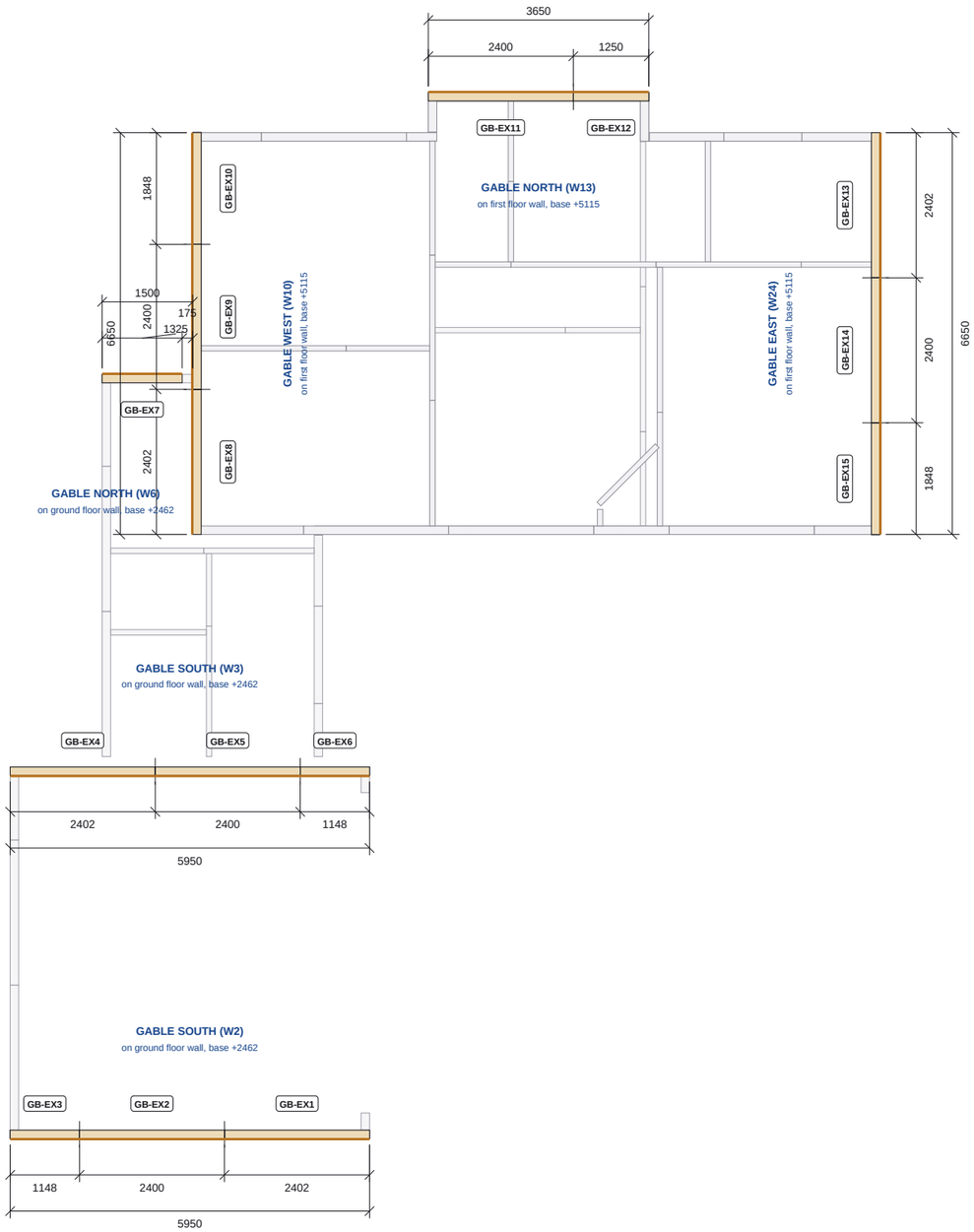
DRAWING TITLE

FIRST FLOOR GENERAL ARRANGEMENT

Panels, openings and steel over

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-GA-02	C03



GABLE PANELS LAYOUT

Scale 1:100 @ A3 | Gable panels heavy; first floor walls grey, ground floor walls light grey

0 1 2 3 4 5m

GABLE PANEL SCHEDULES

GABLE PANELS

Mark	Gable	Width	Height	Wall
GB-EX8	West	2400	2656	W10
GB-EX9	West	2400	3580	W10
GB-EX10	West	1846	2102	W10
GB-EX13	East	2400	2656	W24
GB-EX14	East	2400	3580	W24
GB-EX15	East	1846	2102	W24
GB-EX11	North	2400	2080	W13
GB-EX12	North	1248	1504	W13
GB-EX1	South	2400	2656	W2
GB-EX2	South	2400	3230	W2
GB-EX3	South	1146	1402	W2
GB-EX4	South	2400	2656	W3
GB-EX5	South	2400	3230	W3
GB-EX6	South	1146	1402	W3
GB-EX7	North	1325	1580	W6

Width along the wall, height to the top of the raked rail.

TEMPORARY BRACING (NOT SHOWN)

Item	No.	Order
Brace 4x1 treated rough-sawn, 2.7m	9	6 x 5.4m
Noggin 38x89 C16, 600 (nailed)	9	3 x 4.8m

One brace per gable panel joint. +5%.

NOTES

1. Gable panels: 38x140 C16 studs, 9mm OSB/3 to the external face, raked top rails to the roof pitch. No sole plate or head binder: panels bear on the head binder of the wall below and the truss bottom chords.
2. Set out each gable from the outer edges of its central (ridge) panel to the ends of the wall below, as dimensioned on P4-GA-03.
3. Temporary bracing (not shown): one 2.7m 4x1 treated brace at every gable panel joint, fixed through a 600 38x89 noggin nailed to the deck.
4. Window battens 38x43 to gable openings: head and sill run over the jambs by 38.
5. Opening heads are measured from the underside of the panel bottom rail. Read with P4-GA-02, P4-GA-03 / GA-04 and P4-RF-02.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

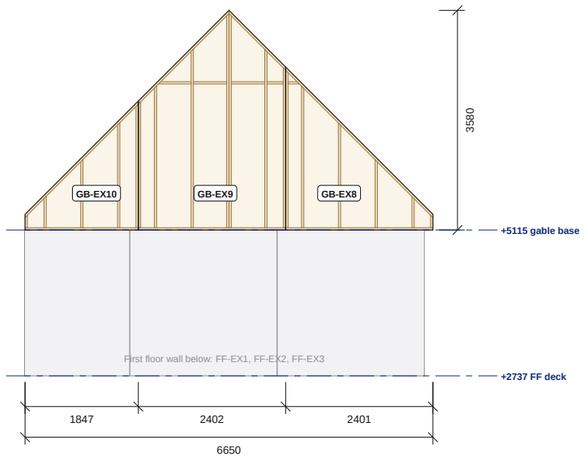
GABLE PANELS LAYOUT

Every gable on plan with its panel marks; elevations on P4-GA-04

SCALE	SIZE	DATE	DRAWN
1:100	A3	11.10.26	TS

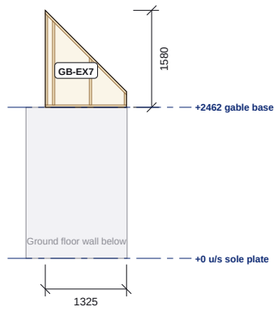
STATUS	DRAWING No.	REV
CONSTRUCTION	P4-GA-03	C03

Do not scale. All dimensions in mm. Check all dimensions on site.



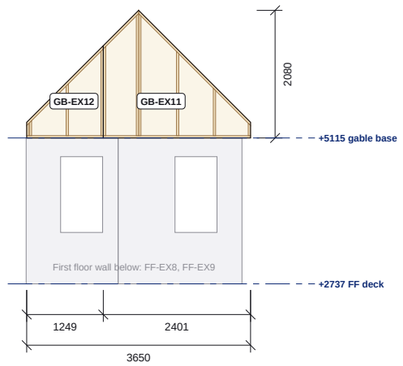
GABLE WEST (WALL 10) - FROM OUTSIDE

Scale 1:100 | Panels GB-EX8, GB-EX9, GB-EX10



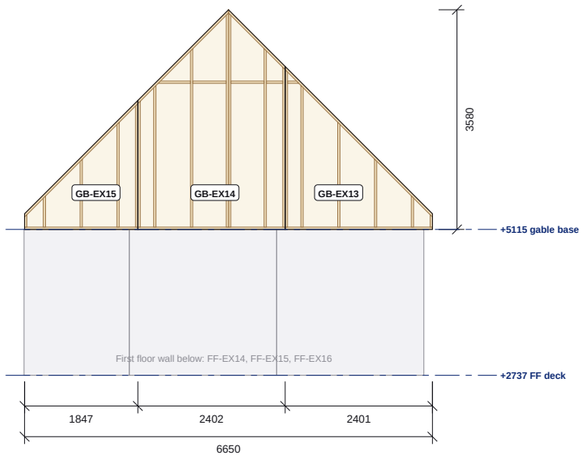
GABLE NORTH (WALL 6) - FROM OUTSIDE

Scale 1:100 | Panels GB-EX7



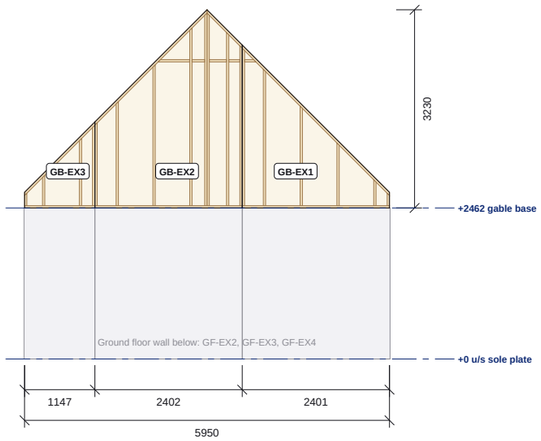
GABLE NORTH (WALL 13) - FROM OUTSIDE

Scale 1:100 | Panels GB-EX11, GB-EX12



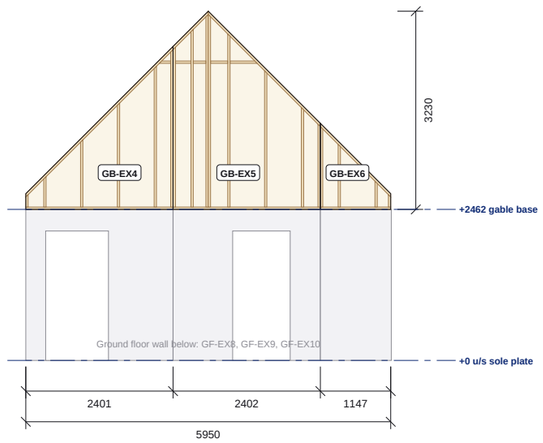
GABLE EAST (WALL 24) - FROM OUTSIDE

Scale 1:100 | Panels GB-EX13, GB-EX14, GB-EX15



GABLE SOUTH (WALL 2) - FROM OUTSIDE

Scale 1:100 | Panels GB-EX1, GB-EX2, GB-EX3



GABLE SOUTH (WALL 3) - FROM OUTSIDE

Scale 1:100 | Panels GB-EX4, GB-EX5, GB-EX6

GABLE ELEVATION SCHEDULES

GABLES AND THE WALLS BELOW

Gable	Stands on	Base
Gable West (wall 10)	first floor wall	+5115
Gable North (wall 6)	ground floor wall	+2462
Gable North (wall 13)	first floor wall	+5115
Gable East (wall 24)	first floor wall	+5115
Gable South (wall 2)	ground floor wall	+2462
Gable South (wall 3)	ground floor wall	+2462

Base = top of the head binder of the wall below, above the underside of the ground floor sole plate.

NOTES

1. Gable panels: 38x140 C16 studs, 9mm OSB/3 to the external face, raked top rails to the roof pitch. No sole plate or head binder: panels bear on the head binder of the wall below and the truss bottom chords.
2. Set out each gable from the outer edges of its central (ridge) panel to the ends of the wall below, as dimensioned on P4-GA-03.
3. Temporary bracing (not shown): one 2.7m 4x1 treated brace at every gable panel joint, fixed through a 600 38x89 noggin nailed to the deck.
4. Window battens 38x43 to gable openings: head and sill run over the jambs by 38.
5. Opening heads are measured from the underside of the panel bottom rail. Read with P4-GA-02, P4-GA-03 / GA-04 and P4-RF-02.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

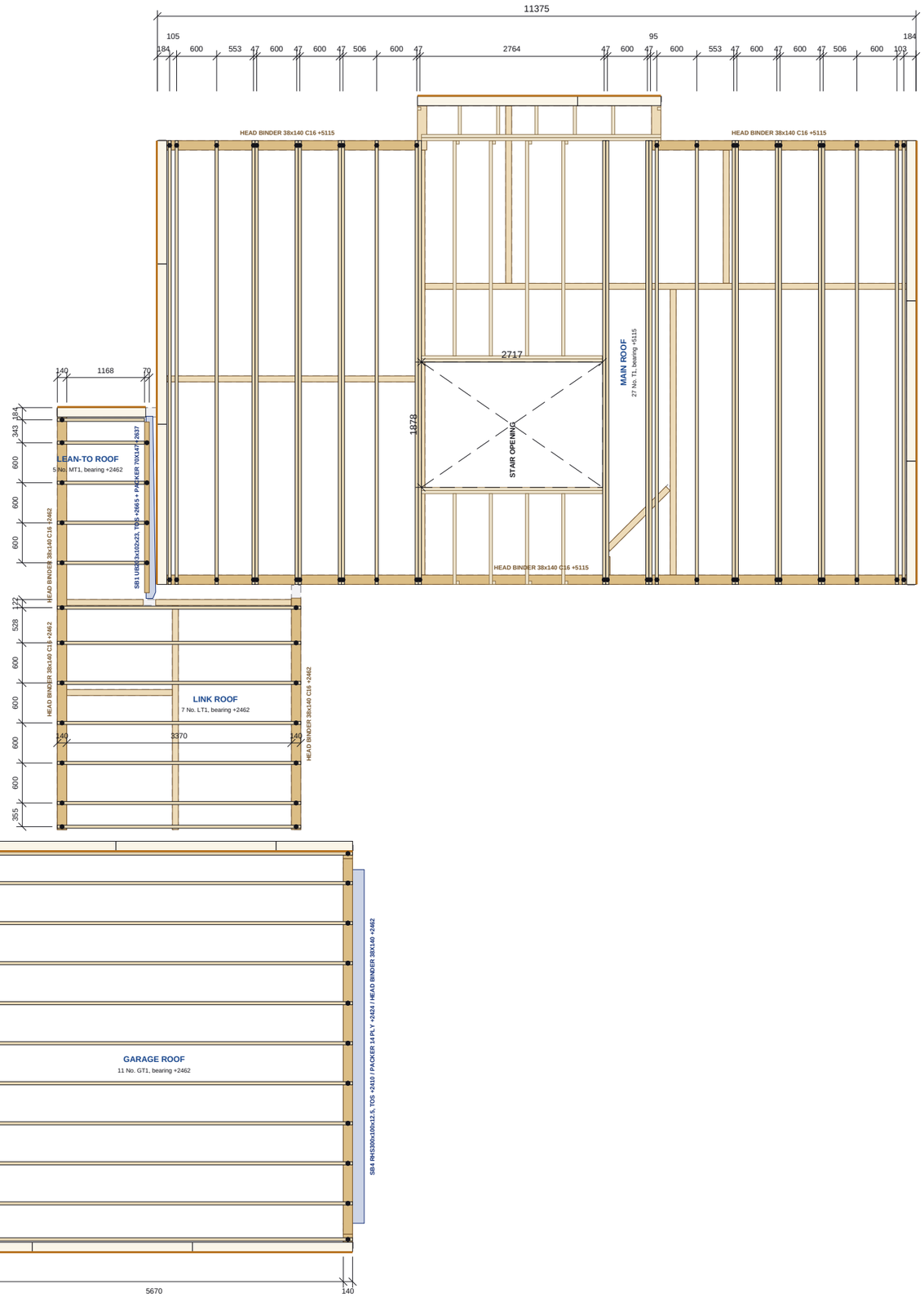
GABLE PANEL ELEVATIONS

Framing from outside, each gable on the wall below it

SCALE	SIZE	DATE	DRAWN
1:100	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-GA-04	C03

Do not scale. All dimensions in mm. Check all dimensions on site.



ROOF - CEILING LEVEL PLAN

Scale 1:75 @ A3 | Plan at +5415; first floor walls below grey dashed

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

CEILING LEVEL SCHEDULE

TRUSSES

Type	No.	Span	Height	Doubles
GT1	11	5950	3238	0
LT1	7	3650	2080	0
MT1	5	1306	1562	0
T1	27	6650	3588	9

Span over external frame faces; height from underside of bottom chord.

TRUSS BEARINGS

Roof	Bearing	Top	Runs
Main	HEAD BINDER 38x140 C16	+5115	3
Garage	SB4 RHS300x100x12.5 S355J2H	+2410	1
Garage	SB4 head binder 38x140 C16	+2462	1
Garage	SB4 packer 14mm marine ply 140	+2424	1
Garage	HEAD BINDER 38x140 C16	+2462	3
Link	HEAD BINDER 38x140 C16	+2462	2
Lean-to	SB1 UB203x102x23 S355JR	+2665	1
Lean-to	SB1 packer 70x147 C24	+2637	1
Lean-to	HEAD BINDER 38x140 C16	+2462	1

Top of bearing above slab; runs = separate lengths on plan. Dots on plan mark each truss bearing.

CEILING-LEVEL ROOF MEMBERS

Item	No.
Cut roof members bearing at ceiling level	29

Pop-out valley and junction framing, indicative.

NOTES

1. Lower roofs (garage roof, link roof and lean-to roof): trusses bear at +2462 above slab on the ground floor head binders and the steel / packers shown solid, set out as dimensioned; dots mark each truss bearing.
2. Ceiling level: attic truss bottom chords bear on the first floor head binder at +5115 above slab.
3. Truss bottom chords form the attic floor zone; double trusses trim the stair void and the pop-out junction.
4. Cut roof members at the pop-out valley shown at their low (ceiling-level) ends; see P4-RF-02.
5. Truss and roof member positions indicative; read with the truss manufacturer's layout and engineer's drawing 26-431-S-0010.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

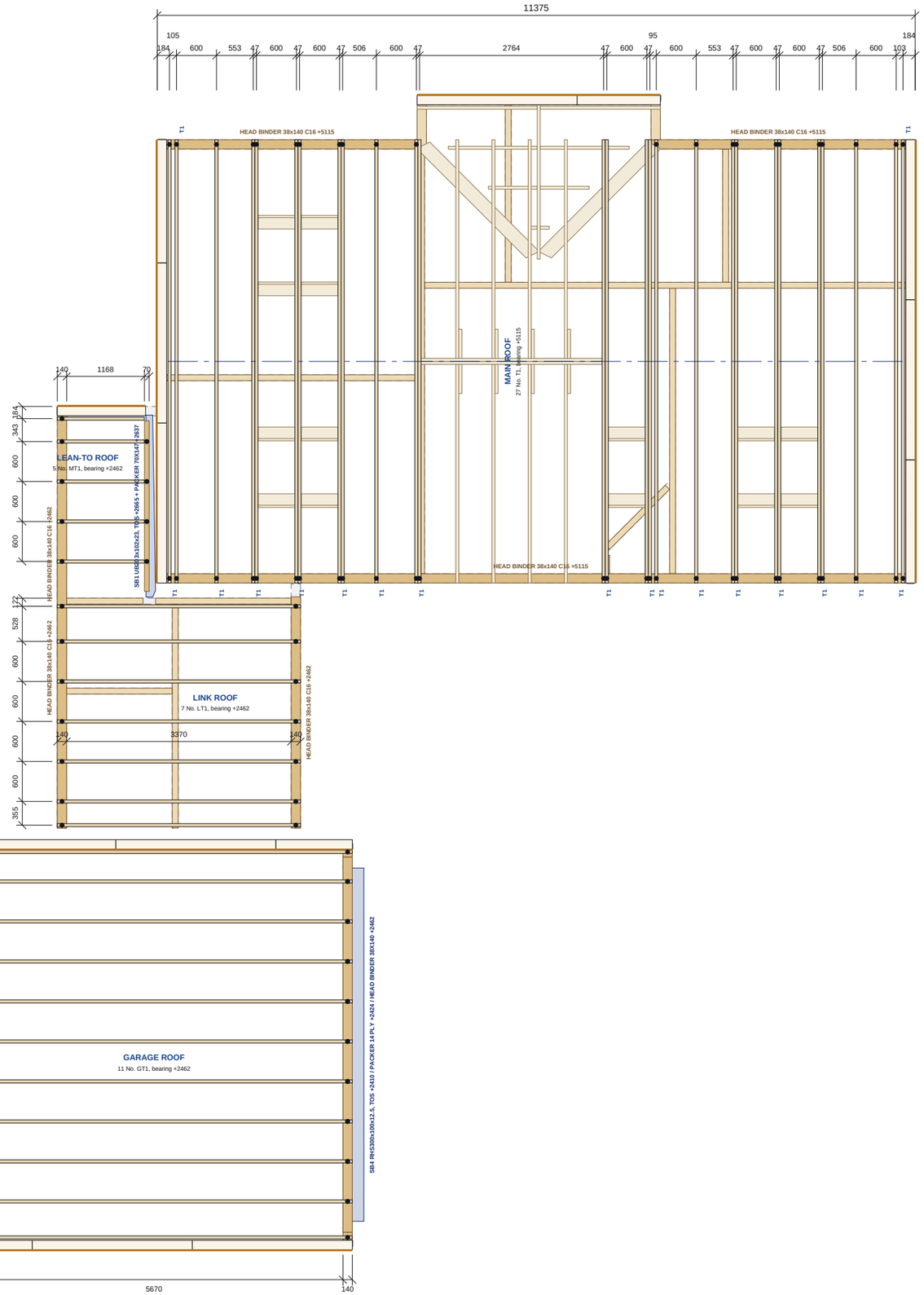
DRAWING TITLE

ROOF - CEILING LEVEL

Attic truss bottom chords and ceiling-level roof members

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-RF-01	C03



ROOF - TRUSS LAYOUT PLAN

Scale 1:75 @ A3 | Ridge lines blue dash-dot; first floor walls below grey dashed

ROOF SCHEDULE

TRUSSES

Type	No.	Span	Height	Doubles
GT1	11	5950	3238	0
LT1	7	3650	2080	0
MT1	5	1306	1562	0
T1	27	6650	3588	9

Span over external frame faces; height from underside of bottom chord.

TRUSS BEARINGS

Roof	Bearing	Top	Runs
Main	HEAD BINDER 38x140 C16	+5115	3
Garage	SB4 RHS300x100x12.5 S355J2H	+2410	1
Garage	SB4 head binder 38x140 C16	+2462	1
Garage	SB4 packer 14mm marine ply 140	+2424	1
Garage	HEAD BINDER 38x140 C16	+2462	3
Link	HEAD BINDER 38x140 C16	+2462	2
Lean-to	SB1 UB203x102x23 S355JR	+2665	1
Lean-to	SB1 packer 70x147 C24	+2637	1
Lean-to	HEAD BINDER 38x140 C16	+2462	1

Top of bearing above slab; runs = separate lengths on plan. Dots on plan mark each truss bearing.

CUT ROOF MEMBERS

Item	No.
Valley / rafter / ridge / purlin members (above ceiling)	53

Indicative, at the pop-out junction.

NOTES

1. Attic trusses T1 (main roof) and GT1 (garage roof); trussed rafters LT1 (link roof) and MT1 (lean-to roof) at the centres dimensioned; doubles where shown.
2. Lower roofs (garage roof, link roof and lean-to roof): trusses bear at +2462 above slab on the ground floor head binders and the steel / packers shown solid, set out as dimensioned; dots mark each truss bearing.
3. Cut roof members (valley, rafters, ridge and purlins) at the pop-out junction shown indicatively.
4. Gable panels see P4-GA-03. Truss and roof member positions indicative; read with the truss manufacturer's layout and engineer's drawing 26-431-S-0010.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

ROOF - TRUSS LAYOUT

Attic trusses, cut roof members and set-out

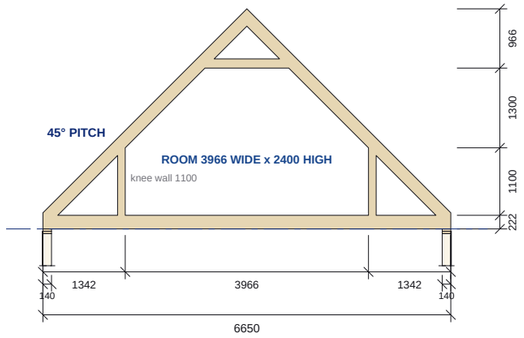
SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-RF-02	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

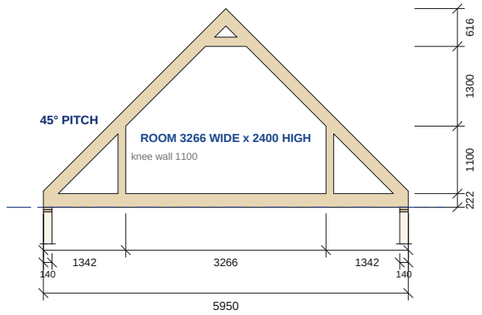
TRUSS T1 PROFILE (27 NO.)

Scale 1:100 | Main roof



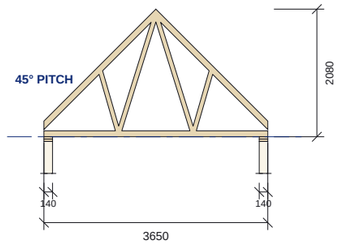
TRUSS GT1 PROFILE (11 NO.)

Scale 1:100 | Garage roof



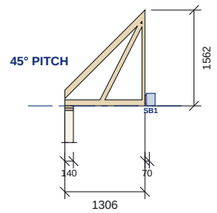
TRUSS LT1 PROFILE (7 NO.)

Scale 1:100 | Link roof



TRUSS MT1 PROFILE (5 NO.)

Scale 1:100 | Lean-to roof



TRUSS SCHEDULE

TRUSSES

Type	No.	Span	Height	Pitch	Room
T1	27	6650	3588	45	3966 x 2400
GT1	11	5950	3238	45	3266 x 2400
LT1	7	3650	2080	45	-
MT1	5	1306	1561	45	-

Span over external frame faces; height from underside of bottom chord; pitch in degrees; room = clear width x height to the collar.

NOTES

1. Attic truss profiles as modelled, for co-ordination. Member sizes, connections and bracing to the truss manufacturer's design.
2. Heights from the underside of the bottom chord (+5115 above slab, top of first floor head binder). Trusses bear on the first floor external wall head binders shown below each profile.
3. Room-in-roof clear sizes are between the inner faces of the attic studs and under the collar tie.
4. Lower roofs (garage roof, link roof and lean-to roof): trusses bear at +2462 above slab on the ground floor head binders and the steel / packers cut below each profile (steel blue).

REVISIONS

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TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

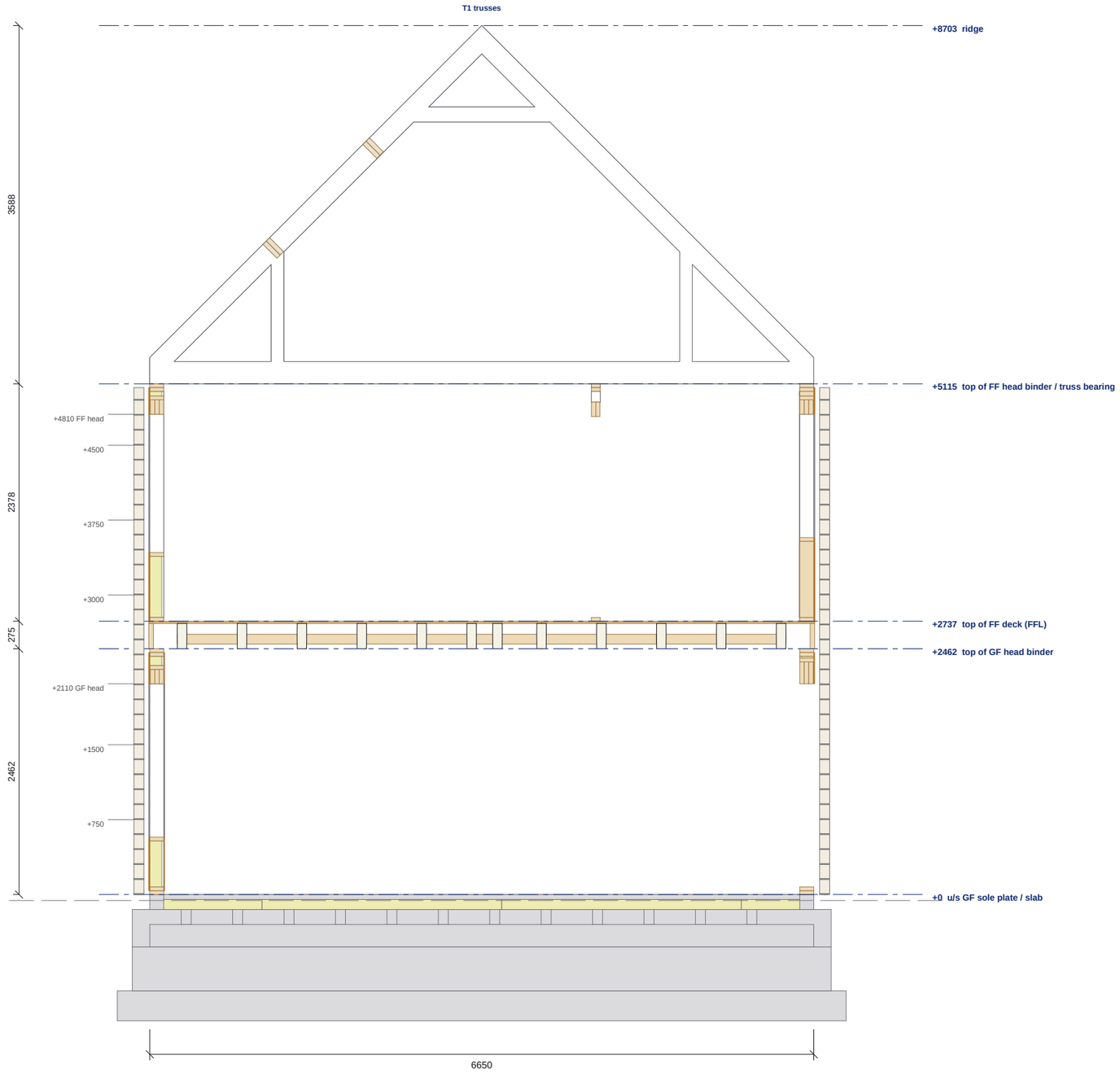
ROOF - TRUSS PROFILES

Attic truss profiles with span, height and room dimensions

SCALE	SIZE	DATE	DRAWN
1:50 / 1:100	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-RF-03	C03

Do not scale. All dimensions in mm. Check all dimensions on site.



SECTION A-A

Scale 1:50 @ A3 | Main building through the lounge and master bedroom, looking west

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

SECTION A-A SCHEDULE

LEVELS

Level	mm
u/s GF sole plate / slab	+0
top of GF head binder	+2462
top of FF deck (FFL)	+2737
top of FF head binder / truss bearing	+5115
ridge	+8703

Above underside of the ground floor sole plate.

ELEMENTS CUT

Element	Pieces cut
Sole plate / head binder	7
Panel framing	47
Edgebinder	2
22mm deck	4
Metal-web joists (Acorn)	12
Cut roof members	4
Substructure (by others)	22

NOTES

1. Section A-A cut through the timber frame model; members up to 700mm beyond the cut shown in outline.
2. Timber frame: 38x140 C16 external walls with 9mm OSB/3 and 120mm PIR; 254 metal-web first floor (Acorn); attic trusses over.
3. Stone coursing 150 (10 bed + 140) from FFL; course lines drawn on the external leaf outside the frame faces.
4. Levels are above the underside of the ground floor sole plate (+0). Substructure by others, shown for reference only.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

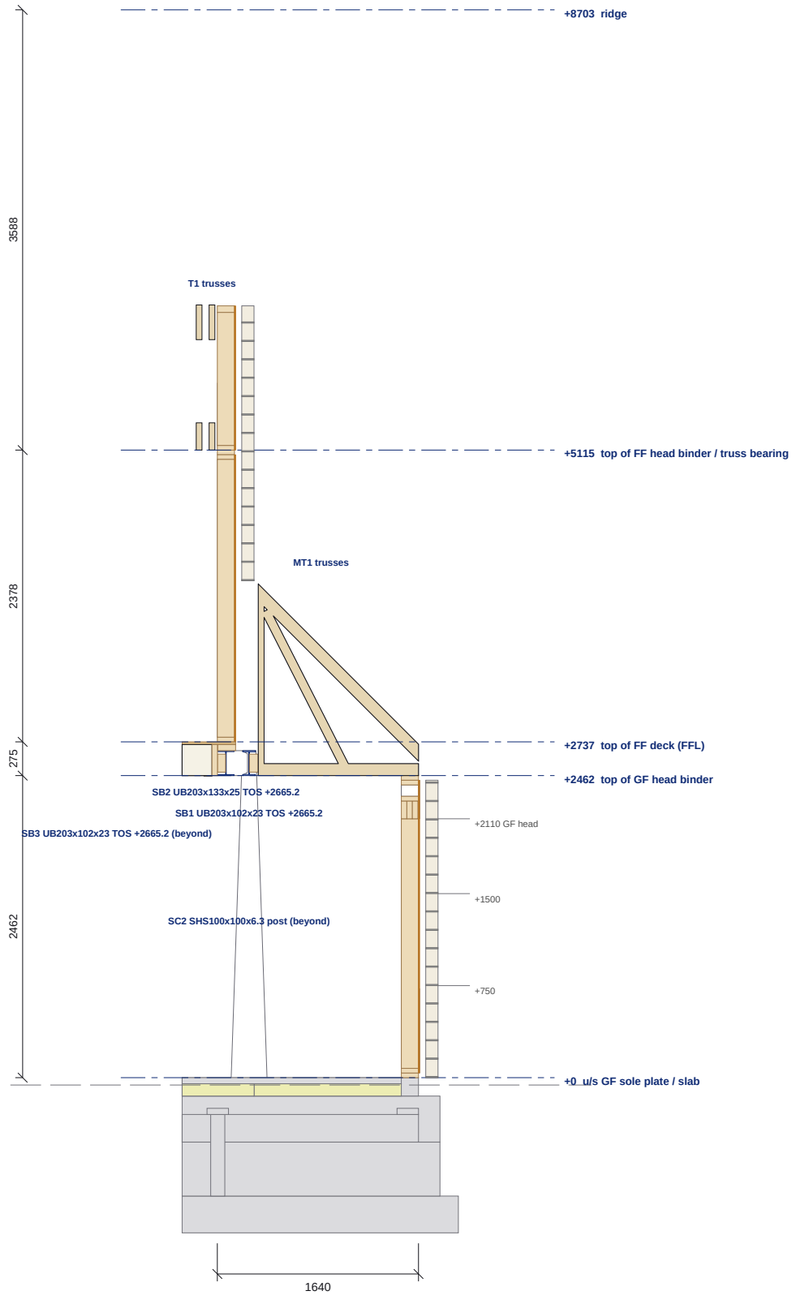
DRAWING TITLE

SECTION A-A

Main building through the lounge and master bedroom, looking west

SCALE	SIZE	DATE	DRAWN
1:50	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-SE-01	C03



SECTION B-B

Scale 1:50 @ A3 | Family room steel and lean-to, looking south

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

SECTION B-B SCHEDULE

LEVELS

Level	mm
u/s GF sole plate / slab	+0
top of GF head binder	+2462
top of FF deck (FFL)	+2737
top of FF head binder / truss bearing	+5115
ridge	+8703

Above underside of the ground floor sole plate.

ELEMENTS CUT

Element	Pieces cut
Sole plate / head binder	3
Panel framing	16
22mm deck	1
Metal-web joists (Acorn)	1
Attic trusses	3
Substructure (by others)	11
Steel beams	2
Steel packers / binders / jamb plates	4

NOTES

1. Section B-B cut through the timber frame model; members up to 700mm beyond the cut shown in outline (trusses and steel up to 1400mm).
2. Timber frame: 38x140 C16 external walls with 9mm OSB/3 and 120mm PIR; 254 metal-web first floor (Acorn); attic trusses over.
3. Stone coursing 150 (10 bed + 140) from FFL; course lines drawn on the external leaf outside the frame faces.
4. Levels are above the underside of the ground floor sole plate (+0). Substructure by others, shown for reference only.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

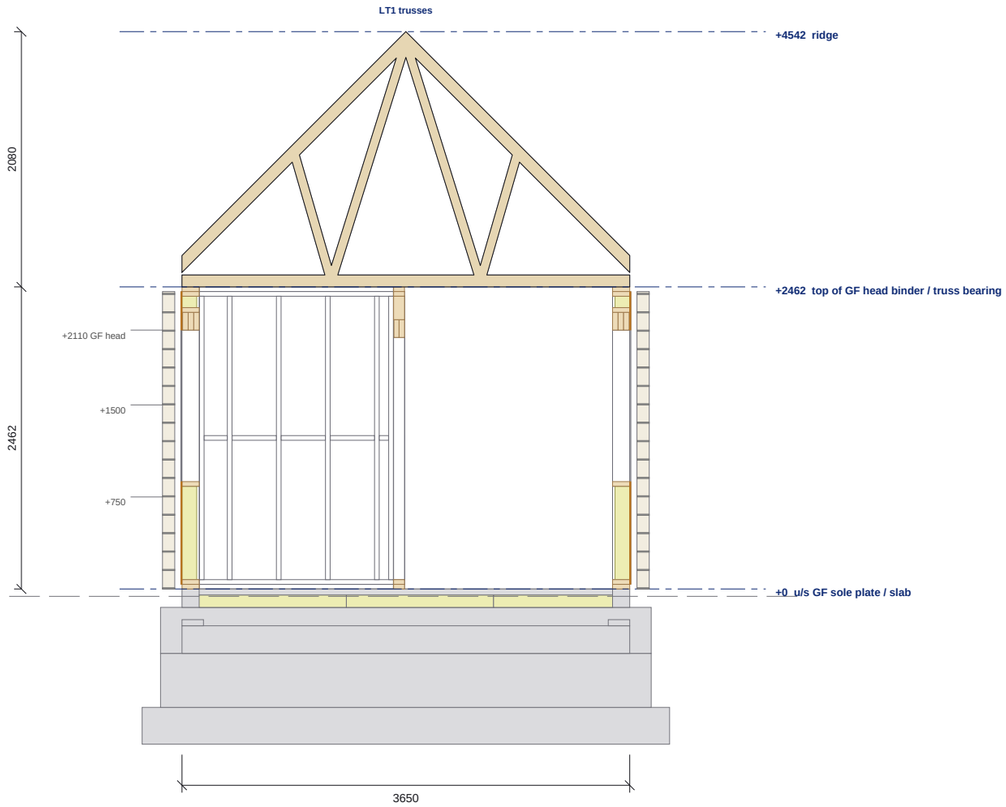
DRAWING TITLE

SECTION B-B

Family room steel and lean-to, looking south

SCALE	SIZE	DATE	DRAWN
1:50	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-SE-02	C03



SECTION C-C

Scale 1:50 @ A3 | Link between garage and house, through the LT1 trusses, looking north

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

SECTION C-C SCHEDULE

LEVELS

Level	mm
u/s GF sole plate / slab	+0
top of GF head binder / truss bearing	+2462
ridge	+4542

Above underside of the ground floor sole plate.

ELEMENTS CUT

Element	Pieces cut
Sole plate / head binder	6
Panel framing	27
Attic trusses	1
Substructure (by others)	11

NOTES

1. Section C-C cut through the timber frame model; members up to 700mm beyond the cut shown in outline.
2. Timber frame: 38x140 C16 external walls with 9mm OSB/3 and 120mm PIR; 254 metal-web first floor (Acorn); attic trusses over.
3. Stone coursing 150 (10 bed + 140) from FFL; course lines drawn on the external leaf outside the frame faces.
4. Levels are above the underside of the ground floor sole plate (+0). Substructure by others, shown for reference only.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

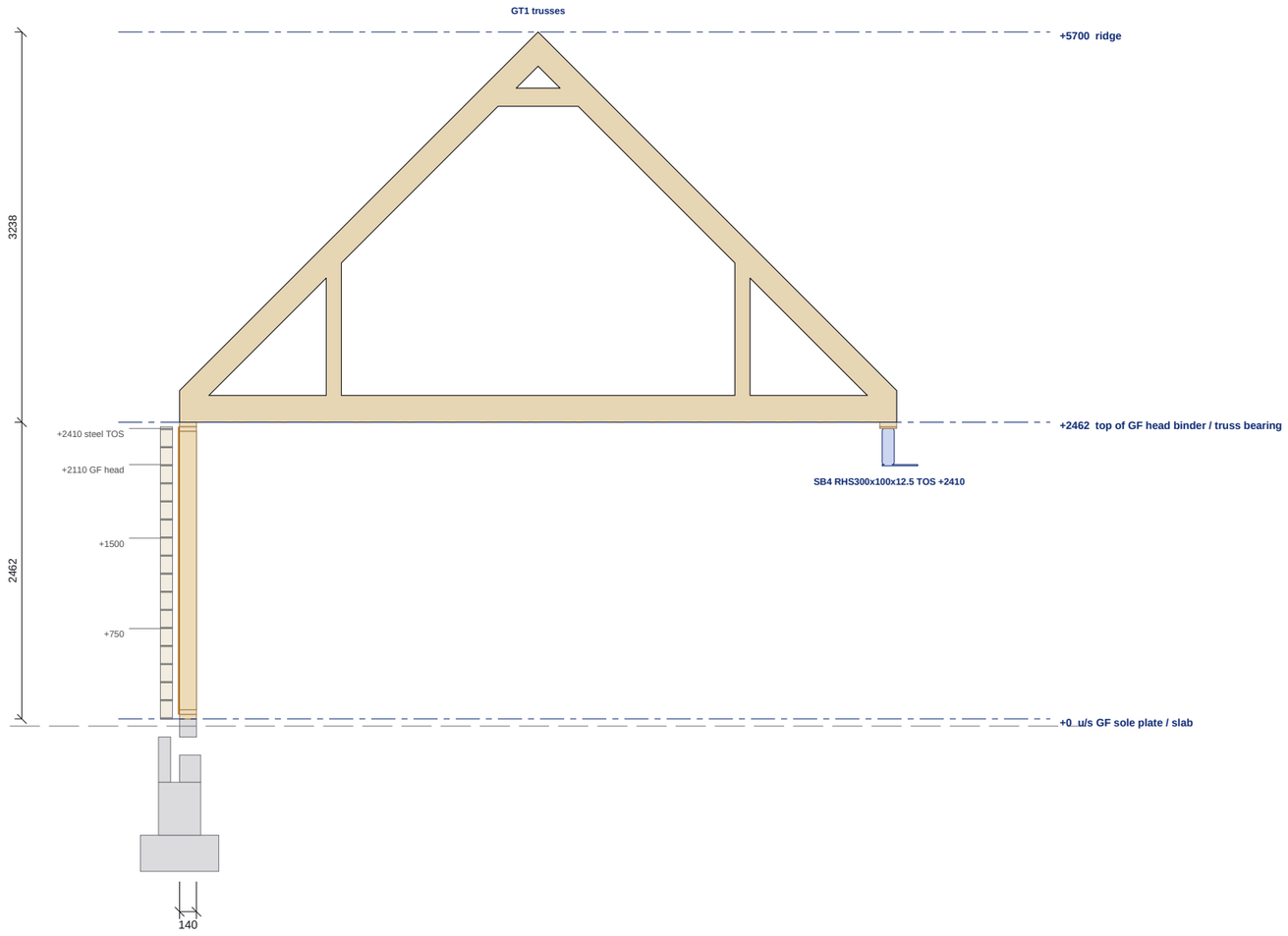
DRAWING TITLE

SECTION C-C

Link between garage and house, through the LT1 trusses, looking north

SCALE	SIZE	DATE	DRAWN
1:50	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-SE-03	C03



SECTION D-D

Scale 1:50 @ A3 | Garage attic trusses and GP1 portal over the garage door, looking north

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

SECTION D-D SCHEDULE

LEVELS

Level	mm
u/s GF sole plate / slab	+0
top of GF head binder / truss bearing	+2462
ridge	+5700

Above underside of the ground floor sole plate.

ELEMENTS CUT

Element	Pieces cut
Sole plate / head binder	2
Panel framing	4
Attic trusses	1
Substructure (by others)	5
Steel beams	1
Steel packers / binders / jamb plates	2

NOTES

1. Section D-D cut through the timber frame model; members up to 700mm beyond the cut shown in outline.
2. Timber frame: 38x140 C16 external walls with 9mm OSB/3 and 120mm PIR; 254 metal-web first floor (Acorn); attic trusses over.
3. Stone coursing 150 (10 bed + 140) from FFL; course lines drawn on the external leaf outside the frame faces.
4. Levels are above the underside of the ground floor sole plate (+0). Substructure by others, shown for reference only.

REVISIONS

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C01	14.09.26	Construction issue
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C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

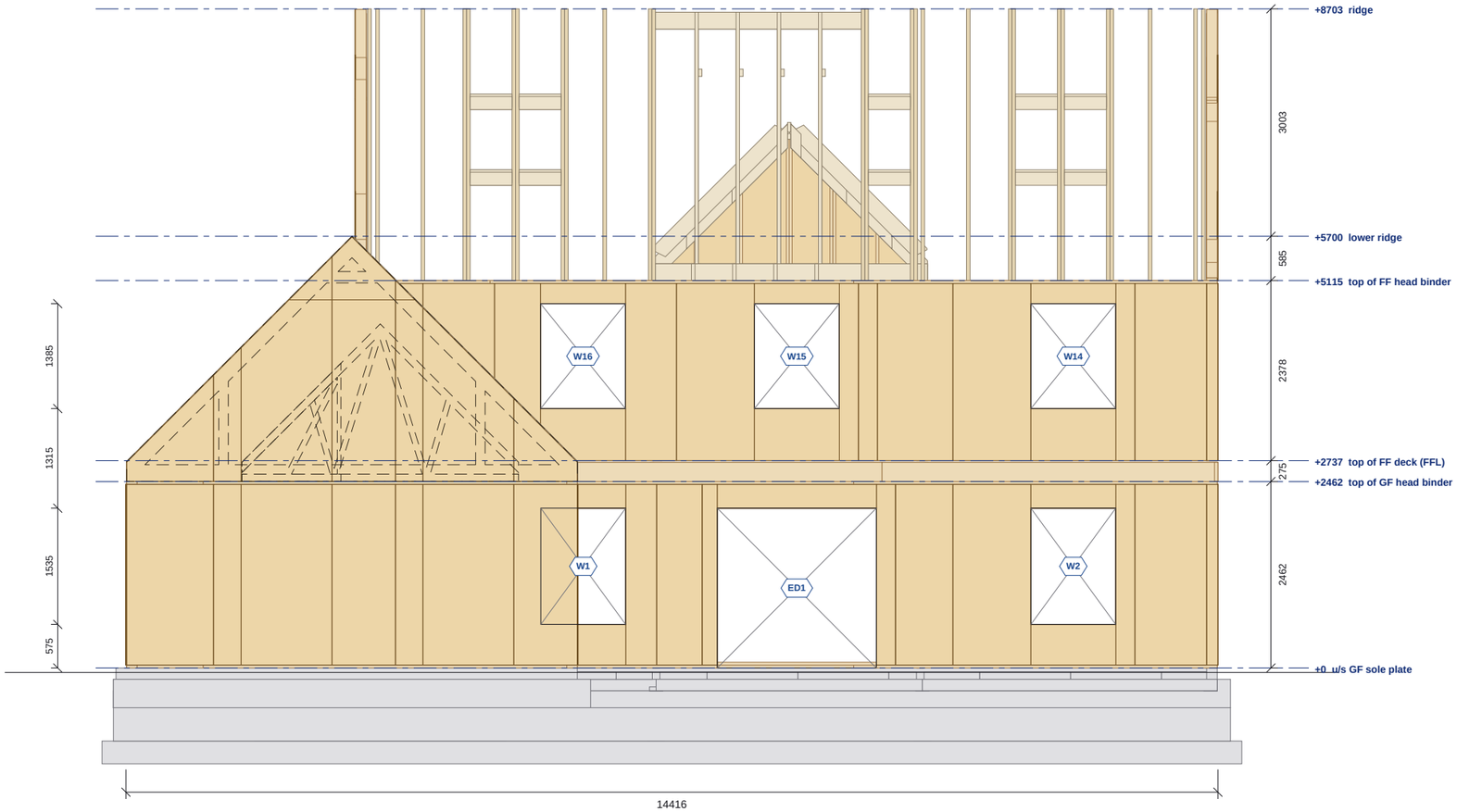
DRAWING TITLE

SECTION D-D

Garage attic trusses and GP1 portal over the garage door, looking north

SCALE	SIZE	DATE	DRAWN
1:50	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-SE-04	C03



FRONT ELEVATION (SOUTH)

Scale 1:75 @ A3 | Levels right: storey levels + visible ridges; left: sills and heads on this face

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

FRONT ELEVATION (SOUTH) SCHEDULE

LEVELS ON THIS FACE

Level	mm
u/s GF sole plate	+0
top of GF head binder	+2462
top of FF deck (FFL)	+2737
top of FF head binder	+5115
lower ridge	+5700
ridge	+8703

Above underside of the ground floor sole plate.

OPENINGS ON THIS FACE

Label	Level	Width	Height	Sill	Head
ED1	GF	2100	2110	0	2110
W1	GF	1120	1535	575	2110
W2	GF	1120	1535	575	2110
W14	FF	1120	1385	3425	4810
W15	FF	1120	1385	3425	4810
W16	FF	1120	1385	3425	4810

Sill and head levels above underside of the GF sole plate.

NOTES

1. Construction elevation of the timber frame: OSB/3 face of external and gable panels, sole plates, head binders and edgebinders. Cladding, roof covering and windows by others, not shown.
2. Level lines shown are only those relevant to this face: storey levels, visible ridges and the sill / head levels of the openings on this face.
3. Roof trusses: edge-on trusses drawn at their centres; trusses behind gable / wall panels shown dashed (hidden detail, nearest truss of each type).
4. Levels above the underside of the ground floor sole plate (+0). Openings are structural openings; see P4-GA-01 / P4-GA-02 / P4-GA-03.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

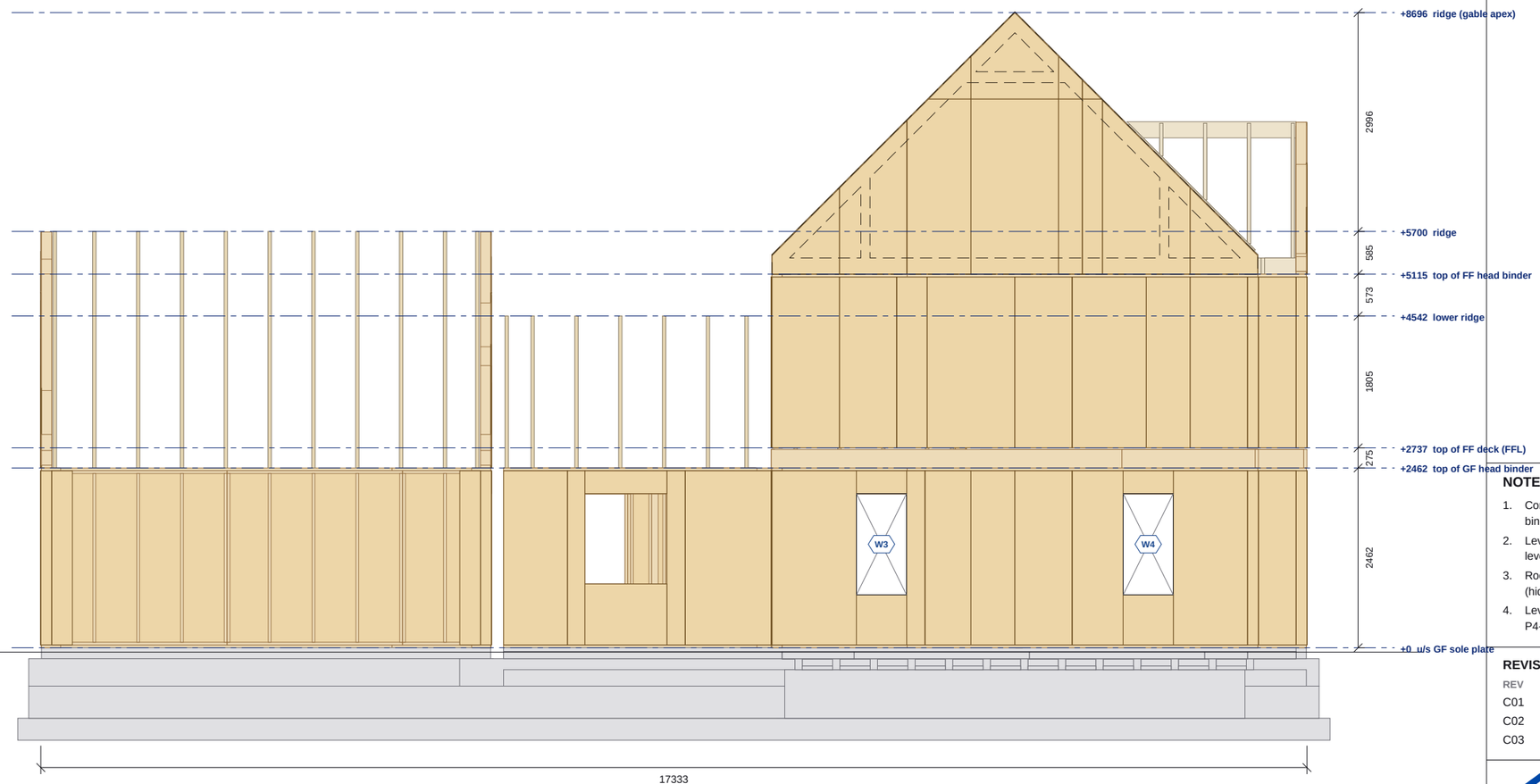
TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE
FRONT ELEVATION (SOUTH)
Timber frame construction elevation

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-EL-01	C03



SIDE ELEVATION (EAST)

Scale 1:75 @ A3 | Levels right: storey levels + visible ridges; left: sills and heads on this face

0 1 2 3 4 5m

SIDE ELEVATION (EAST) SCHEDULE

LEVELS ON THIS FACE

Level	mm
u/s GF sole plate	+0
top of GF head binder	+2462
top of FF deck (FFL)	+2737
lower ridge	+4542
top of FF head binder	+5115
ridge	+5700
ridge (gable apex)	+8696

Above underside of the ground floor sole plate.

OPENINGS ON THIS FACE

Label	Level	Width	Height	Sill	Head
W3	GF	685	1385	725	2110
W4	GF	685	1385	725	2110

Sill and head levels above underside of the GF sole plate.

NOTES

1. Construction elevation of the timber frame: OSB/3 face of external and gable panels, sole plates, head binders and edgebinders. Cladding, roof covering and windows by others, not shown.
2. Level lines shown are only those relevant to this face: storey levels, visible ridges and the sill / head levels of the openings on this face.
3. Roof trusses: edge-on trusses drawn at their centres; trusses behind gable / wall panels shown dashed (hidden detail, nearest truss of each type).
4. Levels above the underside of the ground floor sole plate (+0). Openings are structural openings; see P4-GA-01 / P4-GA-02 / P4-GA-03.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

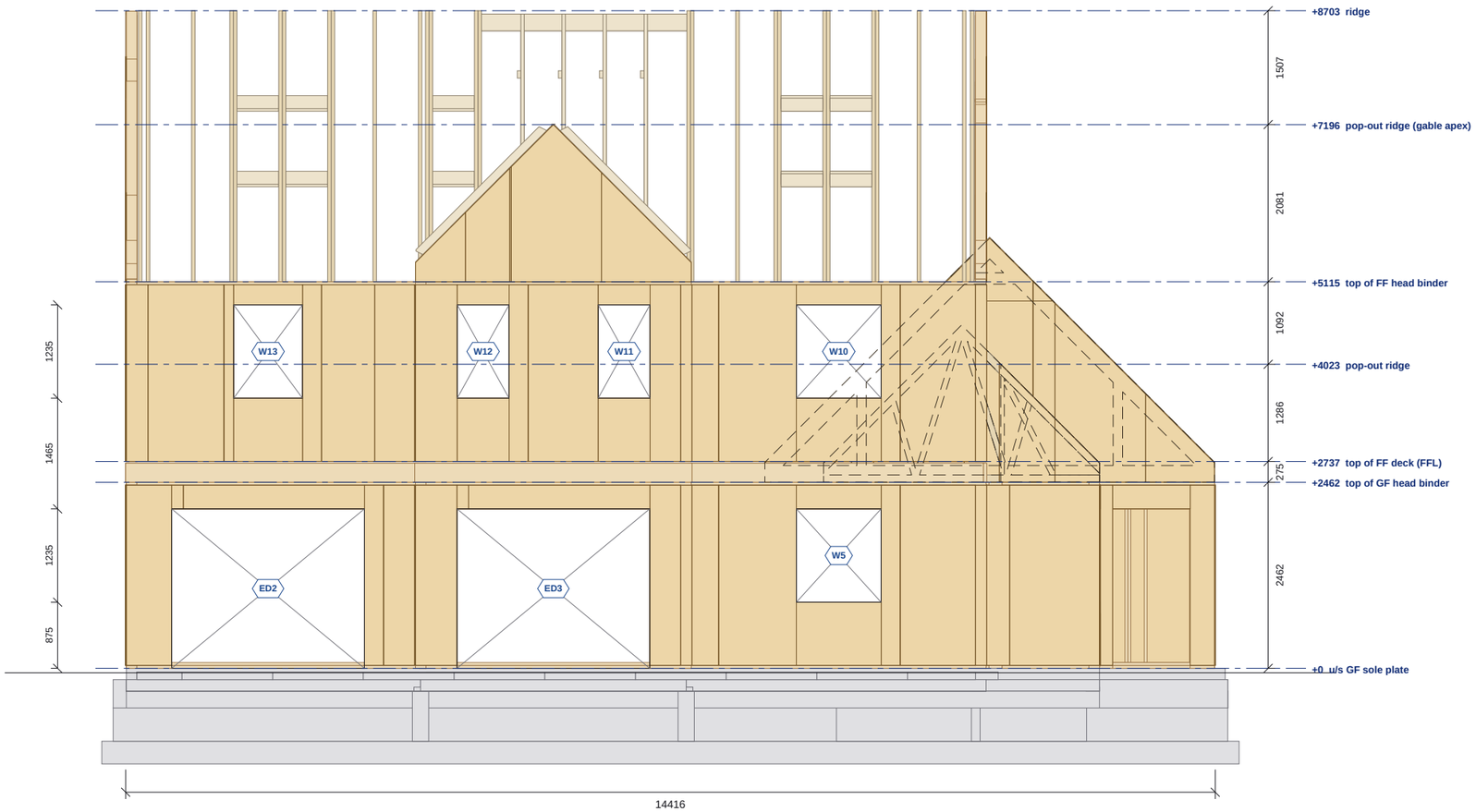
CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE
SIDE ELEVATION (EAST)
Timber frame construction elevation

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-EL-02	C03

Do not scale. All dimensions in mm. Check all dimensions on site.



REAR ELEVATION (NORTH)

Scale 1:75 @ A3 | Levels right: storey levels + visible ridges; left: sills and heads on this face

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

REAR ELEVATION (NORTH) SCHEDULE

LEVELS ON THIS FACE

Level	mm
u/s GF sole plate	+0
top of GF head binder	+2462
top of FF deck (FFL)	+2737
pop-out ridge	+4023
top of FF head binder	+5115
pop-out ridge (gable apex)	+7196
ridge	+8703

Above underside of the ground floor sole plate.

OPENINGS ON THIS FACE

Label	Level	Width	Height	Sill	Head
ED2	GF	2550	2110	0	2110
ED3	GF	2550	2110	0	2110
W5	GF	1120	1235	875	2110
W10	FF	1120	1235	3575	4810
W11	FF	685	1235	3575	4810
W12	FF	685	1235	3575	4810
W13	FF	910	1235	3575	4810

Sill and head levels above underside of the GF sole plate.

NOTES

1. Construction elevation of the timber frame: OSB/3 face of external and gable panels, sole plates, head binders and edgebinders. Cladding, roof covering and windows by others, not shown.
2. Level lines shown are only those relevant to this face: storey levels, visible ridges and the sill / head levels of the openings on this face.
3. Roof trusses: edge-on trusses drawn at their centres; trusses behind gable / wall panels shown dashed (hidden detail, nearest truss of each type).
4. Levels above the underside of the ground floor sole plate (+0). Openings are structural openings; see P4-GA-01 / P4-GA-02 / P4-GA-03.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

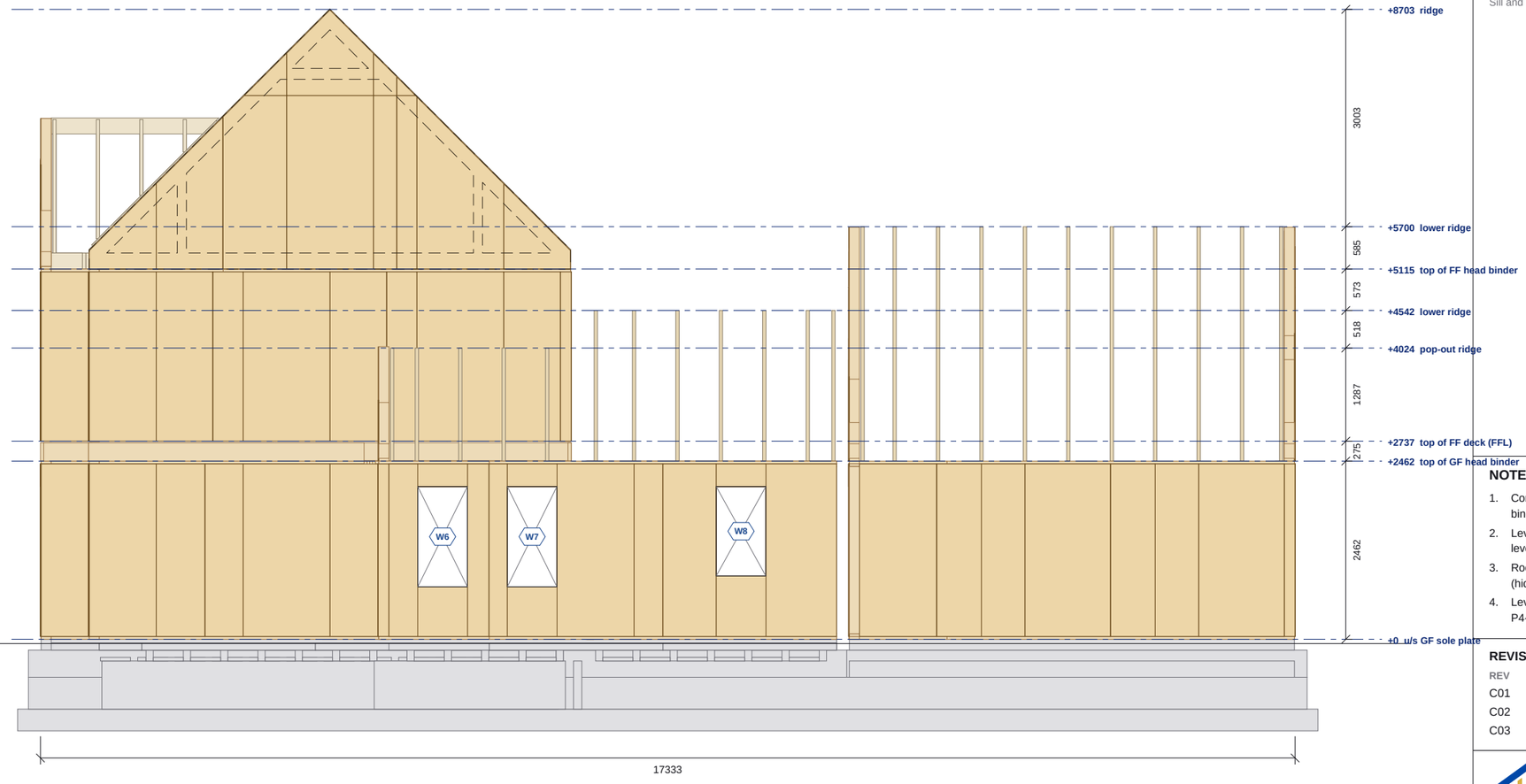
TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE
REAR ELEVATION (NORTH)
Timber frame construction elevation

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-EL-03	C03



SIDE ELEVATION (WEST)

Scale 1:75 @ A3 | Levels right: storey levels + visible ridges; left: sills and heads on this face

0 1 2 3 4 5m

SIDE ELEVATION (WEST) SCHEDULE

LEVELS ON THIS FACE

Level	mm
u/s GF sole plate	+0
top of GF head binder	+2462
top of FF deck (FFL)	+2737
pop-out ridge	+4024
lower ridge	+4542
top of FF head binder	+5115
lower ridge	+5700
ridge	+8703

Above underside of the ground floor sole plate.

OPENINGS ON THIS FACE

Label	Level	Width	Height	Sill	Head
W6	GF	685	1385	725	2110
W7	GF	685	1385	725	2110
W8	GF	685	1235	875	2110

Sill and head levels above underside of the GF sole plate.

NOTES

1. Construction elevation of the timber frame: OSB/3 face of external and gable panels, sole plates, head binders and edgebinders. Cladding, roof covering and windows by others, not shown.
2. Level lines shown are only those relevant to this face: storey levels, visible ridges and the sill / head levels of the openings on this face.
3. Roof trusses: edge-on trusses drawn at their centres; trusses behind gable / wall panels shown dashed (hidden detail, nearest truss of each type).
4. Levels above the underside of the ground floor sole plate (+0). Openings are structural openings; see P4-GA-01 / P4-GA-02 / P4-GA-03.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE
SIDE ELEVATION (WEST)
Timber frame construction elevation

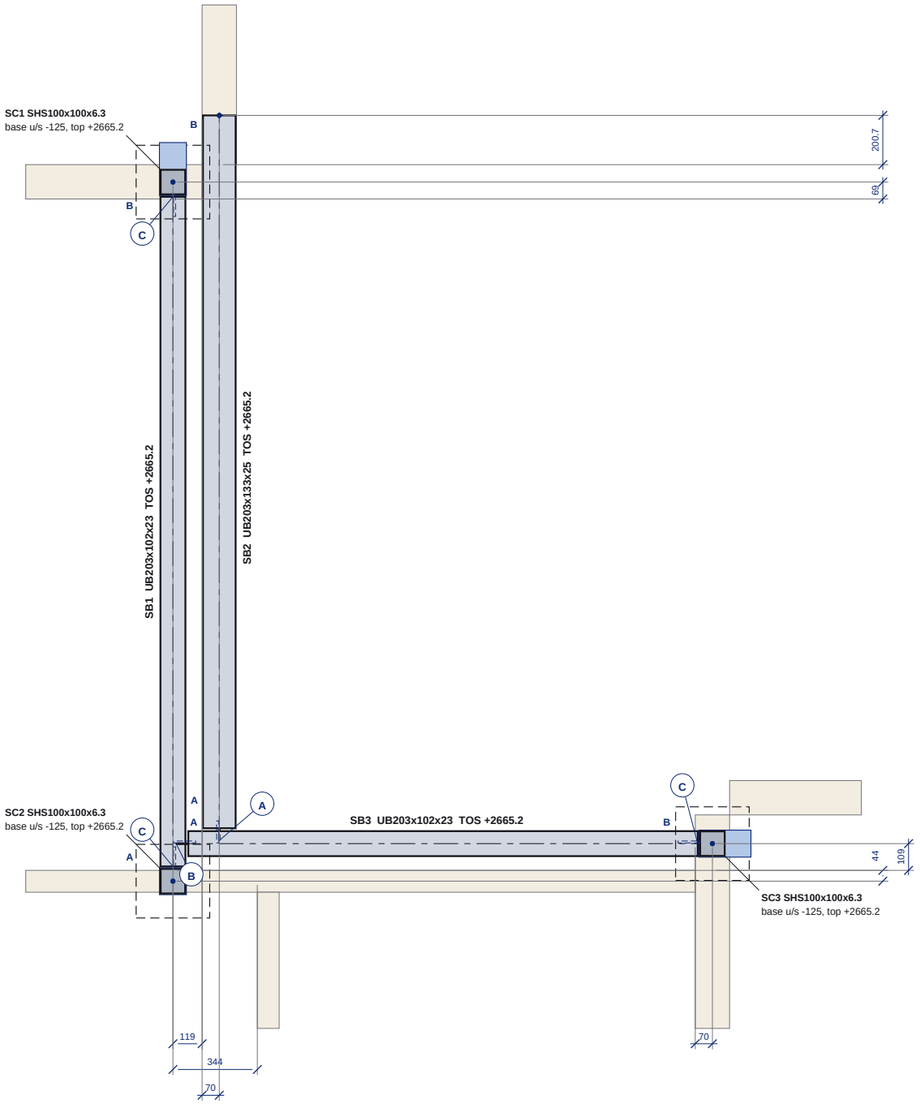
SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-EL-04	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

PLAN

Scale 1:25 | north up

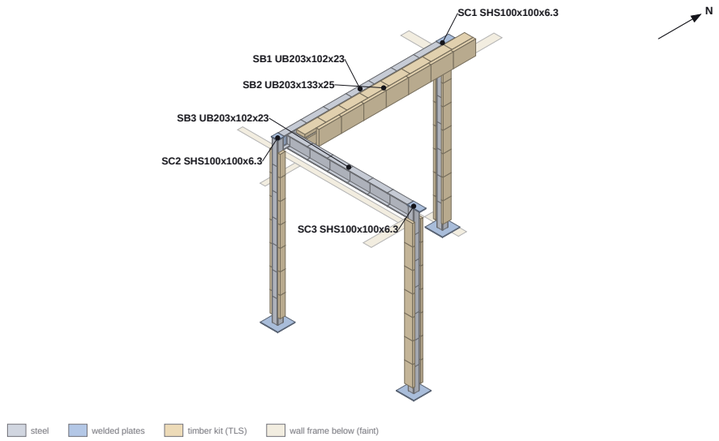


grey = GF timber frame (wall bands)

blue dims: beam end (on its CL) / post centre to the nearest GF frame face

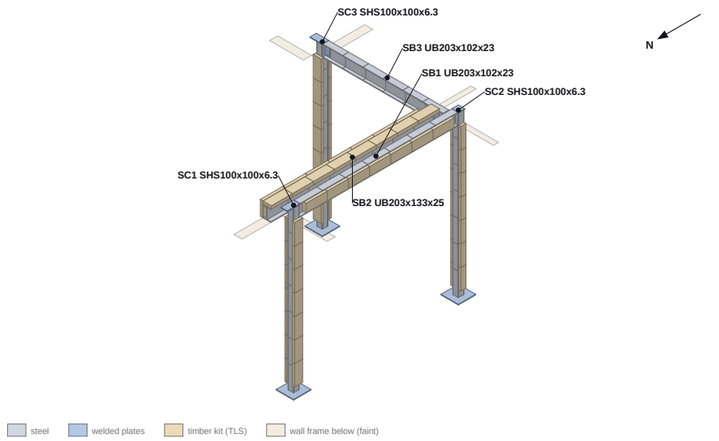
ISOMETRIC FROM THE SOUTH-EAST

Scale 1:75 | axonometric, axis lengths x0.82



ISOMETRIC FROM THE NORTH-WEST

Scale 1:75 | axonometric, axis lengths x0.82



POST SET-OUT (CENTRES, MODEL MM)

POST	X	Y	BASE U/S	SECTION
SC1	2904	12579	-125	SHS100x100x6.3
SC2	2904	9730	-125	SHS100x100x6.3
SC3	5103	9883	-125	SHS100x100x6.3

X / Y from the model origin 0,0 = outside faces of the GF frame at the garage SW corner (Wall 1). Padstones and anchors are set out from the model coordinates / the substructure set-out (402) before the frame is erected; check against the frame faces after.

BEAM END SET-OUT

BEAM	END	X	Y	TOS	USS	BEARING / CONNECTION
SB1	A	2904	9790	+2665.2	+2462	Detail C (SC2): FP1 fin on SC2, 10 gap
	B	2904	12519	+2665.2	+2462	Detail C (SC1): FP2 fin on SC1, 10 gap
SB2	A	3093	9895.7	+2665.2	+2462	Detail A: FP4 fin on SB3, 10 gap
	B	3093	12850.7	+2665.2	+2462	Bears 150 on the GF-EX16 4-ply stud pack end grain
SB3	A	2916.7	9883	+2665.2	+2462	Detail B: FP3 fin on SB1, 10 gap
	B	5043	9883	+2665.2	+2462	Detail C (SC3): FP5 fin on SC3, 10 gap

End points on the beam centre line at the cut end, model mm (origin as the post set-out). Bearing: the end condition in short - full text on the member sheet.

ENGINEER'S DRAWINGS

Skyhooks 26-431-S-0010 P01 (plan, KP, Details A/B/C, Baseplate Detail 2); 26-431-S-0011 P01 (general notes 3.01-3.07, 7.01, 9.02); 26-431-S-0012 P01 (loads)

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. All steelwork to BS EN 1993-1 / BS EN 1090-2 EXC2, CE marked (S-0011 3.01-3.02). Bolts grade 8.8 (3.03), BS EN ISO 4014/4032; HDG bolts, nuts and washers to BS EN ISO 10684 (oversize-tapped nuts, S-0011 3.07).
3. Mark each member with its mark at end A (hard stamp, before galvanising); posts: 'N' on the baseplate.
4. UB S355JR (BS EN 10025-2); SHS / RHS S355J2H (BS EN 10210-1); plates S275JR.
5. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION (S-0010).
6. Welds: full profile and continuous unless noted (S-0011 3.04).
7. Copes: re-entrant corners radiused 10 min (BS EN 1090-2 6.7: no notches, 5 min), edges ground.
8. Post bases stop 25 above the padstone: HDG shim stacks at the mid-sides under the SHS walls (clear of the anchors), then non-shrink cementitious grout. Padstones by others.
9. Fire protection 30 min to all structural steel by others (S-0011 9.02).
10. Stone on SB1 / SB3: DPC on the flange; first course made up 2675 -> 2700 to course (10 bed + 140 stone from FFL).
11. Items marked TLS detail are not on the engineer's drawings - engineer to confirm.
12. Datum: Levels above FFL = model 0 (top of screed / underside of the GF sole plate)

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

MEMBER SCHEDULE (KG INCL. WELDED PLATES)

MARK	SECTION	GRADE	CUT LENGTH	QTY	KG	TOS / BASE	COATING
SB1	UB203x102x23	S355JR	2729	1	63.9	+2665.2	HDG (EN ISO 1461)
SB2	UB203x133x25	S355JR	2955	1	74.2	+2665.2	Shop primer (C2)
SB3	UB203x102x23	S355JR	2126.3	1	49.9	+2665.2	HDG (EN ISO 1461)
SC1	SHS100x100x6.3	S355J2H	2768.2	1	60.4	base -125	HDG (EN ISO 1461)
SC2	SHS100x100x6.3	S355J2H	2768.2	1	59.3	base -125	HDG (EN ISO 1461)
SC3	SHS100x100x6.3	S355J2H	2768.2	1	60.4	base -125	HDG (EN ISO 1461)
TOTAL					**368.1**		

ITEMS FOR THE ENGINEER

1. SC1 / SC3 300x300 baseplates reach −70 into the cavity zone: accept, offset, or reduce (bearing does not govern).
2. Padstone size / depth / grade and R-KEM II edge distances (substructure by others).
3. GLF 140x240 to SB2 connection not detailed on S-0010 (HOLD).
4. SC1 / SC3 cap plates extended to bridge the stone to the GF leaf (12 thk) - confirm.
5. Head binder continuity across the post pockets.
6. Detail C leader reads 'welded to beam' - TLS reads the plate as welded to the SHS face.
7. SC2 (100 SHS) projects 5.5 beyond each face of the 89 GF-IN1 partition: boxed by the dryliner - confirm.
8. Cap plates (12, oversailing 5) and Ø20 galvanising vents are TLS additions - confirm.
9. SB2 coating: C2 shop primer per S-0010, but its end A sits in the external-wall cavity zone - confirm (HDG?).
10. SB2 north bearing: on the GF-EX16 4-ply stud end grain (studs to the steel, rail + binder cut) - confirm.
11. Stone overhangs SB1 by 14 / SB3 by 16 (positions per S-0010).
12. MT1 lean-to trusses must stop at x 2829 (10 clear of the stone face 2839) to hang on the SB1 70 packer (Acorn).
13. GP1 garage portal R4: whole design (loads, deflection 6.9 mm vs the client's 8 mm, knee welds + post face, base fixity, wind share, padstones min 500 x 300 on the garage return foundations).

CONNECTION REFS

C (SC2): SB1 end A on SC2 (FP1)
C (SC1): SB1 end B on SC1 (FP2)
B: SB3 end A on SB1 (FP3)
A: SB2 end A on SB3 (FP4)
C (SC3): SB3 end B on SC3 (FP5)

Do not scale. All dimensions in mm. Check all dimensions on site.

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

GF STEELWORK (FIRST-FLOOR ZONE)

General arrangement SB1-SB3, SC1-SC3, TOS +2665.2: plan, isometric views, set-out, member schedule

SCALE	SIZE	DATE	DRAWN
as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-00	C03

TLS PROPOSAL

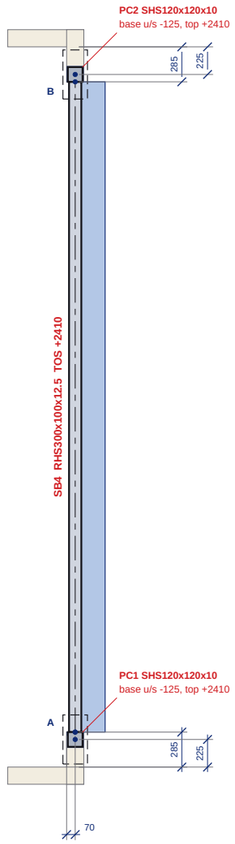
NOT FOR FABRICATION

FOR ENGINEERS REVIEW

until the engineer has reviewed + signed off

PLAN (FOR ENGINEERS REVIEW)

Scale 1:50 | north up

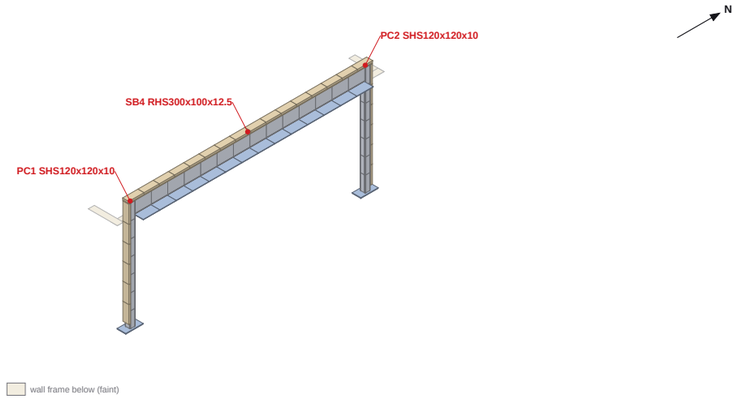


grey = GF timber frame (wall bands)

blue dims: beam end (on its CL) / post centre to the nearest GF frame face

ISOMETRIC FROM THE SOUTH-EAST

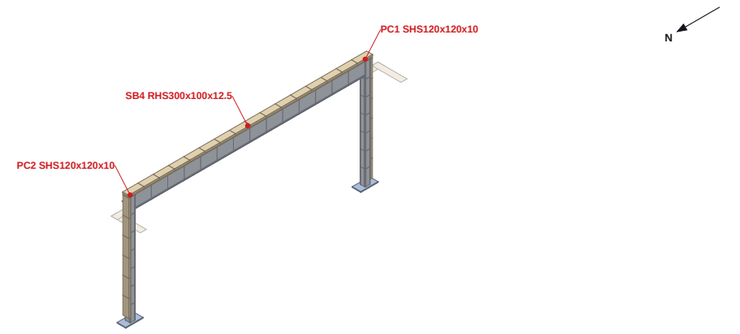
Scale 1:100 | axonometric, axis lengths x0.82



steel | welded plates | timber kit (TLS) | wall frame below (faint)

ISOMETRIC FROM THE NORTH-WEST

Scale 1:100 | axonometric, axis lengths x0.82



steel | welded plates | timber kit (TLS) | wall frame below (faint)

MEMBER SCHEDULE (KG INCL. WELDED PLATES)

MARK	SECTION	GRADE	CUT LENGTH	QTY	KG	TOS / BASE	COATING	
SB4	RHS300x100x12.5	S355J2H	5300	1	505.4	+2410	HDG (EN ISO 1461)	REVIEW
PC1	SHS120x120x10	S355J2H	2510	1	95.2	base -125	HDG (EN ISO 1461)	REVIEW
PC2	SHS120x120x10	S355J2H	2510	1	95.2	base -125	HDG (EN ISO 1461)	REVIEW
TOTAL					**695.8**			

CONNECTION REFS

GP1 knee: SB4 end A/B on PC1/PC2

POST SET-OUT (CENTRES, MODEL MM)

POST	X	Y	BASE U/S	SECTION
PC1	5880	365	-125	SHS120x120x10
PC2	5880	5785	-125	SHS120x120x10

X / Y from the model origin 0,0 = outside faces of the GF frame at the garage SW corner (Wall 1). Padstones and anchors are set out from the model coordinates / the substructure set-out (402) before the frame is erected; check against the frame faces after.

BEAM END SET-OUT

BEAM	END	X	Y	TOS	USS	BEARING / CONNECTION
SB4	A	5880	425	+2410	+2110	Welded all round to the PC1 inner (N) face, 8 FW (GP1 knee, R4)
	B	5880	5725	+2410	+2110	Welded all round to the PC2 inner (S) face, 8 FW (GP1 knee, R4)

End points on the beam centre line at the cut end, model mm (origin as the post set-out). Bearing: the end condition in short - full text on the member sheet.

ENGINEER'S DRAWINGS

Skyhooks 26-431-S-0010 P01 (plan, KP, Details A/B/C, Baseplate Detail 2); 26-431-S-0011 P01 (general notes 3.01-3.07, 7.01, 9.02); 26-431-S-0012 P01 (loads)

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. All steelwork to BS EN 1993-1 / BS EN 1090-2 EXC2, CE marked (S-0011 3.01-3.02). Bolts grade 8.8 (3.03), BS EN ISO 4014/4032; HDG bolts, nuts and washers to BS EN ISO 10684 (oversize-tapped nuts, S-0011 3.07).
3. Mark each member with its mark at end A (hard stamp, before galvanising); posts: 'N' on the baseplate.
4. UB S355JR (BS EN 10025-2); SHS / RHS S355J2H (BS EN 10210-1); plates S275JR.
5. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION (S-0010).
6. Welds: full profile and continuous unless noted (S-0011 3.04).
7. Copes: re-entrant corners radiused 10 min (BS EN 1090-2 6.7: no notches, 5 min), edges ground.
8. Post bases stop 25 above the padstone: HDG shim stacks at the mid-sides under the SHS walls (clear of the anchors), then non-shrink cementitious grout. Padstones by others.
9. Fire protection 30 min to all structural steel by others (S-0011 9.02).
10. Stone on SB1 / SB3: DPC on the flange; first course made up 2675 -> 2700 to course (10 bed + 140 stone from FFL).
11. Items marked TLS detail are not on the engineer's drawings - engineer to confirm.
12. Datum: Levels above FFL = model 0 (top of screed / underside of the GF sole plate)

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

GP1 GARAGE PORTAL

General arrangement SB4, PC1, PC2, TOS +2410: plan, isometric views, set-out, member schedule

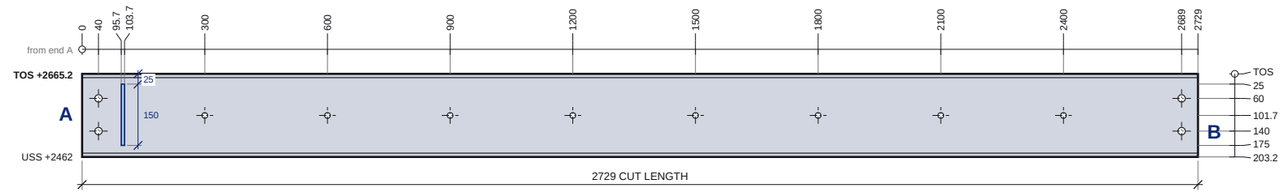
SCALE	SIZE	DATE	DRAWN
as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-01	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

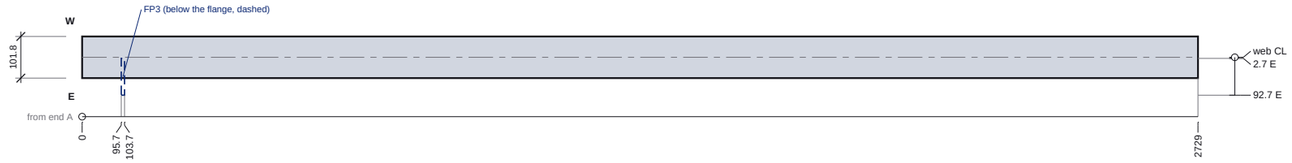
SB1 ELEVATION

Scale 1:15 | viewed from the E side, end A (S) left



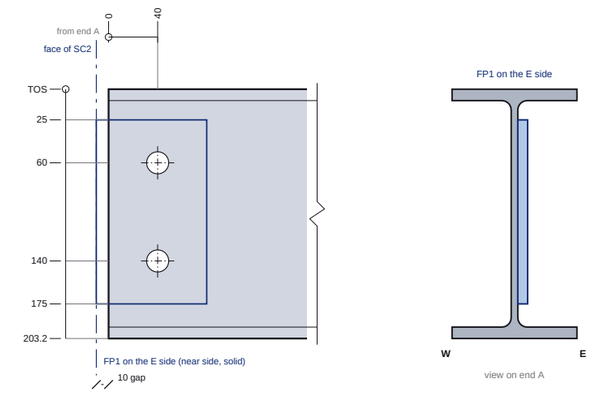
SB1 TOP FLANGE - PLAN

1:15



END A 1:5

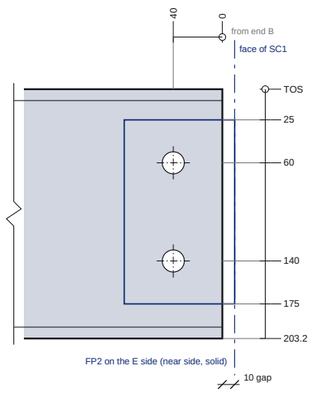
Detail C (SC2): fin plate FP1 welted to SC2

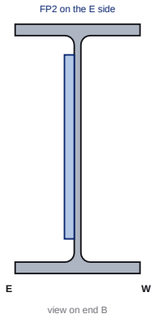


2 no. M16 8.8 bolts in Ø18 holes (bolts loose, site fixed)

END B 1:5

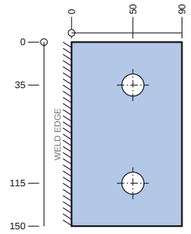
Detail C (SC1): fin plate FP2 welted to SC1





2 no. M16 8.8 bolts in Ø18 holes (bolts loose, site fixed)

FP3 PLATE DETAIL 1:5



150 x 90 x 8 thk S275
2 no. Ø18 holes
0.82 kg qty 1 on SB1
Weld: 8mm double fillet weld (both sides)
SB1 E web face for SB3 end A (B)

HOLE SCHEDULE

No.	FACE	X FROM A	POSITION	HOLE	PURPOSE
H1	web	40	60 below TOS	Ø18	M16 8.8 bolt - C (SC2) (FP1 on SC2)
H2	web	40	140 below TOS	Ø18	M16 8.8 bolt - C (SC2) (FP1 on SC2)
H3	web	2689	60 below TOS	Ø18	M16 8.8 bolt - C (SC1) (FP2 on SC1)
H4	web	2689	140 below TOS	Ø18	M16 8.8 bolt - C (SC1) (FP2 on SC1)
H5	web	300	101.7 below TOS	Ø14	M12 - west packer [TLS detail - engineer to confirm]
H6	web	600	101.7 below TOS	Ø14	M12 - west packer [TLS detail - engineer to confirm]
H7	web	900	101.7 below TOS	Ø14	M12 - west packer [TLS detail - engineer to confirm]
H8	web	1200	101.7 below TOS	Ø14	M12 - west packer [TLS detail - engineer to confirm]
H9	web	1500	101.7 below TOS	Ø14	M12 - west packer [TLS detail - engineer to confirm]
H10	web	1800	101.7 below TOS	Ø14	M12 - west packer [TLS detail - engineer to confirm]
H11	web	2100	101.7 below TOS	Ø14	M12 - west packer [TLS detail - engineer to confirm]
H12	web	2400	101.7 below TOS	Ø14	M12 - west packer [TLS detail - engineer to confirm]

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SB1	UB203x102x23	S355JR	2729	1	63.0
FP3	PL 150x90x8	S275		1	0.82
TOTAL					**63.9 kg**

CONNECTIONS

Detail C (SC2): SB1 end A on SC2 - 150x90x8 S275 fin plate welded to SC2 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10

Detail C (SC1): SB1 end B on SC1 - 150x90x8 S275 fin plate welded to SC1 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10

Detail B: SB3 end A on SB1 - 150x90x8 S275 fin plate welded to SB1 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10; cope 50 long x 25 deep top and bottom

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. UB S355JR (EN 10025-2), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SB1 UB203X102X23 FABRICATION

Under the FF west wall stone leaf, SC2 -> SC1

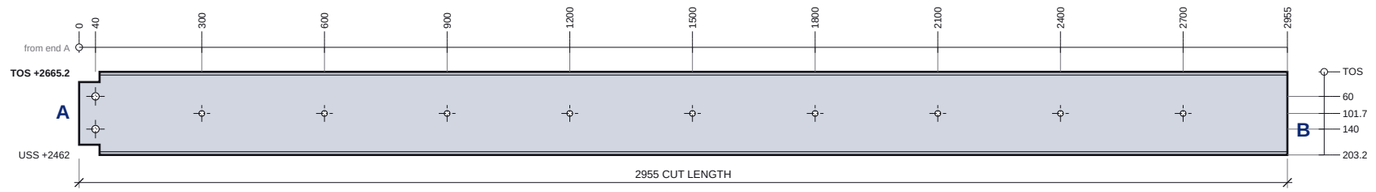
SCALE	SIZE	DATE	DRAWN
1:15 / details as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SB1	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

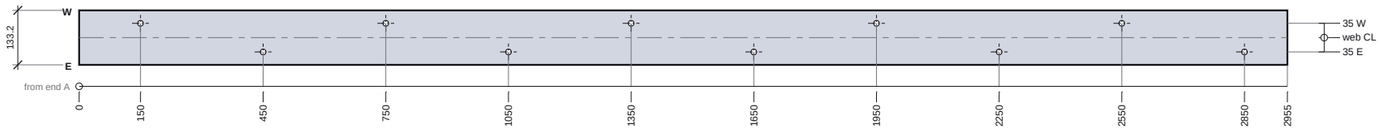
SB2 ELEVATION

Scale 1:15 | viewed from the E side, end A (S) left



SB2 TOP FLANGE - PLAN

1:15



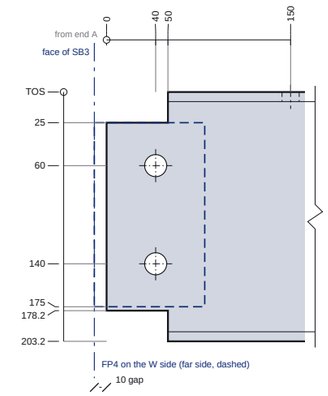
SB2 BOTTOM FLANGE - VIEWED FROM BELOW

1:15



END A 1:5

Detail A: fin plate FP4 welded to SB3



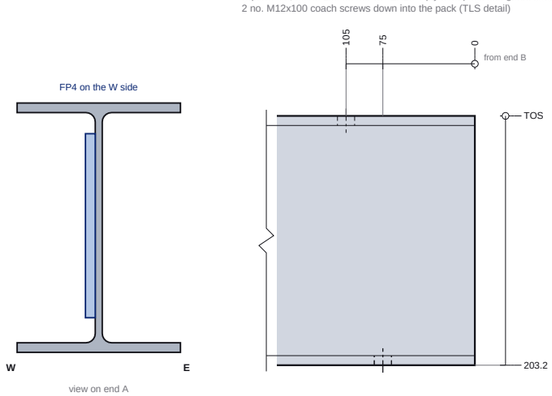
cope 50 long x 25 deep top + bottom

2 no. M16 8.8 bolts in Ø18 holes (bolts loose, site fixed)

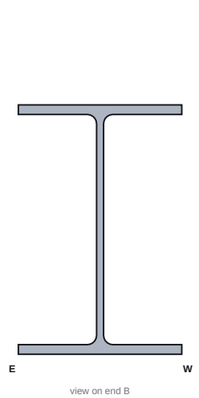
1 no. flange hole(s) in this zone - see the flange plan / schedule

END B 1:5

Square cut; bears 150 on the GF-EX16 4-ply stud pack end grain: studs run up to the steel (+2462), top rail + head binder cut over the pack; 2 no. M12x100 coach screws down into the pack (TLS detail)



3 no. flange hole(s) in this zone - see the flange plan / schedule



HOLE SCHEDULE

No.	FACE	X FROM A	POSITION	HOLE	PURPOSE
H1	web	40	60 below TOS	Ø18	M16 8.8 bolt - A (FP4 on SB3)
H2	web	40	140 below TOS	Ø18	M16 8.8 bolt - A (FP4 on SB3)
H3	web	300	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H4	web	600	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H5	web	900	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H6	web	1200	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H7	web	1500	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H8	web	1800	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H9	web	2100	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H10	web	2400	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H11	web	2700	101.7 below TOS	Ø14	M12 - east packer + binder [TLS detail - engineer to confirm]
H12	tf	150	35 W of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]

HOLE SCHEDULE (CONT.)

No.	FACE	X FROM A	POSITION	HOLE	PURPOSE
H13	tf	450	35 E of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H14	tf	750	35 W of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H15	tf	1050	35 E of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H16	tf	1350	35 W of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H17	tf	1650	35 E of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H18	tf	1950	35 W of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H19	tf	2250	35 E of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H20	tf	2550	35 W of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H21	tf	2850	35 E of web CL	Ø14	M12x50 coach screw up into the 50x145 top plate (39 embed) [TLS detail - engineer to confirm]
H22	bf	2880	35 W of web CL	Ø14	M12x100 coach screw down into the GF-EX16 stud pack [TLS detail - engineer to confirm]
H23	bf	2880	35 E of web CL	Ø14	M12x100 coach screw down into the GF-EX16 stud pack [TLS detail - engineer to confirm]

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SB2	UB203x133x25	S355JR	2955	1	74.2
TOTAL					**74.2 kg**

CONNECTIONS

Detail A: SB2 end A on SB3 - 150x90x8 S275 fin plate welded to SB3 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10; cope 50 long x 25 deep top and bottom

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. UB S355JR (EN 10025-2).
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Blast Sa 2.5 + single-coat shop primer, 80 µm DFT, BS EN ISO 12944-5 C2 (S-0011 3.05) - fabricator to confirm product; 30 min fire protection by others (9.02).
7. Copes: re-entrant corners radiused 10 min (BS EN 1090-2 6.7, 5 min), cut edges ground smooth.
8. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.
9. Install order: bearing coach screws, then the top-plate coach screws, THEN the east packer + binder (the packer leaves ~20 mm inside the section - no screw can be driven after it).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SB2 UB203X133X25 FABRICATION

Under FF-EX1 and the kitchen joist ends, SB3 -> GF-EX16 4-ply pack

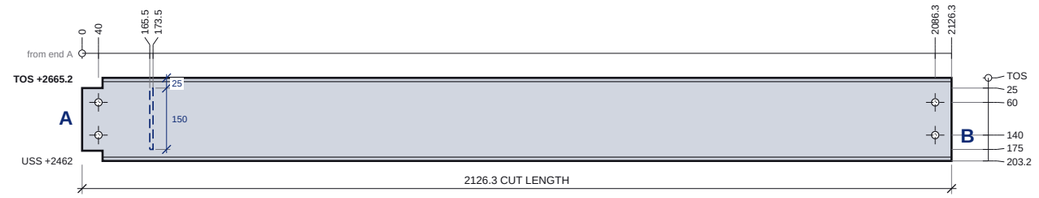
SCALE	SIZE	DATE	DRAWN
1:15 / details as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SB2	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

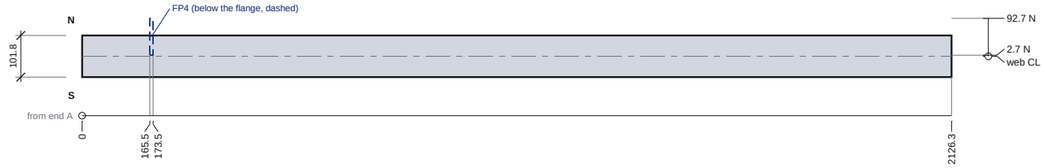
SB3 ELEVATION

Scale 1:15 | viewed from the S side, end A (W) left



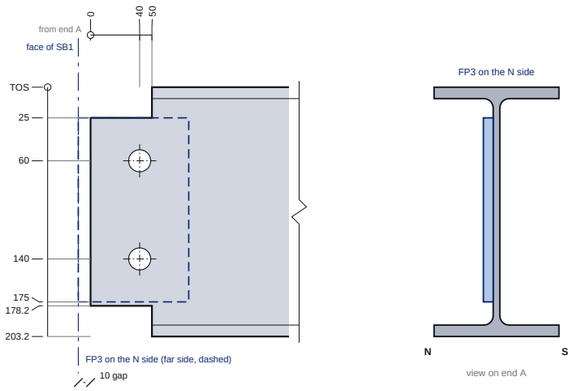
SB3 TOP FLANGE - PLAN

1:15



END A 1:5

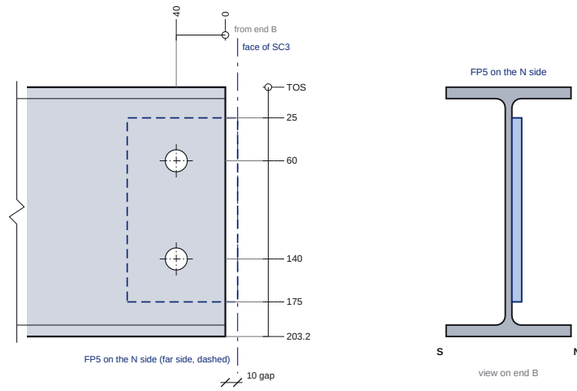
Detail B: fin plate FP3 welted to SB1



cope 50 long x 25 deep top + bottom
2 no. M16 8.8 bolts in Ø18 holes (bolts loose, site fixed)

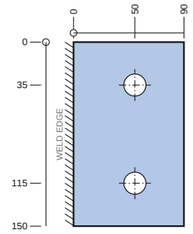
END B 1:5

Detail C (SC3): fin plate FP5 welted to SC3



2 no. M16 8.8 bolts in Ø18 holes (bolts loose, site fixed)

FP4 PLATE DETAIL 1:5



150 x 90 x 8 thk S275
2 no. Ø18 holes
0.82 kg qty 1 on SB3
Weld: 8mm double fillet weld (both sides)
SB3 N web face for SB2 end A (A)

HOLE SCHEDULE

No.	FACE	X FROM A	POSITION	HOLE	PURPOSE
H1	web	40	60 below TOS	Ø18	M16 8.8 bolt - B (FP3 on SB1)
H2	web	40	140 below TOS	Ø18	M16 8.8 bolt - B (FP3 on SB1)
H3	web	2086.3	60 below TOS	Ø18	M16 8.8 bolt - C (SC3) (FP5 on SC3)
H4	web	2086.3	140 below TOS	Ø18	M16 8.8 bolt - C (SC3) (FP5 on SC3)

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SB3	UB203x102x23	S355JR	2126.3	1	49.1
FP4	PL 150x90x8	S275		1	0.82
TOTAL					**49.9 kg**

CONNECTIONS

Detail B: SB3 end A on SB1 - 150x90x8 S275 fin plate welded to SB1 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10; cope 50 long x 25 deep top and bottom

Detail A: SB2 end A on SB3 - 150x90x8 S275 fin plate welded to SB3 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10; cope 50 long x 25 deep top and bottom

Detail C (SC3): SB3 end B on SC3 - 150x90x8 S275 fin plate welded to SC3 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. UB S355JR (EN 10025-2), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Copes: re-entrant corners radiused 10 min (BS EN 1090-2 6.7, 5 min), cut edges ground smooth.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SB3 UB203X102X23 FABRICATION

Under the FF south wall stone leaf, SB1 -> SC3

SCALE	SIZE	DATE	DRAWN
1:15 / details as noted	A3	11.10.26	TS

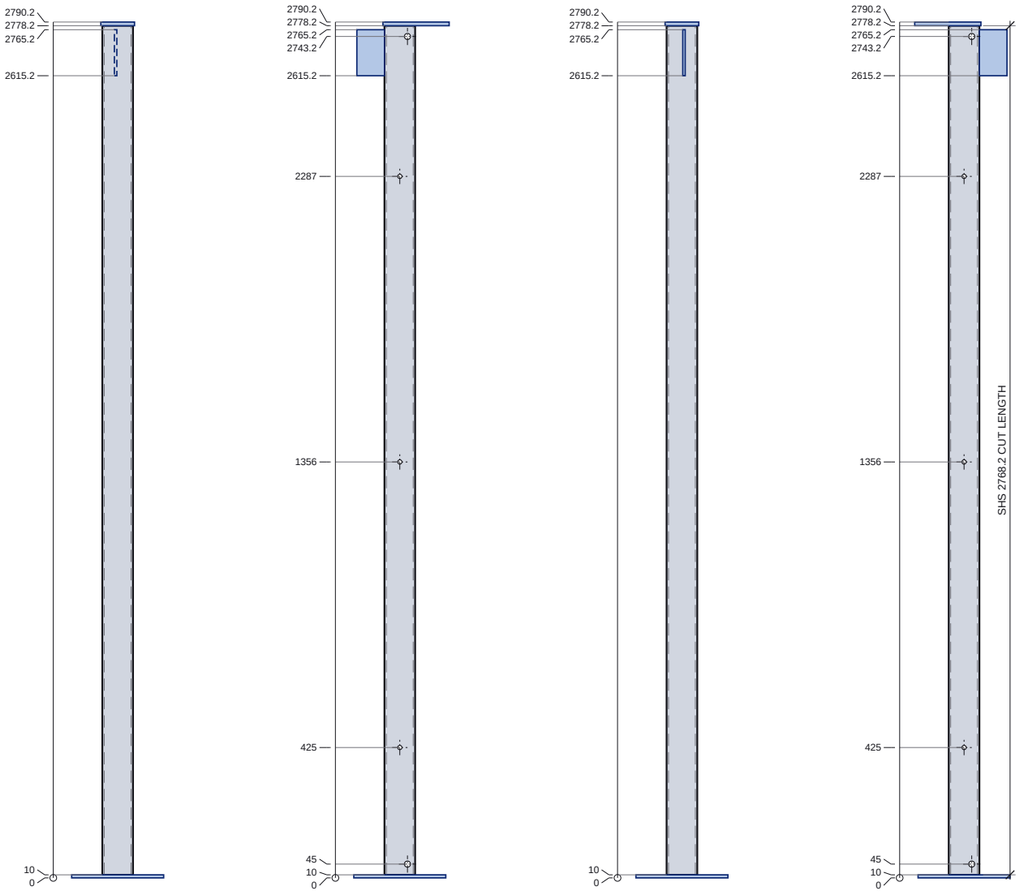
STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SB3	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

SC1 FACE ELEVATIONS

Scale 1:20 | heights above the underside of the baseplate (-125)

Base underside -125 = height 0; top of cap +2665.2 = height 2790.2



N FACE — from N: E left, W right

E FACE — from E: S left, N right

S FACE — from S: W left, E right

W FACE — from W: N left, S right

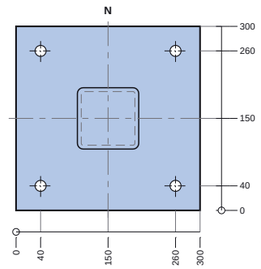
Plates dashed = behind the post in that view. Holes: see the schedule below.

HOLE SCHEDULE

No.	FACE	HEIGHT	POSITION	HOLE	PURPOSE
H1	E	2743.2	25 N	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H2	W	2743.2	25 S	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H3	E	45	25 N	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H4	W	45	25 S	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H5	through-EW	425	on CL	Ø14	M12 rod (165) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]
H6	through-EW	1356	on CL	Ø14	M12 rod (165) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]
H7	through-EW	2287	on CL	Ø14	M12 rod (165) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]

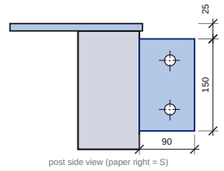
BP1 BASEPLATE PLAN 1:10

from above, N up



PL 300 x 300 x 10 S275 (6.99 kg)
4 no. Ø18 holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 8mm FW all round

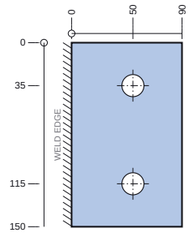
FP2 ON THE S FACE 1:10



post side view (paper right = S)

Plate faces at 2.7 E and 10.7 E of the post CL (8 thk)
top 2765.2 / bottom 2615.2 above base u/s; top 25 below the cap top (= TOS)
Weld: 8mm double fillet weld (both sides)

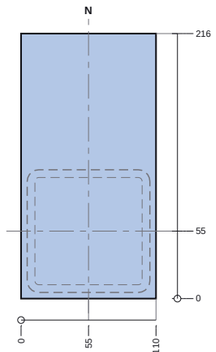
FP2 PLATE DETAIL 1:5



150 x 90 x 8 thk S275
2 no. Ø18 holes
0.82 kg qty 1 on SC1
Weld: 8mm double fillet weld (both sides)
SC1 S face for SB1 end B (C (SC1))

CP1 CAP PLATE PLAN 1:5

from above, N up



PL 216 x 110 x 12 S275 (2.24 kg)
no holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 6mm FW all round (cap oversails 5 min)
ECCENTRIC: extends 55 E / 55 W / 161 N / 55 S of the post CL
TLS detail - engineer to confirm

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SC1	SHS100x100x6.3	S355J2H	2768.2	1	50.4
BP1	PL 300x300x10	S275		1	6.99
CP1	PL 216x110x12	S275		1	2.24
FP2	PL 150x90x8	S275		1	0.82
TOTAL					**60.4 kg**

CONNECTIONS

Detail C (SC1): SB1 end B on SC1 - 150x90x8 S275 fin plate welded to SC1 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. SHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Vent / drain holes: plug after galvanising.
8. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SC1 SHS100X100X6.3 POST FABRICATION

SHS post, base -125 on 25 shims over a padstone at -150 (by others)

SCALE	SIZE	DATE	DRAWN
1:20 / 1:5 / as noted	A3	11.10.26	TS

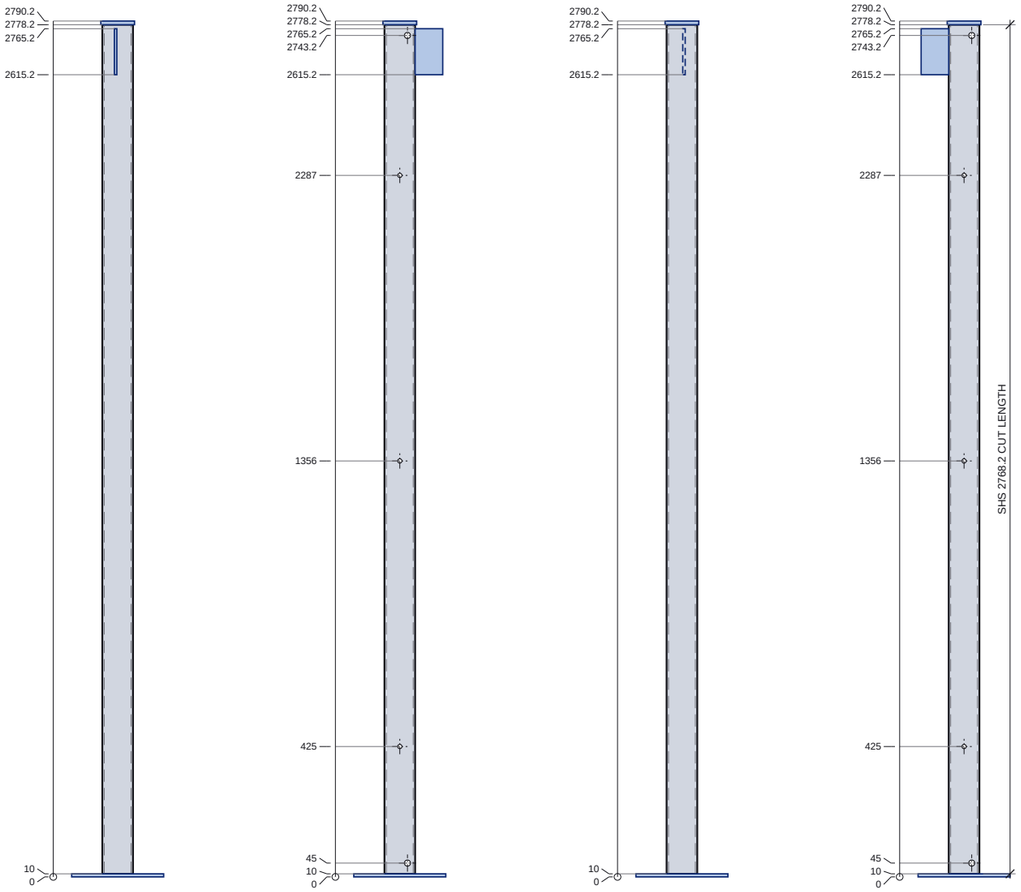
STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SC1	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

SC2 FACE ELEVATIONS

Scale 1:20 | heights above the underside of the baseplate (-125)

Base underside -125 = height 0; top of cap +2665.2 = height 2790.2



N FACE — from N: E left, W right

E FACE — from E: S left, N right

S FACE — from S: W left, E right

W FACE — from W: N left, S right

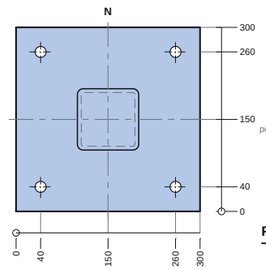
Plates dashed = behind the post in that view. Holes: see the schedule below.

HOLE SCHEDULE

No.	FACE	HEIGHT	POSITION	HOLE	PURPOSE
H1	E	2743.2	25 N	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H2	W	2743.2	25 S	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H3	E	45	25 N	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H4	W	45	25 S	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H5	through-EW	425	on CL	Ø14	M12 rod (173) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]
H6	through-EW	1356	on CL	Ø14	M12 rod (173) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]
H7	through-EW	2287	on CL	Ø14	M12 rod (173) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]

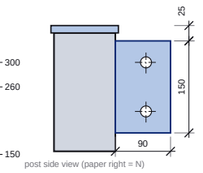
BP2 BASEPLATE PLAN 1:10

from above, N up



PL 300 x 300 x 10 S275 (6.99 kg)
4 no. Ø18 holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 8mm FW all round

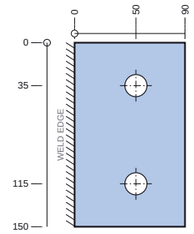
FP1 ON THE N FACE 1:10



post side view (paper right = N)

Plate faces at 2.7 E and 10.7 E of the post CL (8 thk)
top 2765.2 / bottom 2615.2 above base u/s; top 25 below the cap top (= TOS)
Weld: 8mm double fillet weld (both sides)

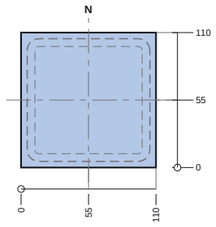
FP1 PLATE DETAIL 1:5



150 x 90 x 8 thk S275
2 no. Ø18 holes
0.82 kg qty 1 on SC2
Weld: 8mm double fillet weld (both sides)
SC2 N face for SB1 end A (C (SC2))

CP2 CAP PLATE PLAN 1:5

from above, N up



PL 110 x 110 x 12 S275 (1.14 kg)
no holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 6mm FW all round (cap oversails 5 min)
TLS detail - engineer to confirm

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SC2	SHS100x100x6.3	S355J2H	2768.2	1	50.4
BP2	PL 300x300x10	S275		1	6.99
CP2	PL 110x110x12	S275		1	1.14
FP1	PL 150x90x8	S275		1	0.82
TOTAL					**59.3 kg**

CONNECTIONS

Detail C (SC2): SB1 end A on SC2 - 150x90x8 S275 fin plate welded to SC2 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. SHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Vent / drain holes: plug after galvanising.
8. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SC2 SHS100X100X6.3 POST FABRICATION

SHS post, base -125 on 25 shims over a padstone at -150 (by others)

SCALE	SIZE	DATE	DRAWN
1:20 / 1:5 / as noted	A3	11.10.26	TS

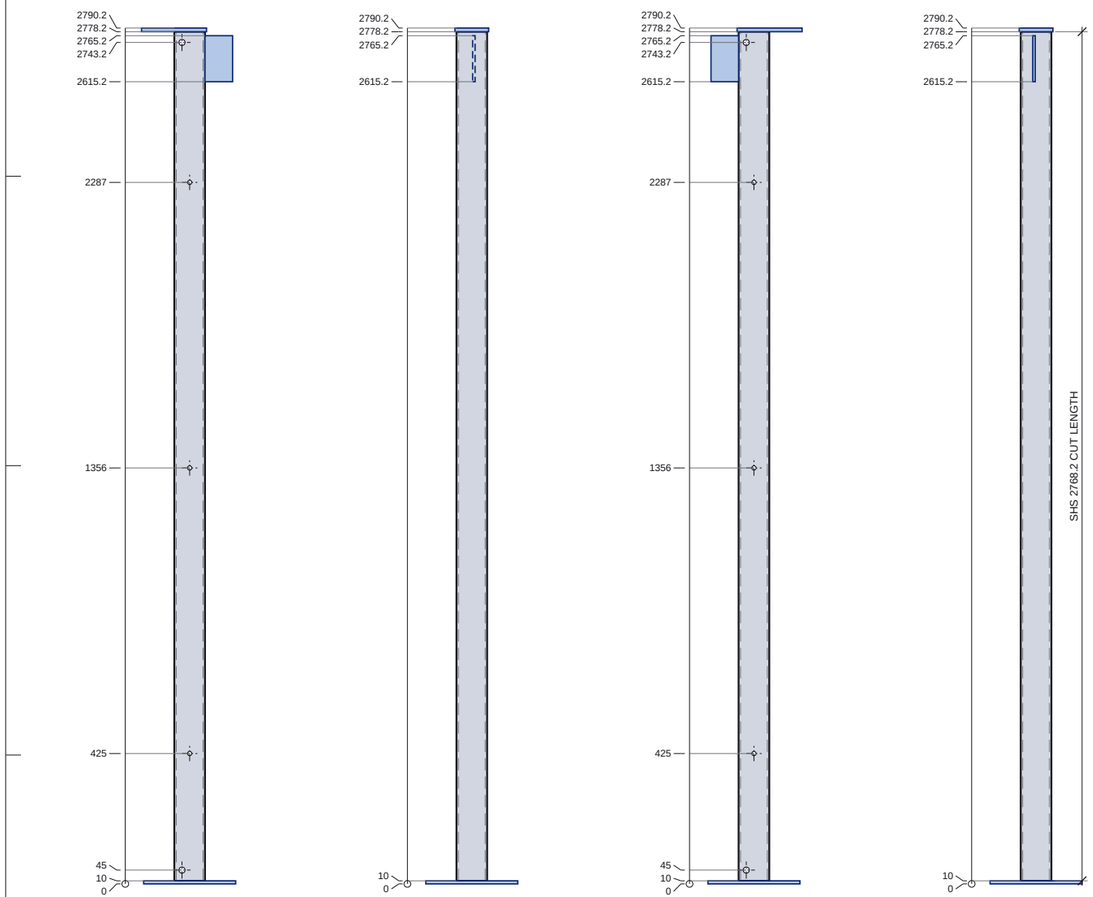
STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SC2	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

SC3 FACE ELEVATIONS

Scale 1:20 | heights above the underside of the baseplate (-125)

Base underside -125 = height 0; top of cap +2665.2 = height 2790.2



N FACE — from N: E left, W right

E FACE — from E: S left, N right

S FACE — from S: W left, E right

W FACE — from W: N left, S right

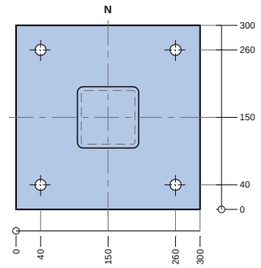
Plates dashed = behind the post in that view. Holes: see the schedule below.

HOLE SCHEDULE

No.	FACE	HEIGHT	POSITION	HOLE	PURPOSE
H1	N	2743.2	25 E	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H2	S	2743.2	25 W	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H3	N	45	25 E	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H4	S	45	25 W	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H5	through-NS	425	on CL	Ø14	M12 rod (173) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]
H6	through-NS	1356	on CL	Ø14	M12 rod (173) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]
H7	through-NS	2287	on CL	Ø14	M12 rod (173) through-bolt for the timber jamb plates (snug tight - do not crush the SHS) [TLS detail - engineer to confirm]

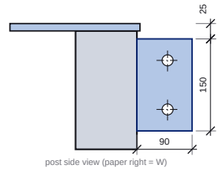
BP3 BASEPLATE PLAN 1:10

from above, N up



PL 300 x 300 x 10 S275 (6.99 kg)
4 no. Ø18 holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 8mm FW all round

FP5 ON THE W FACE 1:10



post side view (paper right = W)

Plate faces at 2.7 N and 10.7 N of the post CL (8 thk)
top 2765.2 / bottom 2615.2 above base u/s; top 25 below the cap top (= TOS)
Weld: 8mm double fillet weld (both sides)

CP3 CAP PLATE PLAN 1:10

from above, N up

PL 212 x 110 x 12 S275 (2.20 kg)
no holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 6mm FW all round (cap oversails 5 min)
ECCENTRIC: extends 157 E / 55 W / 55 N / 55 S of the post CL
TLS detail - engineer to confirm

FP5 PLATE DETAIL 1:5

150 x 90 x 8 thk S275
2 no. Ø18 holes
0.82 kg qty 1 on SC3
Weld: 8mm double fillet weld (both sides)
SC3 W face for SB3 end B (C (SC3))

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SC3	SHS100x100x6.3	S355J2H	2768.2	1	50.4
BP3	PL 300x300x10	S275		1	6.99
CP3	PL 212x110x12	S275		1	2.20
FP5	PL 150x90x8	S275		1	0.82
TOTAL					**60.4 kg**

CONNECTIONS

Detail C (SC3): SB3 end B on SC3 - 150x90x8 S275 fin plate welded to SC3 (8mm double fillet weld (both sides)); 2 no. M16 8.8 bolts, Ø18 holes; bolt line 50 from the support face, 40 to the plate end; pitch 35/80/35; plate top 25 below TOS; beam end gap 10

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. SHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Vent / drain holes: plug after galvanising.
8. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SC3 SHS100X100X6.3 POST FABRICATION

SHS post, base -125 on 25 shims over a padstone at -150 (by others)

SCALE	SIZE	DATE	DRAWN
1:20 / 1:5 / as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SC3	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

TLS PROPOSAL

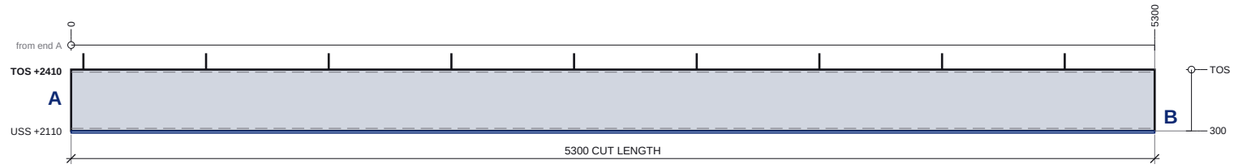
NOT FOR FABRICATION

FOR ENGINEERS REVIEW

until the engineer has reviewed + signed off

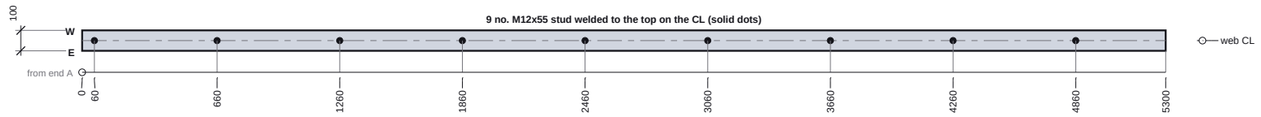
SB4 ELEVATION

Scale 1:30 | viewed from the E side, end A (S) left



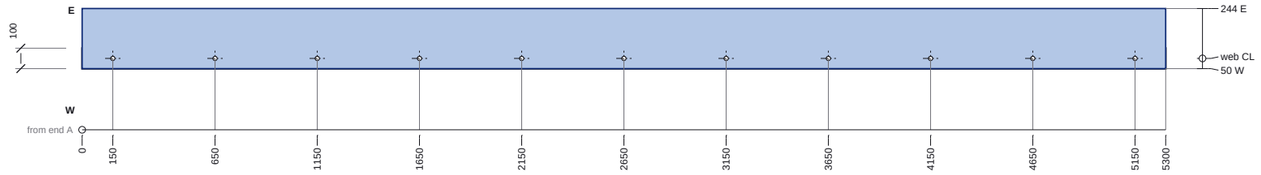
SB4 TOP - PLAN

1:30



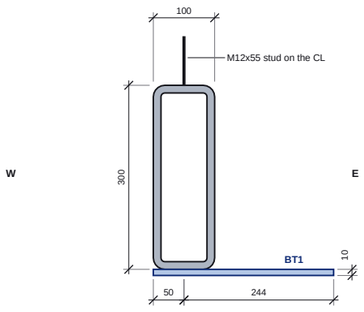
SB4 UNDERSIDE - VIEWED FROM BELOW

1:30



SECTION THROUGH SB4 1:10

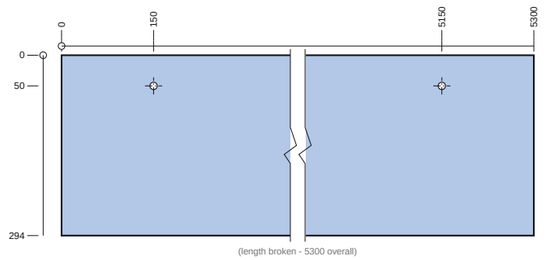
looking from end A to end B: W left, E right



BT1: 50 W / 244 E of the CL, underside 310 below TOS

Knee connections + assembly: see P4-ST-GP1

BT1 PLATE DETAIL 1:10



u along the beam from the south end (left edge), v from the west edge (top edge)

5300 x 294 x 10 thk S275
11 no. Ø12 holes
122.22 kg qty 1 on SB4
Hole u (all): 150, 650, 1150, 1650, 2150, 2650, 3150, 3650, 4150, 4650, 5150 (middle ones in the broken length)
Weld: 6mm FW both toes to the RHS (flare-bevel at the RHS corners, throat 5 min); ends 6 FW to the post inner faces
boot plate under the RHS, 10 back from the stone face; Ø12 vents in the 100 lap under the RHS (sealed lap, BS EN ISO 14713-2), plugged after galvanising - galvaniser to confirm
TLS detail - engineer to confirm

HOLE SCHEDULE

No.	FACE	X FROM A	POSITION	HOLE / STUD	PURPOSE
S1	top	60	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S2	top	660	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S3	top	1260	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S4	top	1860	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S5	top	2460	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S6	top	3060	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S7	top	3660	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S8	top	4260	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S9	top	4860	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
BT1 V1..V11	BT1	150..5150 @500	u / v: BT1 PLATE DETAIL	Ø12	11 no. vent / drain holes in BT1 - plug / seal after galvanising

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SB4	RHS300x100x12.5	S355J2H	5300	1	383.2
BT1	PL 5300x294x10	S275		1	122.22
TOTAL					**505.4 kg**

CONNECTIONS

Detail GP1 knee: SB4 end A/B on PC1/PC2 - Shop-welded knee (R4): RHS square-cut ends welded all round to the post inner faces, 8 FW; BT1 ends 6 FW to the post faces. Frame galvanised as one piece (galvaniser to confirm the bath + dip; temporary bolted spreader between the bases until set; post centres +/-3 after galvanising). FOR ENGINEERS REVIEW

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. RHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.
8. GP1 - TLS PROPOSAL R4 FOR ENGINEERS REVIEW (replaces S-0010 UB203x133x30); calc on the GP1 sheet
9. RHS side walls land on the SHS corner radius (flat +/-45): flare-bevel weld, throat 5 min - engineer to confirm.
10. Studs welded before galvanising: chase the threads / oversize-tapped HDG nuts (BS EN ISO 10684).
11. GP1 R4 - PRELIMINARY, FOR ENGINEER REVIEW (TLS proposal, not the engineer's design). Replaces S-0010's UB203x133x30: carries the 5300 door opening AND ~0.5 m of stone (main-house stone eaves).
12. LOADS (char.): trusses S-0012 DL 15 / IL 10 kN each end -> 5.4 / 3.6 kN/m; stone 0.5x0.1x23 = 1.15 kN/m; boot + timber + self weight 1.0 kN/m; door rails + operator 2.0 kN mid-span; wind 12 kN SLS / 18 kN ULS at the eaves.
13. RHS 300x100x12.5: I 9103 cm4, Wpl 806 cm3. ULS M 61 kNm vs Mpl 286 kNm. Deflection (SLS incl. door, pinned, L 5420) 6.9 mm = L/787 < 8 mm (client limit).
14. SHS 120x120x10 to TOS: N 44 kN + knee M 29.7 kNm vs Mpl 62 kNm; sway 6.8 mm = h/326.
15. KNEE (R4): RHS ends welded all round to the post inner faces, 8 FW: couple 29.7 / 0.288 = 103 kN per RHS wall vs 136 kN; post face (b1/b0 0.83) ~390 kN; torsion from the boot 0.84 kNm per knee.
16. BASE: 400x200x15, 54 clear of the stone leaf; N 44 kN, H 9 kN per base (ULS); pinned base assumed for sway. Boot toe ~0.22 kNm/m -> 13 MPa in the 10 plate.
17. FOR THE ENGINEER: loads, wind share, knee welds + post face (stiffeners in the SHS?), base fixity, padstones on the garage return foundations (min 500 x 300), cavity tray / thermal bridge at the boot, 30 min fire protection (S-0011 9.02).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SB4 RHS300X100X12.5 FABRICATION

GP1 portal beam between the posts over the 5300 garage door; carries GT1 + ~0.5 m stone on the boot

SCALE	SIZE	DATE	DRAWN
1:30 / details as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SB4	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

TLS PROPOSAL

NOT FOR FABRICATION

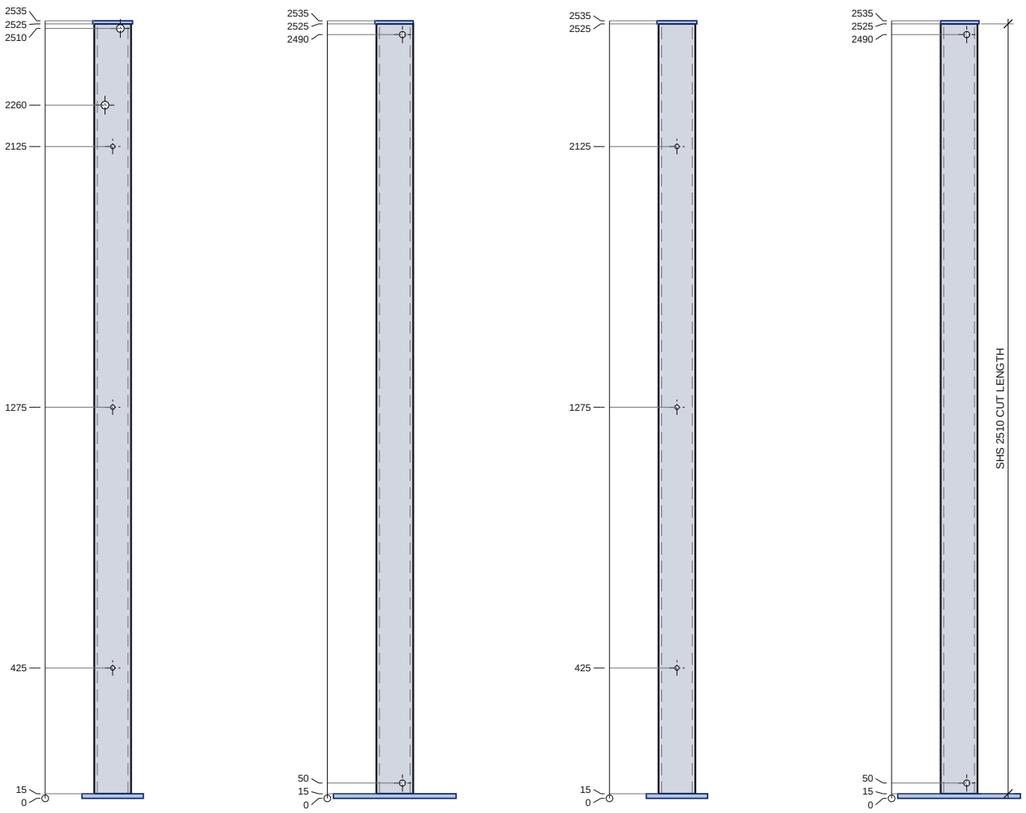
FOR ENGINEERS REVIEW

until the engineer has reviewed + signed off

PC1 FACE ELEVATIONS

Scale 1:20 | heights above the underside of the baseplate (-125)

Base underside -125 = height 0; top of cap +2410 = height 2535



N FACE — from N: E left, W right

E FACE — from E: S left, N right

S FACE — from S: W left, E right

W FACE — from W: N left, S right

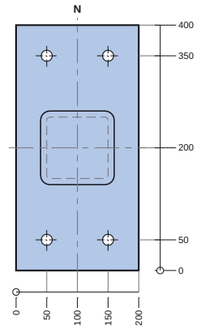
Plates dashed = behind the post in that view. Holes: see the schedule below.

HOLE SCHEDULE

No.	FACE	HEIGHT	POSITION	HOLE	PURPOSE
H1	E	2490	25 N	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H2	W	2490	25 S	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H3	E	50	25 N	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H4	W	50	25 S	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H5	N	2260	25 E	Ø25	Ø25 through the inner face inside the RHS footprint: links + drains the RHS and SHS hollows for galvanising (drill before welding) [TLS detail - engineer to confirm]
H6	N	2510	25 W	Ø25	Ø25 through the inner face inside the RHS footprint: links + drains the RHS and SHS hollows for galvanising (drill before welding) [TLS detail - engineer to confirm]
H7	through-NS	425	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]
H8	through-NS	1275	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]
H9	through-NS	2125	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]

BPPC1 BASEPLATE PLAN 1:10

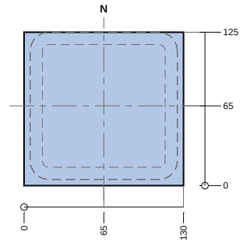
from above, N up



PL 400 x 200 x 15 S275 (9.30 kg)
4 no. Ø18 holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 8mm FW all round
TLS detail - engineer to confirm

KPPC1 CAP PLATE PLAN 1:5

from above, N up



PL 130 x 125 x 10 S275 (1.28 kg)
no holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 6mm FW on the three oversailing sides; inner face flush, sealed by the RHS end weld
ECCENTRIC: extends 65 E / 65 W / 60 N / 65 S of the post CL
TLS detail - engineer to confirm

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
PC1	SHS120x120x10	S355J2H	2510	1	84.6
BPPC1	PL 400x200x15	S275		1	9.30
KPPC1	PL 130x125x10	S275		1	1.28
TOTAL					**95.2 kg**

CONNECTIONS

Detail GP1 knee: SB4 end A/B on PC1/PC2 - Shop-welded knee (R4): RHS square-cut ends welded all round to the post inner faces, 8 FW; BT1 ends 6 FW to the post faces. Frame galvanised as one piece (galvaniser to confirm the bath + dip; temporary bolted spreader between the bases until set; post centres +/-3 after galvanising). FOR ENGINEERS REVIEW

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. SHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Vent / drain holes: plug after galvanising.
8. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.
9. GP1 - TLS PROPOSAL R4 FOR ENGINEERS REVIEW (replaces S-0010 UB203x133x30); calc on the GP1 sheet
10. PC1 and PC2 steel are handed pairs (inner-face holes face each other): jamb plate faces differ, see the timber schedule.
11. GP1 R4 - PRELIMINARY, FOR ENGINEER REVIEW (TLS proposal, not the engineer's design). Replaces S-0010's UB203x133x30: carries the 5300 door opening AND ~0.5 m of stone (main-house stone eaves).
12. LOADS (char.): trusses S-0012 DL 15 / IL 10 kN each end -> 5.4 / 3.6 kN/m; stone 0.5x0.1x23 = 1.15 kN/m; boot + timber + self weight 1.0 kN/m; door rails + operator 2.0 kN mid-span; wind 12 kN SLS / 18 kN ULS at the eaves.
13. RHS 300x100x12.5: I 9103 cm4, Wpl 806 cm3. ULS M 61 kNm vs Mpl 286 kNm. Deflection (SLS incl. door, pinned, L 5420) 6.9 mm = L/787 < 8 mm (client limit).
14. SHS 120x120x10 to TOS: N 44 kN + knee M 29.7 kNm vs Mpl 62 kNm; sway 6.8 mm = h/326.
15. KNEE (R4): RHS ends welded all round to the post inner faces, 8 FW: couple 29.7 / 0.288 = 103 kN per RHS wall vs 136 kN; post face (b1/b0 0.83) ~390 kN; torsion from the boot 0.84 kNm per knee.
16. BASE: 400x200x15, 54 clear of the stone leaf; N 44 kN, H 9 kN per base (ULS); pinned base assumed for sway. Boot toe ~0.22 kNm/m -> 13 MPa in the 10 plate.
17. FOR THE ENGINEER: loads, wind share, knee welds + post face (stiffeners in the SHS?), base fixity, padstones on the garage return foundations (min 500 x 300), cavity tray / thermal bridge at the boot, 30 min fire protection (S-0011 9.02).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

PC1 SHS120X120X10 POST FABRICATION

GP1 portal post to TOS; the RHS frames into its inner face (R4)

SCALE	SIZE	DATE	DRAWN
1:20 / 1:5 / as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-PC1	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

TLS PROPOSAL

NOT FOR FABRICATION

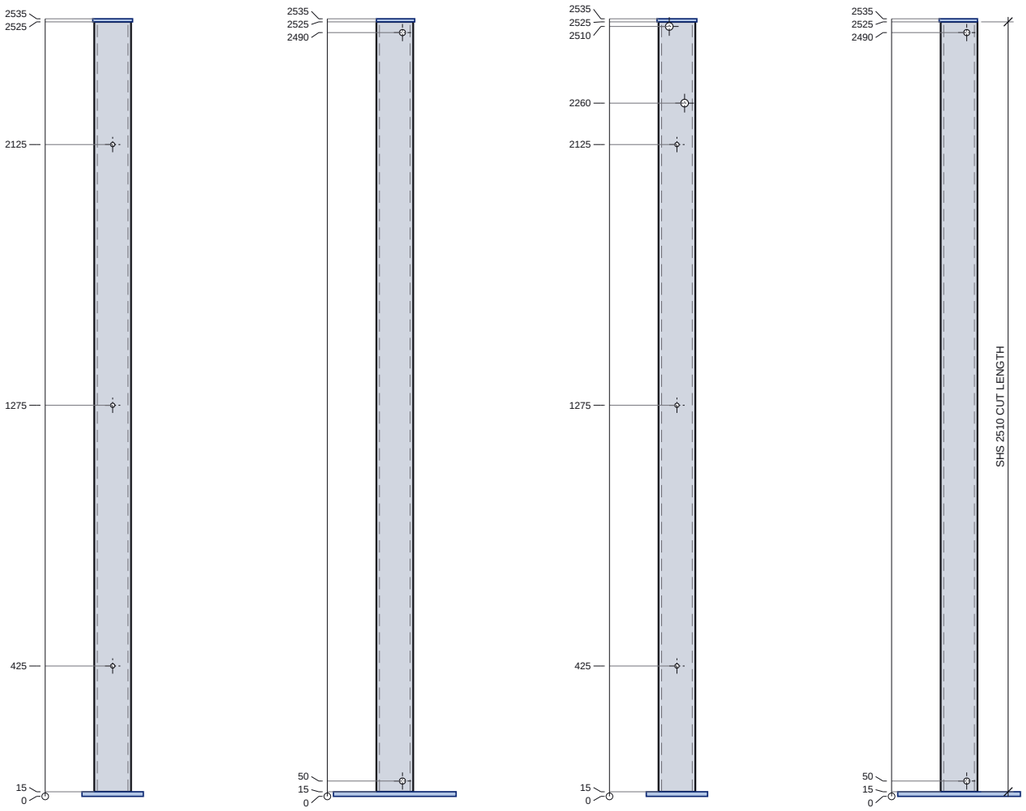
FOR ENGINEERS REVIEW

until the engineer has reviewed + signed off

PC2 FACE ELEVATIONS

Scale 1:20 | heights above the underside of the baseplate (-125)

Base underside -125 = height 0; top of cap +2410 = height 2535



N FACE
from N: E left, W right

E FACE
from E: S left, N right

S FACE
from S: W left, E right

W FACE
from W: N left, S right

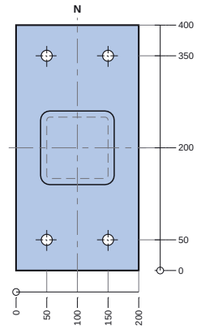
Plates dashed = behind the post in that view. Holes: see the schedule below.

HOLE SCHEDULE

No.	FACE	HEIGHT	POSITION	HOLE	PURPOSE
H1	E	2490	25 N	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H2	W	2490	25 S	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H3	E	50	25 N	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H4	W	50	25 S	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H5	S	2260	25 E	Ø25	Ø25 through the inner face inside the RHS footprint: links + drains the RHS and SHS hollows for galvanising (drill before welding) [TLS detail - engineer to confirm]
H6	S	2510	25 W	Ø25	Ø25 through the inner face inside the RHS footprint: links + drains the RHS and SHS hollows for galvanising (drill before welding) [TLS detail - engineer to confirm]
H7	through-NS	425	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]
H8	through-NS	1275	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]
H9	through-NS	2125	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]

BPPC2 BASEPLATE PLAN 1:10

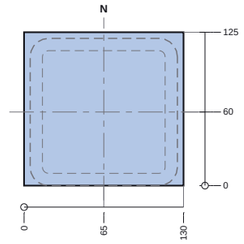
from above, N up



PL 400 x 200 x 15 S275 (9.30 kg)
4 no. Ø18 holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 8mm FW all round
TLS detail - engineer to confirm

KPPC2 CAP PLATE PLAN 1:5

from above, N up



PL 130 x 125 x 10 S275 (1.28 kg)
no holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 6mm FW on the three oversailing sides; inner face flush, sealed by the RHS end weld
ECCENTRIC: extends 65 E / 65 W / 65 N / 60 S of the post CL
TLS detail - engineer to confirm

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
PC2	SHS120x120x10	S355J2H	2510	1	84.6
BPPC2	PL 400x200x15	S275		1	9.30
KPPC2	PL 130x125x10	S275		1	1.28
TOTAL					**95.2 kg**

CONNECTIONS

Detail GP1 knee: SB4 end A/B on PC1/PC2 - Shop-welded knee (R4): RHS square-cut ends welded all round to the post inner faces, 8 FW; BT1 ends 6 FW to the post faces. Frame galvanised as one piece (galvaniser to confirm the bath + dip; temporary bolted spreader between the bases until set; post centres +/-3 after galvanising). FOR ENGINEERS REVIEW

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. SHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Vent / drain holes: plug after galvanising.
8. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.
9. GP1 - TLS PROPOSAL R4 FOR ENGINEERS REVIEW (replaces S-0010 UB203x133x30); calc on the GP1 sheet
10. PC1 and PC2 steel are handed pairs (inner-face holes face each other): jamb plate faces differ, see the timber schedule.
11. GP1 R4 - PRELIMINARY, FOR ENGINEER REVIEW (TLS proposal, not the engineer's design). Replaces S-0010's UB203x133x30: carries the 5300 door opening AND ~0.5 m of stone (main-house stone eaves).
12. LOADS (char.): trusses S-0012 DL 15 / IL 10 kN each end -> 5.4 / 3.6 kN/m; stone 0.5x0.1x23 = 1.15 kN/m; boot + timber + self weight 1.0 kN/m; door rails + operator 2.0 kN mid-span; wind 12 kN SLS / 18 kN ULS at the eaves.
13. RHS 300x100x12.5: I 9103 cm4, Wpl 806 cm3. ULS M 61 kNm vs Mpl 286 kNm. Deflection (SLS incl. door, pinned, L 5420) 6.9 mm = L/787 < 8 mm (client limit).
14. SHS 120x120x10 to TOS: N 44 kN + knee M 29.7 kNm vs Mpl 62 kNm; sway 6.8 mm = h/326.
15. KNEE (R4): RHS ends welded all round to the post inner faces, 8 FW: couple 29.7 / 0.288 = 103 kN per RHS wall vs 136 kN; post face (b1/b0 0.83) ~390 kN; torsion from the boot 0.84 kNm per knee.
16. BASE: 400x200x15, 54 clear of the stone leaf; N 44 kN, H 9 kN per base (ULS); pinned base assumed for sway. Boot toe ~0.22 kNm/m -> 13 MPa in the 10 plate.
17. FOR THE ENGINEER: loads, wind share, knee welds + post face (stiffeners in the SHS?), base fixity, padstones on the garage return foundations (min 500 x 300), cavity tray / thermal bridge at the boot, 30 min fire protection (S-0011 9.02).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

PC2 SHS120X120X10 POST FABRICATION

GP1 portal post to TOS; the RHS frames into its inner face (R4)

SCALE	SIZE	DATE	DRAWN
1:20 / 1:5 / as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-PC2	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

TLS PROPOSAL

NOT FOR FABRICATION

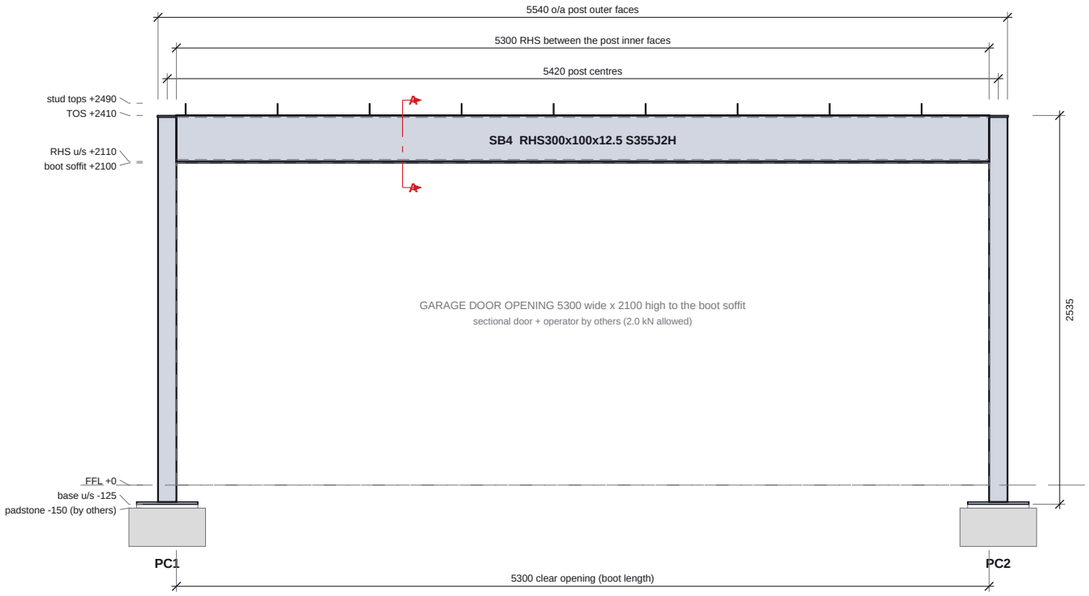
FOR ENGINEERS REVIEW

until the engineer

has reviewed + signed off

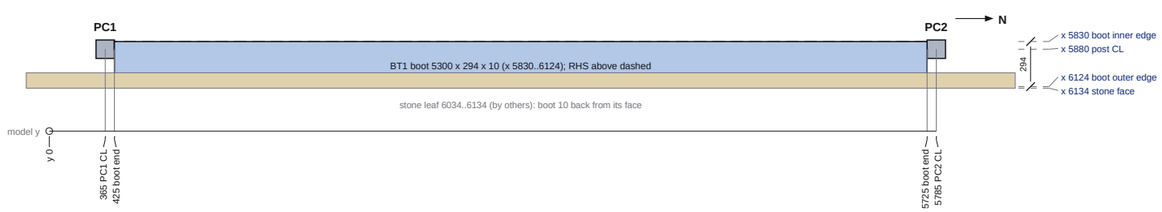
1 GP1 ELEVATION - LOOKING WEST FROM OUTSIDE

Scale 1:40 | levels above FFL; south (PC1) left, north (PC2) right



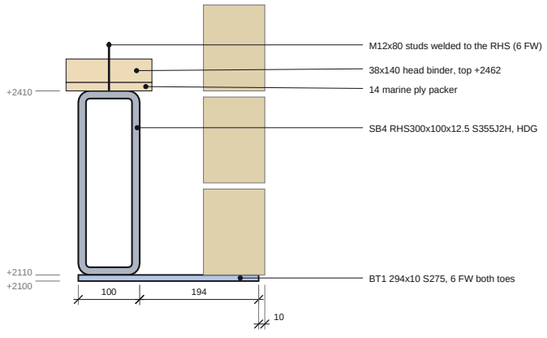
2 GP1 PLAN AT BOOT LEVEL

Scale 1:40 | outside (east) down; north right



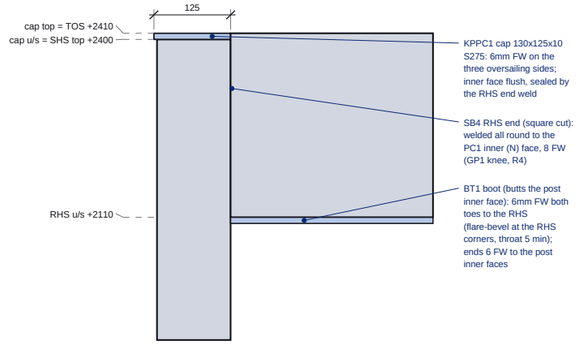
3 SECTION A-A

Scale 1:10 | looking north; inside left



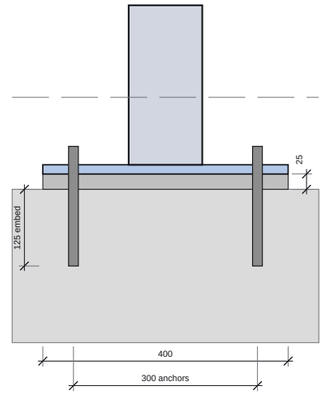
4 KNEE (PC1; PC2 HANDED)

Scale 1:10 | looking west from outside; north right



5 BASE (PC1 / PC2)

Scale 1:10 | elevation



BPPC1/BPPC2 400x200x15 S275 (200 across x 400 along the frame), 8mm FW all round
4 no. M16 8.8 HDG rod, R-KEM II, 125 embed, in Ø18 holes at x ±50 / y ±150 of the post CL (view along the frame: the two rows are in line)
25 HDG shims + non-shrink grout; padstone top -150 by others

GP1 PARTS

MARK	ITEM	LENGTH	KG
SB4	RHS300x100x12.5	5300	383.2
BT1	PL 5300x294x10 S275		122.22
PC1	SHS120x120x10	2510	84.6
BPPC1	PL 400x200x15 S275		9.30
KPPC1	PL 130x125x10 S275		1.28
PC2	SHS120x120x10	2510	84.6
BPPC2	PL 400x200x15 S275		9.30
KPPC2	PL 130x125x10 S275		1.28
GP1	**welded assembly, HDG as one piece**		**695.8**

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. GP1 R4 - PRELIMINARY, FOR ENGINEER REVIEW (TLS proposal, not the engineer's design). Replaces S-0010's UB203x133x30: carries the 5300 door opening AND ~0.5 m of stone (main-house stone eaves).
4. LOADS (char.): trusses S-0012 DL 15 / IL 10 kN each end -> 5.4 / 3.6 kN/m; stone 0.5x0.1x23 = 1.15 kN/m; boot + timber + self weight 1.0 kN/m; door rails + operator 2.0 kN mid-span; wind 12 kN SLS / 18 kN ULS at the eaves.
5. RHS 300x100x12.5: I 9103 cm4, Wpl 806 cm3. ULS M 61 kNm vs Mpl 286 kNm. Deflection (SLS incl. door, pinned, L 5420) 6.9 mm = L/787 < 8 mm (client limit).
6. SHS 120x120x10 to TOS: N 44 kN + knee M 29.7 kNm vs Mpl 62 kNm; sway 6.8 mm = h/326.
7. KNEE (R4): RHS ends welded all round to the post inner faces, 8 FW: couple 29.7 / 0.288 = 103 kN per RHS wall vs 136 kN; post face (b1/b0 0.83) ~390 kN; torsion from the boot 0.84 kNm per knee.
8. BASE: 400x200x15, 54 clear of the stone leaf; N 44 kN, H 9 kN per base (ULS); pinned base assumed for sway. Boot toe ~0.22 kNm/m -> 13 MPa in the 10 plate.
9. FOR THE ENGINEER: loads, wind share, knee welds + post face (stiffeners in the SHS?), base fixity, padstones on the garage return foundations (min 500 x 300), cavity tray / thermal bridge at the boot, 30 min fire protection (S-0011 9.02).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

GARAGE PORTAL GP1 ASSEMBLY

Welded goal-post frame SB4 + PC1 + PC2 with boot BT1 (TLS proposal)

SCALE	SIZE	DATE	DRAWN
1:40 / 1:10	A3	11.10.26	TS

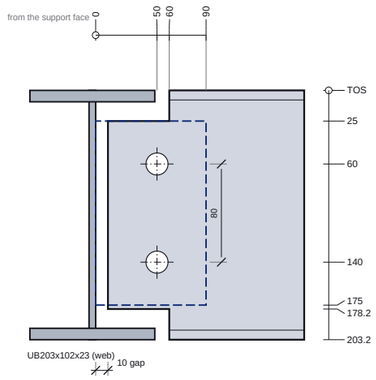
STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-GP1	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

DETAIL A BEAM TO BEAM

Scale 1:5

SB2 end A on SB3

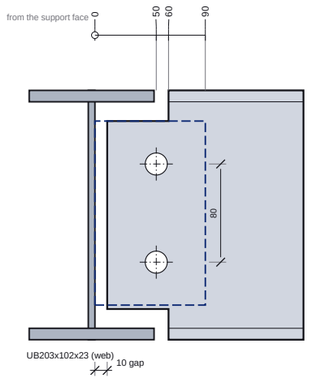


150x90x8 S275 fin plate welded to SB3, 8 FW both sides
2 no. M16 8.8, Ø18; bolt line 50 from the support face, 40 to the plate end
pitch 35/80/35; plate top 25 below TOS; end gap 10
cope 50 long x 25 deep, top + bottom
ref Skyhooks 26-431-S-0010 Detail A

DETAIL B BEAM TO BEAM

Scale 1:5

SB3 end A on SB1

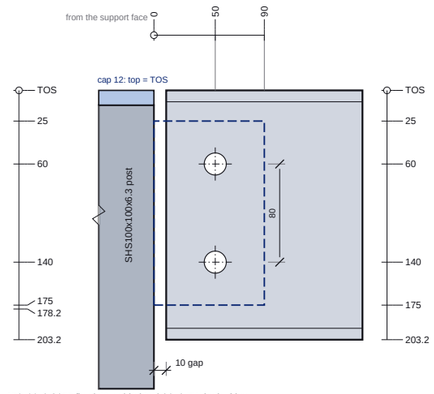


150x90x8 S275 fin plate welded to SB1, 8 FW both sides
2 no. M16 8.8, Ø18; bolt line 50 from the support face, 40 to the plate end
pitch 35/80/35; plate top 25 below TOS; end gap 10
cope 50 long x 25 deep, top + bottom
ref Skyhooks 26-431-S-0010 Detail B

DETAIL C BEAM TO SHS

Scale 1:5

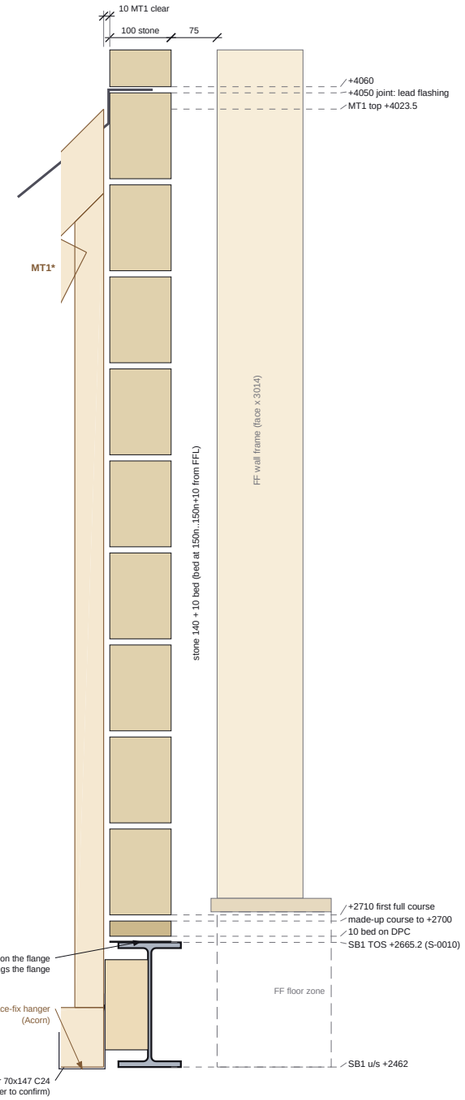
SB1 end A on SC2; SB1 end B on SC1; SB3 end B on SC3



150x90x8 S275 fin plate welded to SC2, 8 FW both sides
2 no. M16 8.8, Ø18; bolt line 50 from the support face, 40 to the plate end
pitch 35/80/35; plate top 25 below TOS; end gap 10
ref Skyhooks 26-431-S-0010 Detail C
post stops under its cap plate (CP1/CP2/CP3); cap top = TOS
S-0010 leader reads 'welded to beam': TLS reads it as welded to the SHS face (RFI)

COURSING SECTION - SB1 AT THE LEAN-TO

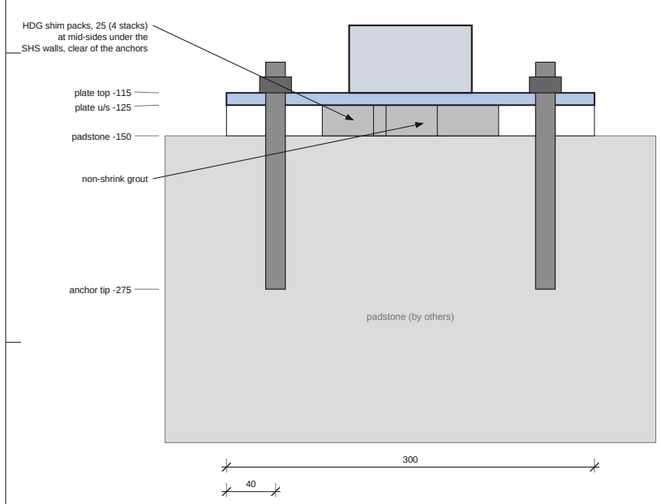
Scale 1:10 | looking north



* MT1 (Acorn) ends at x 2829, 10 clear of the stone face; hangs on the SB1 70 packer face

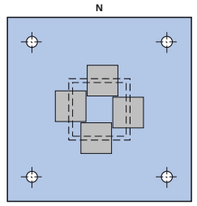
BASEPLATE DETAIL (DETAIL 2)

Scale 1:5 | SC1, SC2, SC3



BP1/BP2/BP3: PL 300x300x10 S275, 8mm FW all round
4 no. M16 8.8 HDG threaded rod x 200, Rawlplug R-KEM II resin, Ø18 x 125 embedment,
1 nut + 1 washer above the plate; Ø18 fixture holes 40 from the plate edges.
Base underside -125: stops 25 above the padstone (top -150) - HDG shim packs
50x50, 4 stacks (consumables), then non-shrink cementitious grout to the 25 gap.
Shim stacks at mid-sides under the SHS walls, clear of the anchors (plan below).
Padstone min 400x400x250 C25/30, anchor edge distance >= 80 (ETA): by others,
engineer to confirm. Seal the SHS at the DPM.

SHIM LAYOUT - PLAN 1:10



grey = shim stacks (4 no.) under the SHS walls (dashed)
anchors 40 from the plate edges
grout the 25 gap after levelling

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. Details redrawn to TLS dimensions from Skyhooks 26-431-S-0010 P01 (Details A / B / C, Baseplate Detail 2). The engineer's drawing governs; TLS additions flagged.
3. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
4. Fin plates S275, welded to the SUPPORTING member, 8 mm double fillet weld. Bolts M16 8.8 + nut + 2 washers, HDG, in Ø18 holes.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

STEEL CONNECTION AND COURSING DETAILS

Details A / B / C, baseplate, coursing section through SB1 at the lean-to abutment

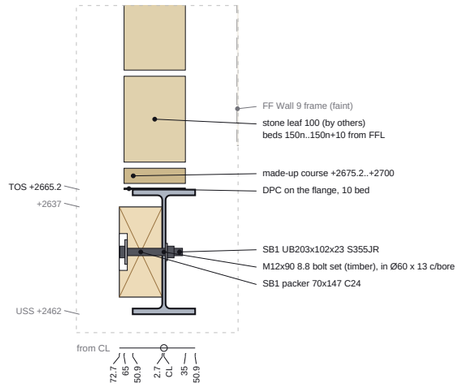
SCALE	SIZE	DATE	DRAWN
1:5 / 1:10	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-DET	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

SECTION SB1-SB1 1:10

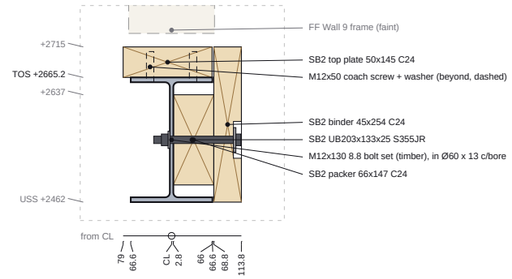
at 1500 from end A, looking from end A to end B: W left, E right



Levels above FFL (model 0). Section sizes as named; dims from the steel CL.

SECTION SB2-SB2 1:10

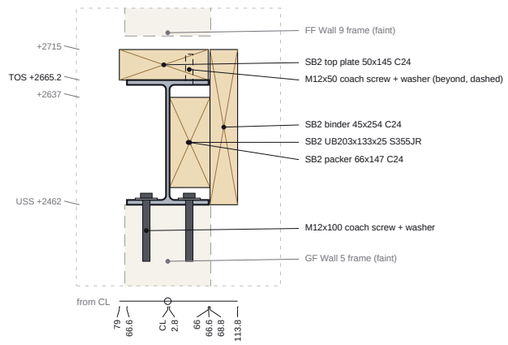
at 1500 from end A, looking from end A to end B: W left, E right



Levels above FFL (model 0). Section sizes as named; dims from the steel CL.

SB2 BEARING, END B 1:10

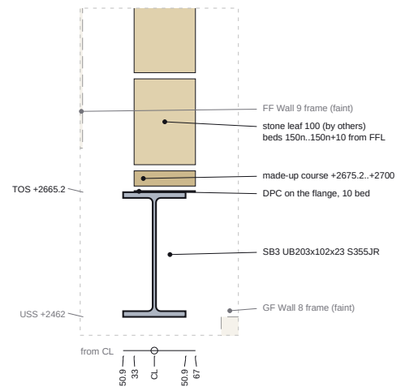
at 2880 from end A, looking from end A to end B: W left, E right



Levels above FFL (model 0). Section sizes as named; dims from the steel CL.

SECTION SB3-SB3 1:10

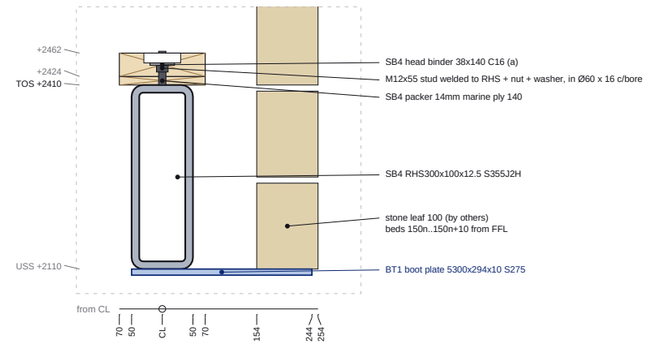
at 1063.2 from end A, looking from end A to end B: N left, S right



Levels above FFL (model 0). Section sizes as named; dims from the steel CL.

SECTION SB4-SB4 1:10

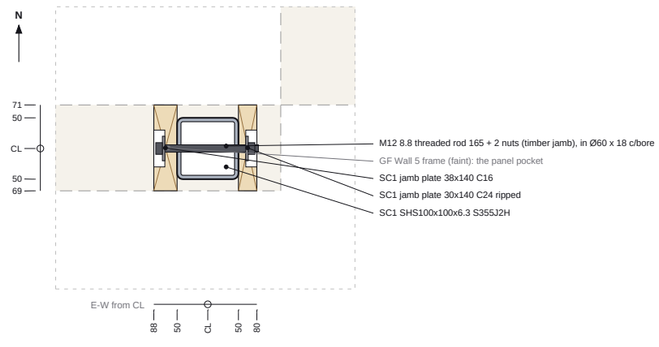
at 2460 from end A, looking from end A to end B: W left, E right



Levels above FFL (model 0). Section sizes as named; dims from the steel CL.

PLAN SECTION SC1 1:10

plan at +1231 (through-bolt height), north up



NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. Cross-sections through each member, drawn from the steel spec: steel true shape, welded plates, the TLS timber kit and its fixings in place. Faint = adjacent timber frame (construction data).
3. Stone coursing from FFL: bed joints 150n..150n+10, 140 stone units. DPC on the flange under the stone; first course made up to the coursing.
4. Counterbores as the fixing schedule (head / nut + washer below the timber face).
5. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

STEEL MEMBER SECTIONS

Sections through each member with the timber kit, fixings, stone and levels

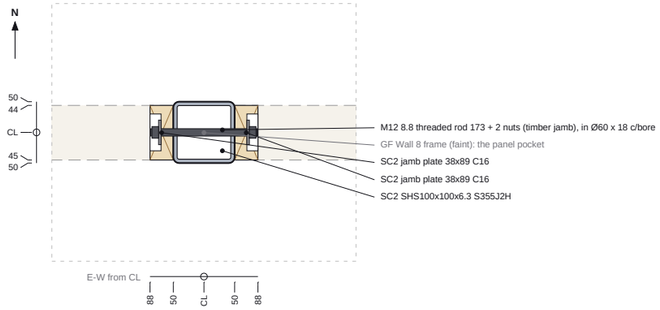
SCALE	SIZE	DATE	DRAWN
1:10 / as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SEC1	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

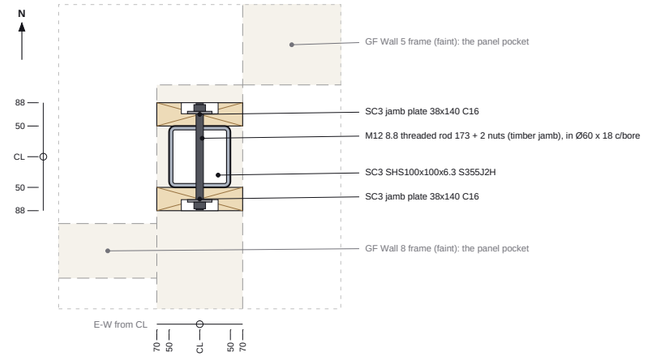
PLAN SECTION SC2 1:10

plan at +1231 (through-bolt height), north up



PLAN SECTION SC3 1:10

plan at +1231 (through-bolt height), north up



PLAN SECTION PC1 1:10

plan at +1150 (through-bolt height), north up

PLAN SECTION PC2 1:10

plan at +1150 (through-bolt height), north up

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. Cross-sections through each member, drawn from the steel spec: steel true shape, welded plates, the TLS timber kit and its fixings in place. Faint = adjacent timber frame (construction data).
3. Stone coursing from FFL: bed joints 150n..150n+10, 140 stone units. DPC on the flange under the stone; first course made up to the coursing.
4. Counterbores as the fixing schedule (head / nut + washer below the timber face).
5. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

STEEL MEMBER SECTIONS

Sections through each member with the timber kit, fixings, stone and levels

SCALE	SIZE	DATE	DRAWN
1:10 / as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SEC2	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

BOLT AND FIXING SCHEDULE

KEY	DESCRIPTION	QTY	UNIT	WHERE	FLAG
M16x50 8.8 bolt set	M16x50 grade 8.8 bolt + nut + 2 washers, hot-dip galvanised	10	sets	A, B, C	
M12x90 8.8 bolt set (timber)	M12x90 grade 8.8 bolt + nut + 40x40x4 timber washer + steel washer, HDG (BS EN ISO 10684); head + washer counterbored Ø60 x 13 into the packer face	8	sets	SB1	TLS
M12x130 8.8 bolt set (timber)	M12x130 grade 8.8 bolt + nut + 40x40x4 timber washer + steel washer, HDG (BS EN ISO 10684); head + washer counterbored Ø60 x 13 into the binder face	9	sets	SB2	TLS
M12x50 coach screw + washer	M12x50 coach screw + 36 washer, zinc; pre-drill Ø12 shank / Ø8 thread	10	no.	SB2	TLS
M12x100 coach screw + washer	M12x100 coach screw + 36 washer, zinc; pre-drill Ø12 shank / Ø8 thread	2	no.	SB2	TLS
M12 8.8 threaded rod 165 + 2 nuts (timber jamb)	M12 grade 8.8 threaded rod cut to 165, 2 nuts + 2 no. 40x40x4 timber washers, HDG (BS EN ISO 10684); both ends in Ø60 x 18 counterbores in the jamb plates (nuts below the plate faces); snug tight	3	sets	SC1	TLS
M12 8.8 threaded rod 173 + 2 nuts (timber jamb)	M12 grade 8.8 threaded rod cut to 173, 2 nuts + 2 no. 40x40x4 timber washers, HDG (BS EN ISO 10684); both ends in Ø60 x 18 counterbores in the jamb plates (nuts below the plate faces); snug tight	6	sets	SC2, SC3	TLS
M12x160 8.8 bolt set (timber)	M12x160 grade 8.8 bolt + nut + 40x40x4 timber washer + steel washer, HDG (BS EN ISO 10684); nut in a Ø60 x 18 counterbore in the jamb plate; head on the door-side face (tell the door supplier); snug tight	6	sets	PC1, PC2	TLS
M12x55 stud welded to RHS + nut + washer	M12x55 grade 4.8 drawn-arc stud (BS EN ISO 13918), welded to the RHS top before galvanising, threads chased; HDG nut (BS EN ISO 10684) + 40x40x4 timber washer in a Ø60 x 16 counterbore in the head binder (nut below the binder top; studs midway between the GT1 trusses)	9	no.	SB4	TLS
M16x200 R-KEM II anchor set	M16x200 grade 8.8 threaded rod, HDG, + nut + washer, Rawlplug R-KEM II resin, Ø18 x 125 hole	20	sets	PC1, PC2, SC1, SC2, SC3	

FLAG TLS = TLS detail - engineer to confirm.

TIMBER KIT BOLTED TO THE STEEL (TLS SUPPLY)

ITEM	SECTION	GRADE	LENGTH	ON	PURPOSE	FLAG
SB1 packer 70x147 C24	70x147	C24	2559	SB1	west face packer (ripped 70 from 75), 21.8 proud of the flange tips, for Acorn's MT1 truss hangers; stops 85 clear of the Detail C bolts each end; Ø14 holes at 215 + 300n from its S end, 73.5 below its top (mid-height), counterbored Ø60 x 13 on the west face	TLS
SB2 packer 66x147 C24	66x147	C24	2866	SB2	east face packer (ripped 66 from 75), 2.25 proud of the flange tips; Ø14 holes at 215.7 + 300n from its S end, 73.5 below its top (mid-height)	TLS
SB2 binder 45x254 C24	45x254	C24	2709	SB2	joist binder: the 5 family-room joists hang off its face (Acorn hangers); Ø14 holes at 53.7 + 300n from its S end, 151.5 below its top, counterbored Ø60 x 13 on the east face	TLS
SB2 top plate 50x145 C24	50x145	C24	2905	SB2	sawn 50 ripped to 145: deck edge + FF-EX1 bearing; starts at the end of the top cope (y 9946), 30 clear of the south stone leaf	TLS
SC1 jamb plate 38x140 C16	38x140	C16	2462	SC1	W face jamb plate: the panel jamb stud nails to it	TLS
SC1 jamb plate 30x140 C24 ripped	30x140	C24	2462	SC1	E face jamb plate: the panel jamb stud nails to it	TLS
SC2 jamb plate 38x89 C16	38x89	C16	2462	SC2	W face jamb plate: the panel jamb stud nails to it	TLS
SC2 jamb plate 38x89 C16	38x89	C16	2462	SC2	E face jamb plate: the panel jamb stud nails to it	TLS
SC3 jamb plate 38x140 C16	38x140	C16	2462	SC3	S face jamb plate: the panel jamb stud nails to it	TLS
SC3 jamb plate 38x140 C16	38x140	C16	2462	SC3	N face jamb plate: the panel jamb stud nails to it	TLS
PC1 jamb plate 38x140 C16	38x140	C16	2400	PC1	jamb plate on the face against the return panel (to the underside of the post cap)	TLS
PC2 jamb plate 38x140 C16	38x140	C16	2400	PC2	jamb plate on the face against the return panel (to the underside of the post cap)	TLS
SB4 packer 14mm marine ply 140	14x140	marine ply	5540	SB4	packer under the head binder, over the RHS and the post caps (3 lengths from 2440 sheets, joints over the RHS)	TLS
SB4 head binder 38x140 C16 (a)	38x140	C16	2813	SB4	head binder over the steel (GT1 bearing)	TLS
SB4 head binder 38x140 C16 (b)	38x140	C16	2813	SB4	head binder over the steel (GT1 bearing)	TLS

CONSUMABLES (NOT MODELLED)

KEY	DESCRIPTION	QTY	UNIT	WHERE	FLAG
Shim set 25mm	HDG steel shims 50x50: per post 4 stacks of 10+5+5+3+1+1 (= 25, 1 mm to adjust), one at each mid-side under the SHS wall	5	post sets	SC1-3, PC1-2 bases (mid-sides under the SHS walls)	
R-KEM II 300ml	Rawlplug R-KEM II 300ml cartridge + mixer nozzles (20 holes Ø18x125 ~ 0.17 L + nozzle waste per base)	2	no.	post bases	
Grout 25kg	Non-shrink cementitious grout 25 kg (5 bases x 25 gap ~ 12 L)	2	bags	post bases	
Vent plugs Ø20	HDG drive-in plugs / caps for Ø20 galvanising vents	20	no.	posts + GP1	TLS
Vent plugs Ø12	HDG drive-in plugs for Ø12 boot lap vents	11	no.	BT1	TLS
DPC 150	DPC 150 wide on the SB1 / SB3 top flanges under the stone (2729 + 2126 + laps)	6	m	SB1, SB3	

STEEL WEIGHT SUMMARY

MARK	SECTION	GRADE	CUT LENGTH	QTY	SECTION KG	PLATES KG	TOTAL KG	COATING
SB1	UB203x102x23	S355JR	2729	1	63.0	0.8	63.9	HDG (EN ISO 1461)
SB2	UB203x133x25	S355JR	2955	1	74.2	0.0	74.2	Shop primer (C2)
SB3	UB203x102x23	S355JR	2126.3	1	49.1	0.8	49.9	HDG (EN ISO 1461)
SC1	SHS100x100x6.3	S355J2H	2768.2	1	50.4	10.1	60.4	HDG (EN ISO 1461)
SC2	SHS100x100x6.3	S355J2H	2768.2	1	50.4	8.9	59.3	HDG (EN ISO 1461)
SC3	SHS100x100x6.3	S355J2H	2768.2	1	50.4	10.0	60.4	HDG (EN ISO 1461)
SB4	RHS300x100x12.5	S355J2H	5300	1	383.2	122.2	505.4	HDG (EN ISO 1461)
PC1	SHS120x120x10	S355J2H	2510	1	84.6	10.6	95.2	HDG (EN ISO 1461)
PC2	SHS120x120x10	S355J2H	2510	1	84.6	10.6	95.2	HDG (EN ISO 1461)
TOTAL					**889.9**	**174.0**	**1063.9**	

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. Bolt sets counted from the model (one token per set). Timber kit: TLS supply, TLS detail - engineer to confirm.
3. Weights: sections at nominal kg/m, plates at 7850 kg/m3 less holes.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

BOLT FIXING AND TIMBER SCHEDULE

Fixings, timber kit, steel weights

SCALE	SIZE	DATE	DRAWN
n/a	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SCH	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

CUTTING LIST

mark	item	on	section	grade	length	qty	cut from (stock x length)	machining	holes	sheet
P4-T1	SB1 packer	SB1	70x147	C24	2559	1	75x147 x 2700	ripped 70 from 75	8	STT-01
P4-T2	SB2 packer	SB2	66x147	C24	2866	1	75x147 x 3000	ripped 66 from 75	9	STT-01
P4-T3	SB2 binder	SB2	45x254	C24	2709	1	47x295 x 3000	ripped 45 from 47, ripped 254 from 295 (next stock size - joiner to confirm)	9	STT-01
P4-T4	SB2 top plate	SB2	50x145	C24	2905	1	50x145 sawn x 3000	ripped to 145	10	STT-02
P4-T5	SC1 jamb plate	SC1	38x140	C16	2462	1	38x140 x 2700	as stock - no ripping	3	STT-02
P4-T6	SC1 jamb plate	SC1	30x140	C24	2462	1	38x140 x 2700	ripped 30 from 38 (next stock size - joiner to confirm)	3	STT-02
P4-T7	SC2 jamb plate	SC2	38x89	C16	2462	2	38x89 x 2700	as stock - no ripping	3	STT-03
P4-T8	SC3 jamb plate	SC3	38x140	C16	2462	2	38x140 x 2700	as stock - no ripping	3	STT-03
P4-T9	SB4 packer	SB4	14x140	marine ply	5540	1	2440 x 1220 sheet, 14 thick	rip 140 strips	9	STT-03
P4-T10	SB4 head binder (a)	SB4	38x140	C16	2813	1	38x140 x 3000	as stock - no ripping	5	STT-04
P4-T11	SB4 head binder (b)	SB4	38x140	C16	2813	1	38x140 x 3000	as stock - no ripping	4	STT-04
P4-T12	PC1 / PC2 jamb plate	PC1, PC2	38x140	C16	2400	2	38x140 x 2700	as stock - no ripping	3	STT-04

Lengths are the finished piece lengths from the model; stock lengths allow 10 for squaring the ends. Machining from the spec where it states it, otherwise the next stock size up (joiner to confirm). Treatment: to the project specification (by others) - treat cut ends and holes.

FIXINGS THROUGH EACH PIECE (PER PIECE X PIECES)

mark	fixing	no.	hole	counterbore
P4-T1	M12x90 8.8 bolt set (timber)	8 x 1	14 through	60 x 13 on the W face
P4-T2	M12x130 8.8 bolt set (timber)	9 x 1	14 through	-
P4-T3	M12x130 8.8 bolt set (timber)	9 x 1	14 through	60 x 13 on the E face
P4-T4	M12x50 coach screw + washer	10 x 1	pilot 8 x 42.2 from underside	-
P4-T5	M12 8.8 threaded rod 165 + 2 nuts (timber jamb)	3 x 1	14 through	60 x 18 on the W face
P4-T6	M12 8.8 threaded rod 165 + 2 nuts (timber jamb)	3 x 1	14 through	60 x 18 on the E face
P4-T7	M12 8.8 threaded rod 173 + 2 nuts (timber jamb)	3 x 2	14 through	60 x 18 on the face away from the steel (E / W)
P4-T8	M12 8.8 threaded rod 173 + 2 nuts (timber jamb)	3 x 2	14 through	60 x 18 on the face away from the steel (N / S)
P4-T9	M12x55 stud welded to RHS + nut + washer	9 x 1	14 through	-
P4-T10	M12x55 stud welded to RHS + nut + washer	5 x 1	14 through	60 x 16 on the top
P4-T11	M12x55 stud welded to RHS + nut + washer	4 x 1	14 through	60 x 16 on the top
P4-T12	M12x160 8.8 bolt set (timber)	3 x 2	14 through	60 x 18 on the face away from the steel (N / S)

EDGE / END DISTANCE AND DATA CHECKS (EN 1995-1-1 8.5: end 7d, edge 4d)

FLAG: SB2 binder 45x254 C24: end 53.7 < 7d 84 - 1 hole at 53.7 from the S end
FLAG: SB2 top plate 50x145 C24: edge 44 < 4d 48 - 5 holes at 99.7, 699.7, 1299.7, 1899.7, 2499.7 from the S end
FLAG: SB2 top plate 50x145 C24: edge 31 < 4d 48 - 5 holes at 399.7, 999.7, 1599.7, 2199.7, 2799.7 from the S end
FLAG: SC2 jamb plate 38x89 C16: edge 44 < 4d 48 - 3 holes at 300, 1231, 2162 from the bottom end

SHEET INDEX

sheet	title
P4-STT-00	Steel timber packing schedule
P4-STT-01	SB1 packer, SB2 packer, SB2 binder
P4-STT-02	SB2 top plate, SC1 jamb plate
P4-STT-03	SC2 jamb plate, SC3 jamb plate, SB4 packer
P4-STT-04	T10 to T12

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL TIMBER SUBJECT TO SITE MEASURE AGAINST THE ERECTED STEEL BEFORE DRILLING.
3. Timber: strength-graded to BS EN 14081, C16 / C24 as scheduled, regularised; preservative treatment to the project specification (by others). Treat all cut ends and drilled holes.
4. Holes: bolts, rods and studs drilled d + 2 (M12 = 14), square to the face. Counterbores flat-bottomed (Forstner), concentric with the hole. Coach screws pre-drilled per EN 1995-1-1 8.7.
5. Running dimensions from the named reference end; across dimensions from the named edge. Dimensions in mm.
6. Bolt spacings checked against EN 1995-1-1 8.5 as set by the client: end 7d, edge 4d (M12: 84 / 48). Shortfalls are listed in red on STT-00.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel timber packing drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE
STEEL TIMBER PACKING SCHEDULE

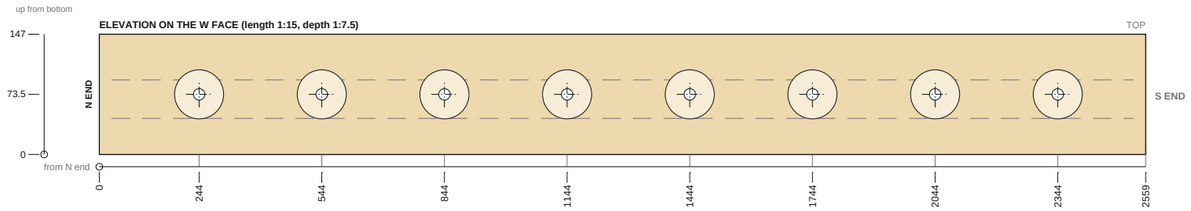
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NTS	A3	11.10.26	TS

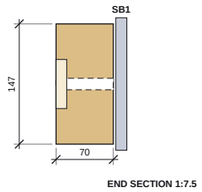
STATUS	DRAWING No.	REV
FOR APPROVAL	P4-STT-00	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

P4-T1 SB1 PACKER 70x147 C24 x L 2559 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM



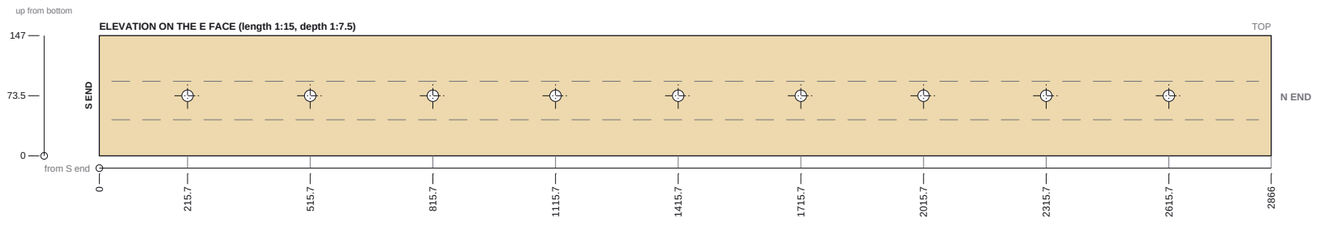


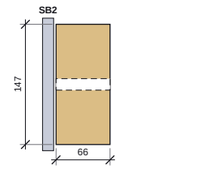
END SECTION 1:7.5

PIECE	P4-T1 - SB1 packer 70x147 C24
FIXES TO	SB1 - its E face against the steel
CUT FROM	75x147 x 2700; ripped 70 from 75
HOLES	8 No. - 14 through
COUNTERBORE	60 x 13 deep on the W face
FIXINGS	8 x M12x90 8.8 bolt set (timber)
PURPOSE	west face packer (ripped 70 from 75), 21.8 proud of the flange tips, for Acorn's MT1 truss hangers; stops 85 clear of the Detail C bolts each end; Ø14 holes at 215 + 300n from its S end, 73.5 below its top (mid-height), counterbored Ø60 x 13 on the west face

P4-T2 SB2 PACKER 66x147 C24 x L 2866 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM



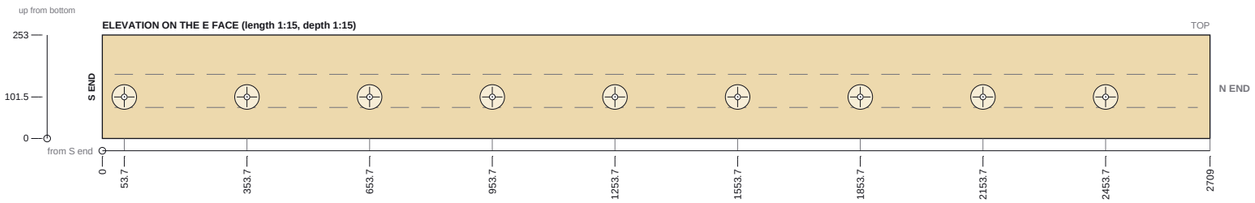


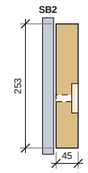
END SECTION 1:7.5

PIECE	P4-T2 - SB2 packer 66x147 C24
FIXES TO	SB2 - its W face against the steel
CUT FROM	75x147 x 3000; ripped 66 from 75
HOLES	9 No. - 14 through
COUNTERBORE	none
FIXINGS	9 x M12x130 8.8 bolt set (timber)
PURPOSE	east face packer (ripped 66 from 75), 2.25 proud of the flange tips; Ø14 holes at 215.7 + 300n from its S end, 73.5 below its top (mid-height)

P4-T3 SB2 BINDER 45x254 C24 x L 2709 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM





END SECTION 1:12.5

PIECE	P4-T3 - SB2 binder 45x254 C24
FIXES TO	SB2 - its W face against the steel
CUT FROM	47x295 x 3000; ripped 45 from 47, ripped 254 from 295 (next stock size - joiner to confirm)
HOLES	9 No. - 14 through
COUNTERBORE	60 x 13 deep on the E face
FIXINGS	9 x M12x130 8.8 bolt set (timber)
PURPOSE	joist binder: the 5 family-room joists hang off its face (Acorn hangers); Ø14 holes at 53.7 + 300n from its S end, 151.5 below its top, counterbored Ø60 x 13 on the east face

P4-T1 HOLES

#	from N	up bottom	dn top	hole	c'bore	fixing
1	244	73.5	73.5	14 thru	60x13 W	M12 bolt
2	544	73.5	73.5	14 thru	60x13 W	M12 bolt
3	844	73.5	73.5	14 thru	60x13 W	M12 bolt
4	1144	73.5	73.5	14 thru	60x13 W	M12 bolt
5	1444	73.5	73.5	14 thru	60x13 W	M12 bolt
6	1744	73.5	73.5	14 thru	60x13 W	M12 bolt
7	2044	73.5	73.5	14 thru	60x13 W	M12 bolt
8	2344	73.5	73.5	14 thru	60x13 W	M12 bolt

P4-T2 HOLES

#	from S	up bottom	dn top	hole	c'bore	fixing
1	215.7	73.5	73.5	14 thru	-	M12 bolt
2	515.7	73.5	73.5	14 thru	-	M12 bolt
3	815.7	73.5	73.5	14 thru	-	M12 bolt
4	1115.7	73.5	73.5	14 thru	-	M12 bolt
5	1415.7	73.5	73.5	14 thru	-	M12 bolt
6	1715.7	73.5	73.5	14 thru	-	M12 bolt
7	2015.7	73.5	73.5	14 thru	-	M12 bolt
8	2315.7	73.5	73.5	14 thru	-	M12 bolt
9	2615.7	73.5	73.5	14 thru	-	M12 bolt

P4-T3 HOLES

#	from S	up bottom	dn top	hole	c'bore	fixing
1	53.7	101.5	151.5	14 thru	60x13 E	M12 bolt
2	353.7	101.5	151.5	14 thru	60x13 E	M12 bolt
3	653.7	101.5	151.5	14 thru	60x13 E	M12 bolt
4	953.7	101.5	151.5	14 thru	60x13 E	M12 bolt
5	1253.7	101.5	151.5	14 thru	60x13 E	M12 bolt
6	1553.7	101.5	151.5	14 thru	60x13 E	M12 bolt
7	1853.7	101.5	151.5	14 thru	60x13 E	M12 bolt
8	2153.7	101.5	151.5	14 thru	60x13 E	M12 bolt
9	2453.7	101.5	151.5	14 thru	60x13 E	M12 bolt

NOTES

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2. ALL TIMBER SUBJECT TO SITE MEASURE AGAINST THE ERECTED STEEL BEFORE DRILLING.
3. Timber: strength-graded to BS EN 14081, C16 / C24 as scheduled, regularised; preservative treatment to the project specification (by others). Treat all cut ends and drilled holes.
4. Holes: bolts, rods and studs drilled d + 2 (M12 = 14), square to the face. Counterbores flat-bottomed (Forstner), concentric with the hole. Coach screws pre-drilled per EN 1995-1-1 8.7.
5. Running dimensions from the named reference end; across dimensions from the named edge. Dimensions in mm.
6. Bolt spacings checked against EN 1995-1-1 8.5 as set by the client: end 7d, edge 4d (M12: 84 / 48). Shortfalls are listed in red on STT-00.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel timber packing drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

TIMBER PACKING - SB1 PACKER, SB2 PACKER, SB2 BINDER

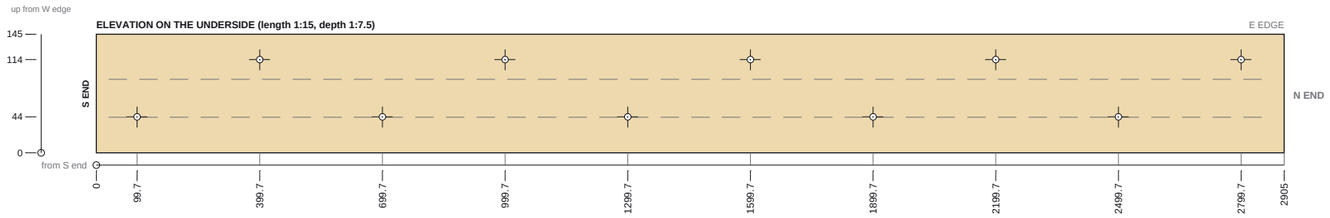
SCALE	SIZE	DATE	DRAWN
AS SHOWN	A3	11.10.26	TS

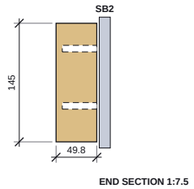
STATUS	DRAWING No.	REV
FOR APPROVAL	P4-STT-01	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

P4-T4 SB2 TOP PLATE 50x145 C24 x L 2905 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM





END SECTION 1:7.5

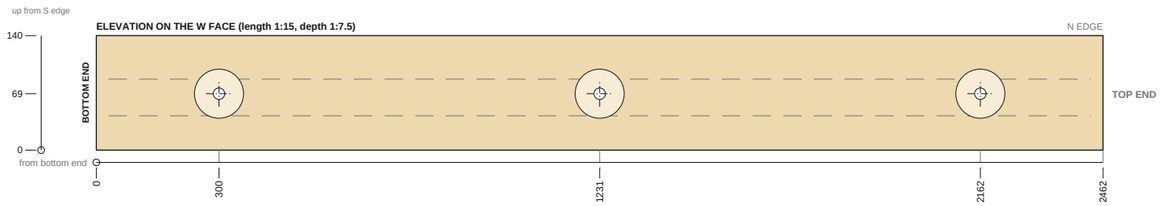
PIECE	P4-T4 - SB2 top plate 50x145 C24
FIXES TO	SB2 - its underside against the steel
CUT FROM	50x145 sawn x 3000; ripped to 145
HOLES	10 No. - pilot 8 x 42.2 deep from the underside
COUNTERBORE	none
FIXINGS	10 x M12x50 coach screw + washer
PURPOSE	sawn 50 ripped to 145: deck edge + FF-EX1 bearing; starts at the end of the top cope (y 9946), 30 clear of the south stone leaf

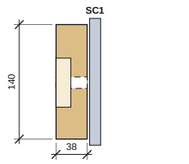
P4-T4 HOLES

#	from S	up W	dn E	hole	c'bore	fixing
1	99.7	44	101	8 x 42.2 pilot	-	M12 coach screw
2	399.7	114	31	8 x 42.2 pilot	-	M12 coach screw
3	699.7	44	101	8 x 42.2 pilot	-	M12 coach screw
4	999.7	114	31	8 x 42.2 pilot	-	M12 coach screw
5	1299.7	44	101	8 x 42.2 pilot	-	M12 coach screw
6	1599.7	114	31	8 x 42.2 pilot	-	M12 coach screw
7	1899.7	44	101	8 x 42.2 pilot	-	M12 coach screw
8	2199.7	114	31	8 x 42.2 pilot	-	M12 coach screw
9	2499.7	44	101	8 x 42.2 pilot	-	M12 coach screw
10	2799.7	114	31	8 x 42.2 pilot	-	M12 coach screw

P4-T5 SC1 JAMB PLATE 38x140 C16 x L 2462 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM





END SECTION 1:7.5

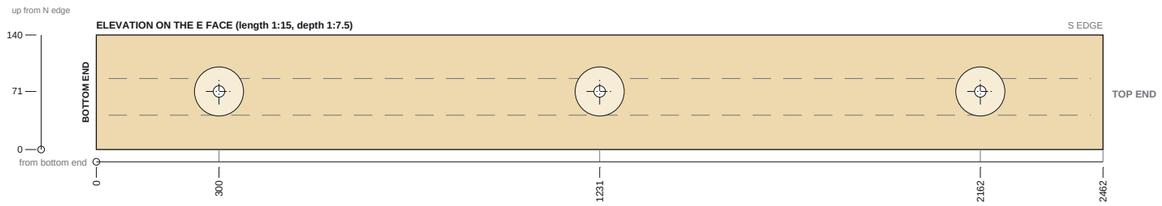
PIECE	P4-T5 - SC1 jamb plate 38x140 C16
FIXES TO	SC1 - its E face against the steel
CUT FROM	38x140 x 2700; as stock - no ripping
HOLES	3 No. - 14 through
COUNTERBORE	60 x 18 deep on the W face
FIXINGS	3 x M12 8.8 threaded rod 165 + 2 nuts (timber jamb)
PURPOSE	W face jamb plate: the panel jamb stud nails to it

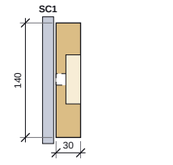
P4-T5 HOLES

#	from bottom	up S	dn N	hole	c'bore	fixing
1	300	69	71	14 thru	60x18 W	M12 rod
2	1231	69	71	14 thru	60x18 W	M12 rod
3	2162	69	71	14 thru	60x18 W	M12 rod

P4-T6 SC1 JAMB PLATE 30x140 C24 x L 2462 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM





END SECTION 1:7.5

PIECE	P4-T6 - SC1 jamb plate 30x140 C24 ripped
FIXES TO	SC1 - its W face against the steel
CUT FROM	38x140 x 2700; ripped 30 from 38 (next stock size - joiner to confirm)
HOLES	3 No. - 14 through
COUNTERBORE	60 x 18 deep on the E face
FIXINGS	3 x M12 8.8 threaded rod 165 + 2 nuts (timber jamb)
PURPOSE	E face jamb plate: the panel jamb stud nails to it

P4-T6 HOLES

#	from bottom	up N	dn S	hole	c'bore	fixing
1	300	71	69	14 thru	60x18 E	M12 rod
2	1231	71	69	14 thru	60x18 E	M12 rod
3	2162	71	69	14 thru	60x18 E	M12 rod

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL TIMBER SUBJECT TO SITE MEASURE AGAINST THE ERECTED STEEL BEFORE DRILLING.
3. Timber: strength-graded to BS EN 14081, C16 / C24 as scheduled, regularised; preservative treatment to the project specification (by others). Treat all cut ends and drilled holes.
4. Holes: bolts, rods and studs drilled d + 2 (M12 = 14), square to the face. Counterbores flat-bottomed (Forstner), concentric with the hole. Coach screws pre-drilled per EN 1995-1-1 8.7.
5. Running dimensions from the named reference end; across dimensions from the named edge. Dimensions in mm.
6. Bolt spacings checked against EN 1995-1-1 8.5 as set by the client: end 7d, edge 4d (M12: 84 / 48). Shortfalls are listed in red on STT-00.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel timber packing drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

TIMBER PACKING - SB2 TOP PLATE, SC1 JAMB PLATE

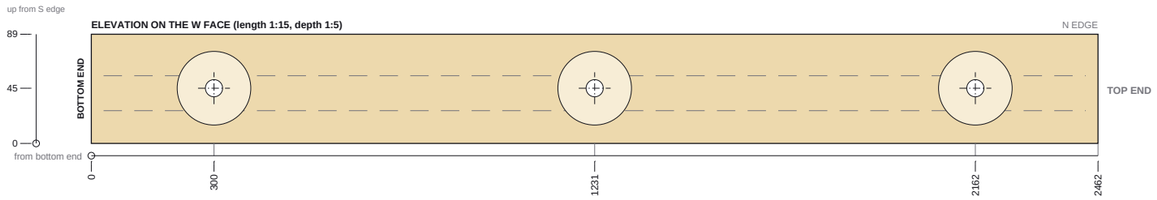
SCALE	SIZE	DATE	DRAWN
AS SHOWN	A3	11.10.26	TS

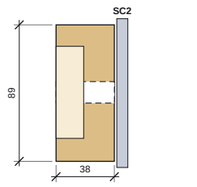
STATUS	DRAWING No.	REV
FOR APPROVAL	P4-STT-02	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

P4-T7 SC2 JAMB PLATE — 38x89 C16 x L 2462 - 2 No.

TLS DETAIL - ENGINEER TO CONFIRM





PIECE	P4-T7 - SC2 jamb plate 38x89 C16, SC2 jamb plate 38x89 C16
FIXES TO	SC2 - its E face against the steel (E face, W face)
CUT FROM	38x89 x 2700; as stock - no ripping
HOLES	3 No. - 14 through
COUNTERBORE	60 x 18 deep on the face away from the steel (E / W)
FIXINGS	3 x M12 8.8 threaded rod 173 + 2 nuts (timber jamb)
PURPOSE	W face jamb plate: the panel jamb stud nails to it

P4-T7 HOLES

#	from bottom	up S	dn N	hole	c'bore	fixing
1	300	45	44	14 thru	60x18 outer	M12 rod
2	1231	45	44	14 thru	60x18 outer	M12 rod
3	2162	45	44	14 thru	60x18 outer	M12 rod

P4-T8 HOLES

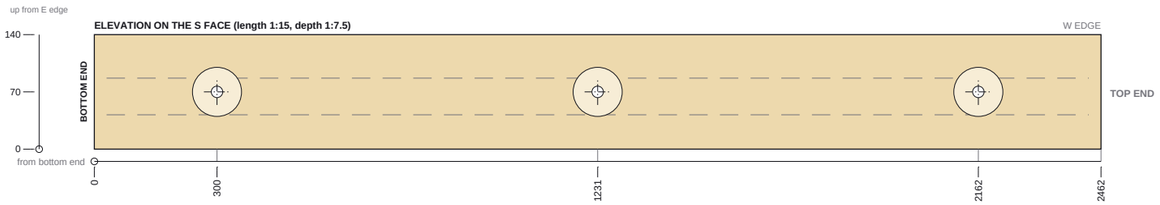
#	from bottom	up E	dn W	hole	c'bore	fixing
1	300	70	70	14 thru	60x18 outer	M12 rod
2	1231	70	70	14 thru	60x18 outer	M12 rod
3	2162	70	70	14 thru	60x18 outer	M12 rod

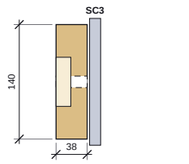
P4-T9 HOLES

#	from S	up E	dn W	hole	c'bore	fixing
1	180	70	70	14 thru	-	M12 stud
2	780	70	70	14 thru	-	M12 stud
3	1380	70	70	14 thru	-	M12 stud
4	1980	70	70	14 thru	-	M12 stud
5	2580	70	70	14 thru	-	M12 stud
6	3180	70	70	14 thru	-	M12 stud
7	3780	70	70	14 thru	-	M12 stud
8	4380	70	70	14 thru	-	M12 stud
9	4980	70	70	14 thru	-	M12 stud

P4-T8 SC3 JAMB PLATE — 38x140 C16 x L 2462 - 2 No.

TLS DETAIL - ENGINEER TO CONFIRM

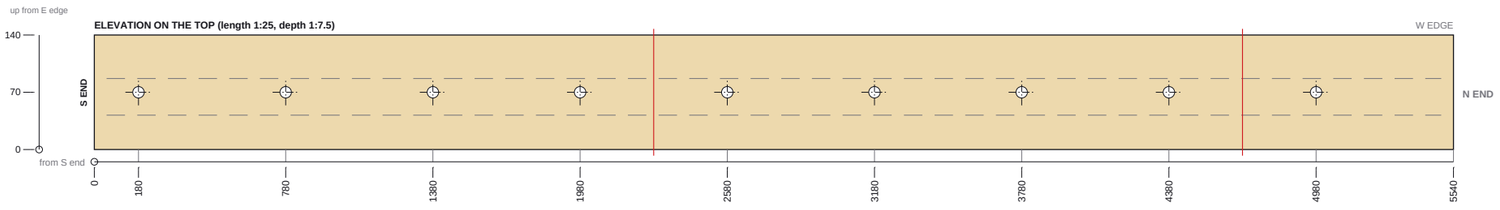


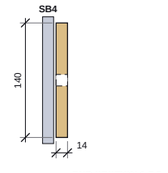


PIECE	P4-T8 - SC3 jamb plate 38x140 C16, SC3 jamb plate 38x140 C16
FIXES TO	SC3 - its N face against the steel (N face, S face)
CUT FROM	38x140 x 2700; as stock - no ripping
HOLES	3 No. - 14 through
COUNTERBORE	60 x 18 deep on the face away from the steel (N / S)
FIXINGS	3 x M12 8.8 threaded rod 173 + 2 nuts (timber jamb)
PURPOSE	S face jamb plate: the panel jamb stud nails to it

P4-T9 SB4 PACKER — 14x140 marine ply x L 5540 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM





PIECE	P4-T9 - SB4 packer 14mm marine ply 140
FIXES TO	SB4 - its underside against the steel
CUT FROM	2440 x 1220 sheet, 14 thick; rip 140 strips
LENGTHS	3 lengths 2280 + 2400 + 860 (joints at 2280, 4680 from the S end, midway between studs; red on the elevation)
HOLES	9 No. - 14 through
COUNTERBORE	none
FIXINGS	9 x M12x55 stud welded to RHS + nut + washer
PURPOSE	packer under the head binder, over the RHS and the post caps (3 lengths from 2440 sheets, joints over the RHS)

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL TIMBER SUBJECT TO SITE MEASURE AGAINST THE ERECTED STEEL BEFORE DRILLING.
3. Timber: strength-graded to BS EN 14081, C16 / C24 as scheduled, regularised; preservative treatment to the project specification (by others). Treat all cut ends and drilled holes.
4. Holes: bolts, rods and studs drilled d + 2 (M12 = 14), square to the face. Counterbores flat-bottomed (Forstner), concentric with the hole. Coach screws pre-drilled per EN 1995-1-1 8.7.
5. Running dimensions from the named reference end; across dimensions from the named edge. Dimensions in mm.
6. Bolt spacings checked against EN 1995-1-1 8.5 as set by the client: end 7d, edge 4d (M12: 84 / 48). Shortfalls are listed in red on STT-00.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel timber packing drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

TIMBER PACKING - SC2 JAMB PLATE, SC3 JAMB PLATE, SB4 PACKER

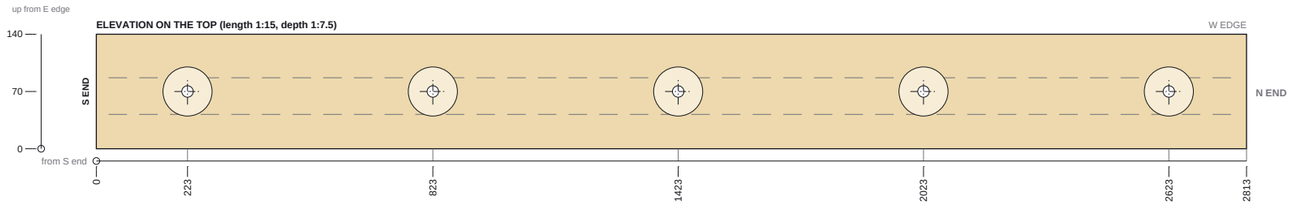
SCALE	SIZE	DATE	DRAWN
AS SHOWN	A3	11.10.26	TS

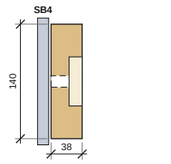
STATUS	DRAWING No.	REV
FOR APPROVAL	P4-STT-03	C03

Do not scale. All dimensions in mm. Check all dimensions on site.

P4-T10 SB4 HEAD BINDER (A) 38x140 C16 x L 2813 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM



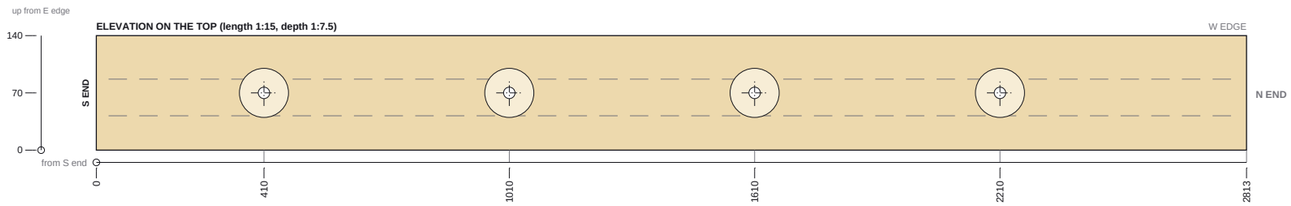


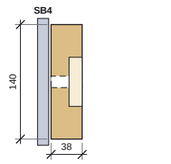
END SECTION 1:7.5

PIECE	P4-T10 - SB4 head binder 38x140 C16 (a)
FIXES TO	SB4 - its underside against the steel
CUT FROM	38x140 x 3000; as stock - no ripping
HOLES	5 No. - 14 through
COUNTERBORE	60 x 16 deep on the top
FIXINGS	5 x M12x55 stud welded to RHS + nut + washer
PURPOSE	head binder over the steel (GT1 bearing)

P4-T11 SB4 HEAD BINDER (B) 38x140 C16 x L 2813 - 1 No.

TLS DETAIL - ENGINEER TO CONFIRM



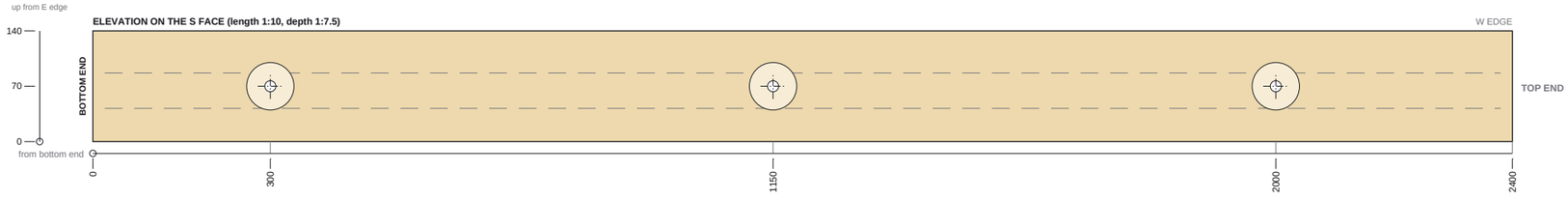


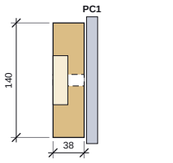
END SECTION 1:7.5

PIECE	P4-T11 - SB4 head binder 38x140 C16 (b)
FIXES TO	SB4 - its underside against the steel
CUT FROM	38x140 x 3000; as stock - no ripping
HOLES	4 No. - 14 through
COUNTERBORE	60 x 16 deep on the top
FIXINGS	4 x M12x55 stud welded to RHS + nut + washer
PURPOSE	head binder over the steel (GT1 bearing)

P4-T12 PC1 / PC2 JAMB PLATE 38x140 C16 x L 2400 - 2 No.

TLS DETAIL - ENGINEER TO CONFIRM





END SECTION 1:7.5

PIECE	P4-T12 - PC1 jamb plate 38x140 C16, PC2 jamb plate 38x140 C16
FIXES TO	PC1 - its N face against the steel (N face, S face)
CUT FROM	38x140 x 2700; as stock - no ripping
HOLES	3 No. - 14 through
COUNTERBORE	60 x 18 deep on the face away from the steel (N / S)
FIXINGS	3 x M12x160 8.8 bolt set (timber)
PURPOSE	jamb plate on the face against the return panel (to the underside of the post cap)

P4-T10 HOLES

#	from S	up E	dn W	hole	c'bore	fixing
1	223	70	70	14 thru	60x16 top	M12 stud
2	823	70	70	14 thru	60x16 top	M12 stud
3	1423	70	70	14 thru	60x16 top	M12 stud
4	2023	70	70	14 thru	60x16 top	M12 stud
5	2623	70	70	14 thru	60x16 top	M12 stud

P4-T11 HOLES

#	from S	up E	dn W	hole	c'bore	fixing
1	410	70	70	14 thru	60x16 top	M12 stud
2	1010	70	70	14 thru	60x16 top	M12 stud
3	1610	70	70	14 thru	60x16 top	M12 stud
4	2210	70	70	14 thru	60x16 top	M12 stud

P4-T12 HOLES

#	from bottom	up E	dn W	hole	c'bore	fixing
1	300	70	70	14 thru	60x18 outer	M12 bolt
2	1150	70	70	14 thru	60x18 outer	M12 bolt
3	2000	70	70	14 thru	60x18 outer	M12 bolt

NOTES

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3. Timber: strength-graded to BS EN 14081, C16 / C24 as scheduled, regularised; preservative treatment to the project specification (by others). Treat all cut ends and drilled holes.
4. Holes: bolts, rods and studs drilled d + 2 (M12 = 14), square to the face. Counterbores flat-bottomed (Forstner), concentric with the hole. Coach screws pre-drilled per EN 1995-1-1 8.7.
5. Running dimensions from the named reference end; across dimensions from the named edge. Dimensions in mm.
6. Bolt spacings checked against EN 1995-1-1 8.5 as set by the client: end 7d, edge 4d (M12: 84 / 48). Shortfalls are listed in red on STT-00.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel timber packing drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

TIMBER PACKING - T10 TO T12

SCALE	SIZE	DATE	DRAWN
AS SHOWN	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-STT-04	C03

Do not scale. All dimensions in mm. Check all dimensions on site.